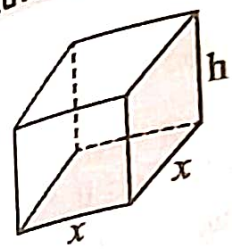
Illustrative Example

103. A manufacture wants to design open box having as square based and a surface area of $108m^2$. What dimension will produce a box with maximum volume?

Solution



Surface area of box:

A = (Area of base) + (Area of 4 faces)

$A = x^2 + 4xh = 108m^2$, solving for h,

$$\Rightarrow h = \frac{108 - x^2}{4x}$$

Volume: V = (Based area) (height) $= x^2 h$

$$V(x) = x^2\left(\frac{108 - x^2}{4x}\right) = 27x - \frac{1}{4}x^3$$

$$V'(x) = 27 - \frac{3}{4}x^2,\ V'(x) = 0 \Rightarrow 27 - \frac{3}{4}x^2 = 0$$

$$x^2 = \frac{108}{3} = 36$$

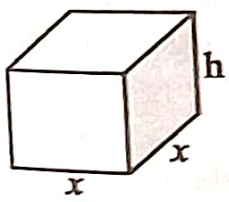
$$\therefore \quad x = 6 \text{ and } h = \frac{108 - 6^2}{4(6)} = \frac{72}{24} = 3$$

$\therefore$ Volume is maximum when height is h = 3m and base edge of 6m.

104. An open top square based a rectangular box contain is to have a volume of 32 cm^3. What should be height (h) of the box to obtain the minimum possible surface area?

A. h = 2cm B. h = 8cm C. h = 4cm D. h = 16cm

Solution



Volume; V = (height)

$$V = x^2 h = 32, h = \frac{32}{x^2}$$

Surface area = $x^2 + 4xh$

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$$A(x) = x^2 + 4x\left(\frac{32}{x^2}\right) = x^2 + \frac{128}{x}$$

$$A'(x) = 2x - \frac{128}{x^2}$$

$$A'(x) = 0,\ 2x^3 = 128,\ x^3 = 64,\ x = 4$$

$$\therefore \quad h = \frac{32}{x^2} = \frac{32}{16} = 2cm$$

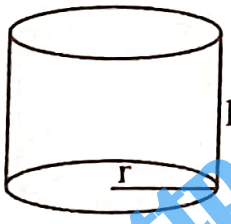
105. A cylindrical can is to be to hold 256π cm^3 of oil. Find the dimension that will minimize the cost of the metal to manufacture the can. Answer: A

Solution

In order to minimize the cost of the metal, we minimize the total surface area of the cylinder.

$$\text{Volume} = \pi r^2 h = 256\pi cm^3$$

$$h = \frac{256}{r^2}$$



Surface area: $A = 2\pi r^2 + 2\pi rh$

$$A(r) = 2\pi r^2 + 2\pi r\left(\frac{256}{r^2}\right) = 2\pi r^2 + \frac{512\pi}{r}$$

$$A'(r) = 4\pi r - \frac{512\pi}{r}$$

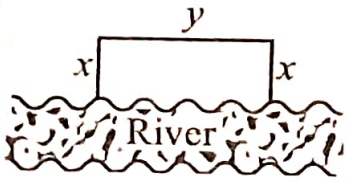
$$A'(r) = 0 \text{ when } r^3 = \frac{512}{4} = 128$$

$$\therefore \quad r = \sqrt[3]{128} = 4\sqrt[3]{2}$$

$$\therefore \quad h = \frac{256}{r^2} = \frac{16}{\sqrt[3]{4}}$$

106. A farmer has 3200m of fencing and wants to fence off a rectangular field that borders a straight rivers. He needs no fence along the river. What are the dimensions of the field that has the largest area?

Solution



Let x = Width of the rectangle.

x = the depth of the rectangle.

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Perimeter: $2x + y = 3200$, and $y = 3200 - 2x$

Area: $A = xy$ and $A(x) = x(3200 - 2x)$.

$A(x) = 3200x - 2x^2$ and $A'(x) = 3200 - 4x$

$A'(x) = 0$, when $3200 - 4x = 0$,

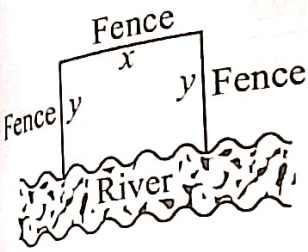
$x = 800$ and $y = 1600$

$\therefore$ The rectangular field should be 800m deep and 1600m wide.

107. **The short fence:** A farmer plans to fence a rectangular pasture adjacent to a river. The pasture must contain $180{,}000m^2$ in order to provide enough grass for the herd. What dimensions would requires the smallest amount of fencing if no fencing is needed along river.

Solution

We need to minimize the fencing martial.



Area: $xy = 180{,}000m^2$, $y = \dfrac{180{,}000}{x}$

Perimeter: $y + y + x$ = minimum value

$P(x) = \dfrac{360{,}000}{x} + x$ and $P'(x) = 1 - \dfrac{360{,}000}{x^2}$

$P'(x) = 0$, when $x^2 = 360{,}000$, $x = 600$, $y = 300$

$\therefore$ The smallest amount of fencing used when $x = 600$m and $y = 300$m

108. A farmer wants to make three identical rectangular enclosure along a straight river. If he has a fencing material of 1200 meter and the side along the river need no fence. What should be the dimension (length ℓ and width) of each enclosure if the total area of the enclosure is to be maximized?

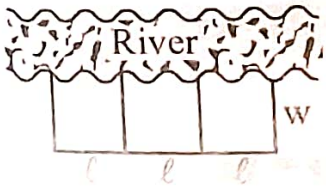
A. $\ell = 100$m, W = 225m

B. $\ell = 300$m, W = 75m

C. $\ell = 200$m, W = 200m

D. $\ell = 200$m, W = 150m

Solution



Let ℓ = length of each enclosure

W = width of each enclosure

Length of fence: $L = 3\ell + 4W = 1200$m

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$4W = 1200 - 3\ell$, $W = 300 - \frac{3}{4}\ell$

Area: $A = (3\ell)W = 3\ell\left(300 - \frac{3}{4}\ell\right)$

$A(\ell) = 900\ell - \frac{9\ell^2}{4}$ and $A'(\ell) = 900 - \frac{9\ell}{2}$

$A'(\ell) = 0\ell = 200$, and $W = 300 - \frac{3}{4}(200) = 150$ **Answer: D**

109. Find two numbers whose product is 192, the sum of the second number and three times the first is minimum.

Solution

Let x = the first number and y = the second number.

Then product of number: $xy = 192$, and $y = \frac{192}{x}$.

Sum: $S = y + 3x$ is to be minimized

$S(x) = 3x + \frac{192}{x}$, So that $S'(x) = 3 - \frac{192}{x^2}$

$\Rightarrow S'(x) = 0$ when $\frac{192}{x^2} = 3$

$\Rightarrow x = 8$ and $y = \frac{192}{8} = 24$.

$\therefore$ The two numbers are $x = 8$ and $y = 24$

110. Find two non negative number when whose is 24 and whose product is a maximum.

Solution

Let x and denote the two non negative numbers:

$x + y = 24$ and product: $P = xy$

$P(x) = x(24 - x) = 24x - x^2$ and $P'(x) = 24 - 2x$

$P'(x) = 0$, when $x = 12$.

$\therefore$ The two numbers are $x = 12$ and $y = 12$

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111. Find two non negative numbers whose product is 50 and whose sum minimum.

Solution

Let x and y denote the two non negative numbers

Product: $P = xy = 50$ and $y = \frac{50}{x}$

Sum: $S = x + y = x + \frac{50}{x}$, $S'(x) = 1 - \frac{50}{x}$

$S'(x) = 0$, when $x^2 = 50$, and $x = 5\sqrt{2}$

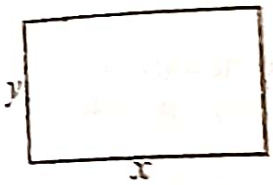
$\therefore \quad x = 5\sqrt{2}$ and $y = 5\sqrt{2}$

112. Find the length and width of a rectangle with perimeter 100 that maximize area.

Solution: Let x = the length and y = width perimeter $P = 2x + 2y = 100$

$y = 50 - x$

Area: $A = xy$



$A(x) = x(50 - x) = 50x - x^2$

$A'(x) = 50 - 2x$, we set $A'(x) = 0$

$50 - 2x = 0$, implies $x = 25$ and $y = 25$

$\therefore$ The length : $x = 25$ and width : $y = 25$ that gives the maximum area.

113. Find the length and width of a rectangle with area 64m^2 that minimized the perimeter.

Solution: Let x = the length and y = width

Area: $A = xy = 64$, $y = \frac{64}{x}$

Perimeter: $P = 2x + 2y$

$P(x) = 2x + 2\left(\frac{64}{x}\right) = 2x + \frac{128}{x}$

$P'(x) = 2 - \frac{128}{x^2}$ and $P'(x) = 0$

$2x^2 = 128$, and $x^2 = 64$

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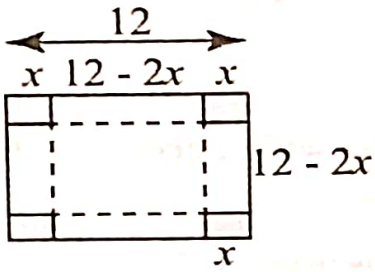
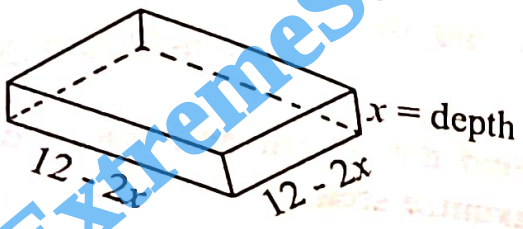
$$\therefore \quad x = 8m \text{ and } y = \frac{64}{8} = 8m$$

114. Maximizing volume: An open box is it be made by cutting a square from each corner of 12cm by 12cm, piece of metal and then folding up the sides. What size square should be cut from each corner to produce a box of maximum volume?

Solution

Let x = Length of a side of the square that is cut from each corner.





Volume of the box = Bh = $(12 - 12x)(12 - 2x)x$

$$V(x) = 144x - 48x^2 + 4x^3$$

Since neither the length nor the width can be negative.
Domain, of V is $0 \leq x \leq 6$.

$V'(x) = 144 - 96x + 12x^2 = 12(x^2 - 8x + 12)$

Set $V'(x) = 0$: $12(x - 2)(x - 6) = 0$

$\therefore \quad x = 2 \quad$ or $\quad x = 6.$

$\therefore$ The box will have maximum volume when $x = 2$ and the volume: $V = 128cm^3$.

115. If box with square based and open top is made from $1200cm^2$ material, what is the largest volume of the box in cm^3?

A. 4000 B. 8000 C. 1500 D. 3000

Solution

Let x = side of the square and h = height of the box

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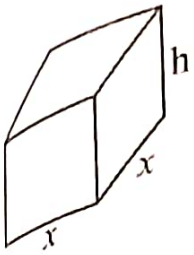
Surface area: $A = \begin{pmatrix}\text{Area of}\\ \text{based}\end{pmatrix} + \begin{pmatrix}\text{Area of}\\ \text{4 faces}\end{pmatrix}$

$$A = x^2 + 4xh = 1200cm^2$$



$$h = \frac{1200 - x^2}{4x} = \frac{300}{x} - \frac{1}{4}x$$

Now, the volueme $v = Bh = x^2 h$

$$V(x) = x^2\left(\frac{300}{x} - \frac{x}{4}\right) = 300x - \frac{x^3}{4}$$

$$V'(x) = 300 - \frac{3x^2}{4} \text{ and } V'(x) = 0$$

$$\text{if } \frac{3x^2}{4} = 300, \text{ then } x^2 = \frac{300 \times 4}{3} = 400$$

This implies $x = 20$cm and $h = 10$cm

$\therefore$ Volume $= Bh = (400)(10) = 4000\ cm^3$

Answer: A

11.5.1 Related Rates

Application of Derivative in finding Rates of change.

If a quantity y is a function of time t, the rate change of y with respect to time is given by $\frac{dy}{dt}$

- If a quantity y is a function of x, and x is a function of time t, the rate of change of y with respect to time is given by $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$
- The rates of change $\frac{dy}{dt}$ and $\frac{dx}{dt}$ are related, for this reason they are called related rate.

For example: a $y = x^2, \Rightarrow \frac{dx}{dt} = 2x\frac{dx}{dt}$

b. $A = \pi r^2 \Rightarrow \frac{dA}{dt} = 2\pi r^2 \frac{dr}{dt}$

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Illustrative Example

116. A point movies a long the curve $5y = x^2$. If the x - coordinate increases at a constant rate of 5cm/sec, at what rate the y - coordinate is increasing when $x = 4$.

Solution

$$\text{At any time } t = \frac{dx}{dt} = 5\,{}^{m}/_{sec}$$

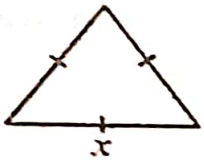
$$y = \frac{1}{5}x^2 \text{ then } \frac{dy}{dt} = \frac{2}{5} \times \frac{dx}{dt}.$$

$$\therefore \quad \frac{dy}{dt} = \frac{2}{5}(4)\left(5\,{}^{m}/_{sec}\right) = 8\,{}^{cm}/_{sec}$$

117. The side of an equilateral triangle is 4cm, and is increasing at the rate of 2cm/sec. At what rate the area is increasing?

Solution

Area of an equilateral triangle of side x

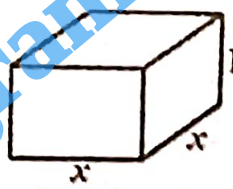


$$A = \frac{1}{2}x^2\frac{\sqrt{3}}{2} = \frac{\sqrt{3}\,x^2}{4} \text{ and } \frac{dx}{dt} = 2\,{}^{cm}/_{sec}$$

$$\frac{dA}{dt} = \frac{2\sqrt{3}}{4}x\,\frac{dx}{dt} = \frac{\sqrt{3}}{2}(4\text{cm})\left(2\,{}^{cm}/_{sec}\right) = 4\sqrt{3}\,{}^{cm^2}/_{sec}$$

118. The edge of a cube is increasing at the rate of 0.4cm/sec. At what rate is the surface area increasing when the edge of the cube is 10cm.

Solution



Surface area of a cube with edge x : $A = 6x^2$, $\frac{dx}{dt} = 0.4\,{}^{cm}/_{sec}$

$$\frac{dA}{dt} = 12x\,\frac{dx}{dt} = 12(10\,cm)\left(0.4\,{}^{cm}/_{sec}\right) = 48\,{}^{cm^2}/_{sec}$$

119. The edge of a cube is 5cm long and volume is increasing at the rate of $\frac{2cm^3}{min}$. How fast the surface of a cube is increasing?

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A. $\frac{2}{5}\,cm^2/min$ B. $\frac{8}{5}\,cm^2/min$ C. $\frac{25}{5}\,cm^2/min$ D. $8\,cm^2/min$

Solution

Volume of a cube is $V = x^3$, and $\frac{dv}{dt} = 2\,m^3/min$

$$\frac{dv}{dt} = 3x^2\,\frac{dx}{dt} \Rightarrow 2\,cm^3/min = 3(5)^2\,\frac{dx}{dt}$$

$$\frac{dx}{dt} = \frac{2}{75}\,cm/min$$

Surface: $A = 6x^2$, $\frac{dA}{dt} = 12x\,\frac{dx}{dt} = 12(5)\left(\frac{2}{75}\right) = \frac{8}{5}$

Answer: B

120. The side of square increases uniformly at the rate of 2cm per second at what rate is the area increasing when the side of the square is 6cm.

Solution : $\frac{dx}{dt} = 2\,cm/sec$

Let the side of square $= x$

Area of a square: $A = x^2$, then $\frac{dA}{dt} = 2x\,\frac{dx}{dt}$

$$\therefore \quad \frac{dA}{dt} = 2(6cm)\left(2\,cm/sec\right) = 24\,cm^2/sec$$

121. The radius of a circle is increasing at the rate of 3cm/sec. Find the rate at which the area is increasing when the radius is 4cm.

Solution

Area of a circle: $A = \pi r^2$

Rate of change of area of a circle with respect to time t is given by:

$$\frac{dA}{dt} = 2\pi r\,\frac{dr}{dt} = 2\pi\,(4\ cm)\left(3\,cm/sec\right) = 24\pi\ cm^2/sec$$

122. The radius of a sphere is r where the time is t - seconds. Find the radius when the rate of change of the surface area and the rate of change of the radius are equal.

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Solution

The surface area: $A = 4\pi r^2$ and $\frac{dA}{dt} = 8\pi r \frac{dr}{dt}$

We have given: $\frac{dA}{dt} = \frac{dr}{dt}$

$\Rightarrow 8\pi r \frac{dr}{dt} = \frac{dr}{dt}$, $r = \frac{1}{8\pi}$

123. The volume of a spherical baloon is increasing at the of 24 cm^3/sec Find the rate of change of its surface area when its radius is 4cm.

Solution

Volume $= \frac{4}{3}\pi r^3$ and surface area: $A = 4\pi r^2$

Rate of change of volume with respspct to time t

$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$

$24\ cm^3/_{sec} = 4\pi\ (16cm^2)\frac{dr}{dt}$, and $\frac{dr}{dt} = \frac{24}{64\pi} = \frac{3\,cm/s}{8\pi}$

Again: $\frac{dA}{dt} = 8\pi r \frac{dr}{dt} = 8\pi\,(4)\left(\frac{3}{8\pi}\right) = 12\ cm^2/_{sec}$

124. Water is poured into a cylinder tanker of radius 5metrs at a rate of 10m^3/min. What is the rate of change of height of the level of water when it risen to 3 meters?

A. $\frac{3}{25\pi}$ m/min B. $\frac{2}{5\pi}$ m/min C. $\frac{1}{2\pi}$ m/min D. $\frac{2}{5\pi}$ m/min

Solution

Volume of Water $V = \pi r^2 h = \pi(5)^2 h = 25\pi h$

$\frac{dV}{dt} = 25\pi \frac{dh}{dt}$, since r is constant

$10\ m^3/_{min} = 25\pi \frac{dh}{dt}$

$\therefore \frac{dh}{dt} = \frac{10}{25\pi} = \frac{2}{5\pi}\ m/_{min}$

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Answer: B

125. Air is being pumped into a spherical balloon at a rate of $6m^3/min$. At what rate is the radius changing when the radius is 2m.

A. $\frac{3}{8}$ m/min B. $\frac{3}{8\pi}$ m/min C. $\frac{9}{4\pi}$ m/min D. $\frac{4}{9\pi}$ m/min

Solution

"Air is being pumped ... at a rate of $6\,m^3/min$"

interpreted as: $\frac{dv}{dt} = 6\,m^3/min$

We know: $V = \frac{4}{3}\pi r^3$ and $\frac{dv}{dt} = 4\pi r^2 \frac{dr}{dt}$

Now, $\frac{dr}{dt} = \frac{6}{4\pi r^2}$,

$\therefore \quad \left.\frac{dr}{dt}\right|_{r=2} = \frac{6}{4\pi(2)^2} = \frac{6}{16\pi} = \frac{3}{8\pi}\,m/min$

Answer: B

126. Gas is escaping from a spherical balloon at the rate of $3m^3/min$. How fast is the surface area when the radius is 3m?

Solution

A sphere of radius has aVolume: $V = \frac{4}{3}\pi r^3$

$\frac{dv}{dt} = 4\pi r^2 \frac{dr}{dt}$

$2\frac{m^3}{min} = 4\pi(3)^2 \frac{dr}{dt}$ then $\frac{dr}{dt} = \frac{2}{36\pi} = \frac{1}{18\pi}$

A surface area: $A = 4\pi r^2$

$\therefore \quad \frac{dA}{dt} = 8\pi r \frac{dr}{dt} = 8\pi(3)\frac{1}{18\pi} = \frac{4}{3}\,m^2/min$

$\therefore$ The surface area is shrinking at the rate of $\frac{4}{3}\,m^2/min$

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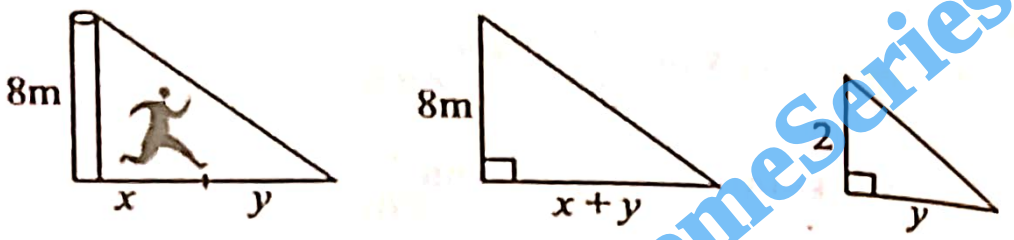
127. Obang walks at the rate of 5m/sec to ward a street light whose lamp is 8m high. If Obang is 2m tall, determine the rate of change of the length of Obang's shadow:

Solution

At any time t:

Let x = the distance between Obang and the lamp post

y = the length of Obang's shadow



By the properties of similar triangle:

$$\frac{8}{x+y} = \frac{2}{y}, \text{ this implies } y = \frac{1}{3}x$$

$$\frac{dy}{dt} = \frac{1}{3}\frac{dx}{dt} = \frac{1}{3}\left(5\,{}^{m}/_{sec}\right) = \frac{5}{3}\,{}^{m}/_{sec}$$

11.5.2 Application of Derivative Economics

The function $f'(x)$ represents the instantaneous rate of change of $y = f(x)$ with respect to x.

This instantaneous rate of change could be interpreted as **marginal cost, marginal revenue,** or **marginal profit** if the original function represented cost, revenue or profit.

The second has to do with minimizing the cost. Three functions of importance to an economist or a manufacture are **cost, revenue and profit.**

C (x) = total cost of producing the x - units (items)

R (x) = total revenue from selling x - units (items).

P (x) = total profit obtained by selling x - units (item)

These are called, respectively, the **cost function, revenue function,** and **profit function.** These three functions are related by: P(x) = R(x) - C (x).

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Note: when a company sell x - item at a price P - Birr per unit, then the revenue: $R = xP$.

Marginal Functions

In economics the term **marginal function** refers the derivative of cost, revenue and profit function. The **marginal cost, marginal revenue and marginal profit** when producing and selling x - items are

$\frac{dR}{dx} = R'(x)$ = marginal revenue (Derivative of revenue function)

$\frac{dc}{dx} = C'(x)$ = marginal cost (Derivative of cost function)

$\frac{dP}{dx} = P'(x)$ = marginal profit (Derivative of profit function)

Illustrative Example

128. Given $R(x) = -x^2 + 400x$ and $C(x) = x^2 + 40x + 100$ are revenue are cost function respectively for producing x - units of a commodity.

a. Find the maximum profit
b. Find the maximum revenue
c. Find the marginal revenue at $x = 10$
d. Find the marginal cost at $x = 10$
e. Find the actual cost of producing the 10^{th} unit.

Solution

a. $R'(x) = -2x + 400$, and $R'(x) = 0$, at $x = 200$

$\therefore$ Maximum revenue: $R(200) = -(200)^2 + 400(200) = 40{,}000$

b. $P(x) = R(x) - C(x) = (x^2 + 400x) - (x^2 + 40x + 100)$

$P(x) = -2x^2 + 360x - 100$

$P'(x) = -4x + 360$ and $P(x) = 0$, at $x = 90$

$\therefore$ maximum profit: $P(90) = -2(90)^2 + 360(90) - 100 = 16100$

c. Marginal revenue is the derivative or revenue function.

$R'(x) = -2x + 400$

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The revenue at the moment of producing 10 unites is

$R'(10) = -2(10) + 400 = 380$

d. Marginal cost is the derivative of cost function.

$C'(x) = 2x + 40$

The cost at the moment of producing 10 units is

$C'(10) = 2(10) + 40 = 60$

e. The actual cost of producing the 10^{th} units is: $C(10) - c(9)$

$(10^2 + 40(10) + 100) - (9^2 + 40(9) + 100) = 59$

129. Suppose that $C(x) = \frac{x^2}{3} - 4x^2 + 16x$ and $R(x) = 4x$ are total cost and revenue function, respectively, for producing x units of a certain product.

a. Find the total profit

b. Find minimum profit

c. Find marginal cost when $x = 5$

Solution

a. $P(x) = R(x) - C(x)$

$$= 4x - \left(\frac{x^3}{3} - 4x^2 + 16x\right)$$

$$= \frac{-x^3}{3} + 4x^2 - 12x$$

b. $P'(x) = -x^2 + 8x - 12 = (2 - x)(x - 6)$

$P'(x) = 0$, when $x = 2$ and $x = 6$

$\therefore$ maximum profit: $P(6) = \frac{-108}{3} + 144 - 72 = 36$

c. $C'(x) = x^2 - 8x + 16$

$C'(5) = 25 - 40 + 16 = 1$

130. The cost in ETB to manufacture x garment products is given by $C(x) = -0005x^2 + 30x + 200$ when $0 \leq x \leq 300$.

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Find the rate of change of cost with respect to the number manufactured when 150 clothes when 1000 clothes are made.

Solution: The rate of change of cost is given by the derivative of the cost function $C'(x) = -0.01x + 30$

When $x = 150$, $C'(150) = -0.01(150) + 30 = 28.5$

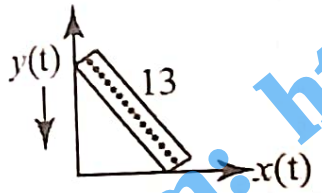
This rate of change of cost per clothes gives the marginal cost at $x = 150$ which means the approximate cos of producing 151st cloth is ETB 28.50. When 1000 clothes are made, the marginal cost is

$$C'(1000) = (-0.01)(1000) + 30 = 20$$

131. **A sliding ladder:** A 25*ft* ladder is leaning a gainst a house when its base starts to slide away. By the time the base is 15*ft* from the house, the base is moving at the rate of 8 *ft*/sec. How fast is the ladder sliding down the wall then? At what rate is the area of the triangle formed by the ladder, wall, and ground changing then?

Solution :- $x = 15\,ft$, $\frac{dx}{dt} = 8\,^{ft}/_{sec}$

a. $x^2 + y^2 = 25^2$, $y = \sqrt{25^2 - 15^2} = 20$



$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

$$2(15)(8) + 2(20)\frac{dy}{dt} = 0$$

$$\frac{dy}{dt} = -6\,^{ft}/_{sec}$$

Area: $A = \frac{1}{2}xy$

$$\frac{dA}{dt} = \frac{1}{2}\left(x\frac{dy}{dt} + y\frac{dx}{dt}\right)$$

$$= \frac{1}{2}(15(-6) + 20(8)) = 35\,^{ft^2}/_{sec}$$

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11.6 Introduction to Integration

The basic problem in integral calculus is finding the area under curve. The term " integration" is used in two ways. It refers to:

(i) The process of finding area (computing of definite integral)

(ii) The process of finding anti - derivative

11.6.1 The Area Problem

In this section, we will consider the problem of calculating areas of plane regions with curve linear boundaries. There are two basic methods for finding the area of the region. The rectangle method and the anti - derivative method.

1. The rectangle method for finding Area.

Approximating Area by Rectangles method.

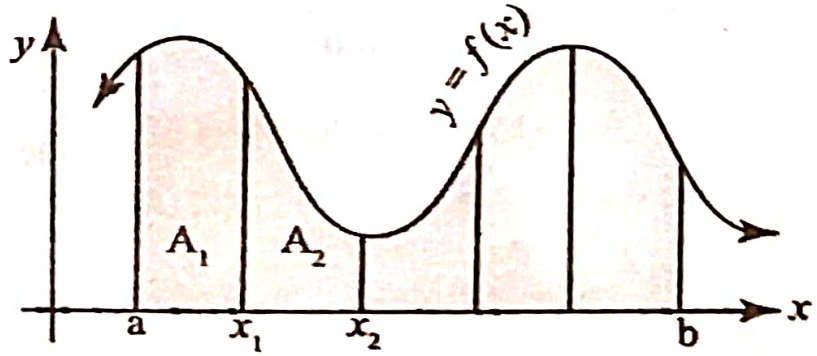
Our goal is to compute the area under the graph of a function $f(x) \geq 0$ over the interval [a, b] that lies above the x - axis.

To begin, the interval [a, b] into n subintervals of equal width.

- The full interval [a, b] has width b - a.
- Each subinterval has width $\Delta x = \dfrac{b-a}{n}$
- a = left endpoint of interval [a, b]
- b = right endpoint of interval [a, b]
- n = number of subintervals in [a, b]

The right end points of the subintervals are

$$x_1 = a + \Delta x, x^2 = a + 2\Delta x, \ldots\ldots x_{i-1} = a + (i-1)\Delta x \text{ and } x_n = a + n\Delta x = b$$



The sum of the areas of these rectangles provides an approximation to the area under the graph.

Dividing [a, b] into

n - subinterval.

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- The first rectangle has base Δx and height $f(x_1)$, so its area is $f(x_1)\,\Delta x$.
- Second rectangle has height $f(x_2)$, so its area is $f(x_2)\,\Delta x$.
- The i^{th} rectangle has height $f(x_i)$, so its area is $f(x_i)\,\Delta x$.
- The area of all n - rectangle $= \sum_{i=1}^{n} f(x_i)\,\Delta x$
- There are three rectangular area approximation option:

(i) Right end point approximation

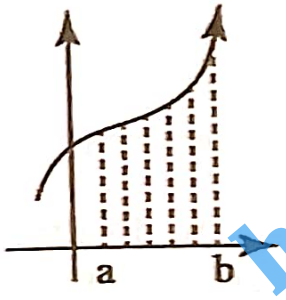
- The height of the rectangles are the values of $f(x)$ at the right end point.

(ii) Left endpoint approximation.

- The height of the rectangles are the values of $f(x)$ at the left of $f(x)$ at the left endpoint.

(iii) Midpoint approximation.

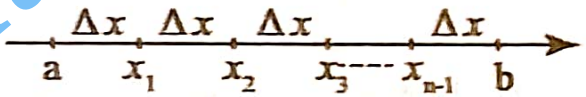
- The height of the rectangles, are the values of $f(x)$ at the midpoints of the subinterval rather than the end points.



To approximate the area described above:

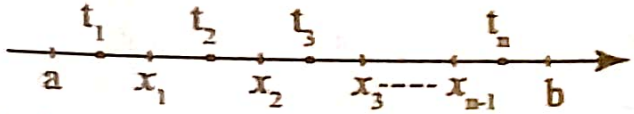
STEP 1: Partition [a, b] into n - equal subintervals each with width:

$$\Delta x = \frac{b-a}{n}$$



STEP 2: Select a sample point t_k from each subinterval $[x_{k-1}, x_k]$.

(Usually the choice is made in a systemic way, such as, right end point, left endpoint or midpoint).



STEP 3: Calculate sum of the areas of all n - rectangles.

$$\sum_{k=1}^{n} f(t_k)\Delta x = f(t_1)\Delta x + f(t_2)\Delta x + f(t_3)\Delta x + \ldots + f(t_n)\Delta x$$

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Where $f(t_k)$ is the height of kth rectangle and Δx is the wide.

Illustrative Example

132. Let $f(x) = x^3 + x$ defined on the interval [2, 8]. Then find the approximate area of the function by dividing the interval [2, 8] into 3 subintervals [2, 8] into 3 subintervals, (by using midpoint approximation).

Solution

$$\text{For } n = 3,\ \Delta x = \frac{8 - 2}{3} = 2$$

We use the midpoint of each subinterval for sample point t_k.

Subinterval	t_k	$f(t_k)$	$f(t_k)\,\Delta x$
[2, 4]	$t_1 = 3$	30	60
[4, 6]	$t_2 = 5$	130	260
[6, 8]	$t_3 = 7$	350	700

$\therefore$ The approximate area of the region is

$$A = 60 + 260 + 700 = 1020$$

133. Use midpoint approximation :

Let $f(x) = 20x - x^2$

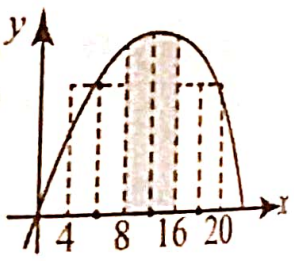
Find the approximate area of the function by dividing the interval [4, 16] into 3 subintervals of equal width. (use midpoints as ample points)

Solution

$$\text{For } n = 3,\ \Delta x = \frac{16 - 4}{3} = 4$$

Use the midpoint of each subinterval for sample points (t_k)

Subinterval	Sample point, t_k	$f(t_k)$	$f(t_k)\,\Delta x$
[4, 8]	$t_1 = 6$	84	336
[8 12]	$t_2 = 10$	100	400
[12 16]	$t_3 = 14$	84	336



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The area is approximately:

$A = 336 + 400 + 336 = 1072$

134. Use the **right endpoints** as sample points t_k to approximate the area under the function $f(x) = x^2$ on the interval [1 3], by dividing into four subintervals of equal width.

Solution

To calculate, divide [1, 3] into n = 4 subintervals of width

$$\Delta x = \frac{3 - 1}{4} = \frac{2}{4} = \frac{1}{2}$$

K	Subinterval	t_k	$f(t_k)$	$f(t_k)\,\Delta x$
1	$\left[1, \frac{3}{2}\right]$	$t_k = \frac{3}{2}$	$f\left(\frac{3}{2}\right) = \frac{9}{4}$	$\frac{9}{2}$
2	$\left[\frac{3}{2}, 2\right]$	$t_2 = 2$	$f(2) = 4$	2
3	$\left[2, \frac{5}{2}\right]$	$t_3 = \frac{5}{2}$	$f\left(\frac{5}{2}\right) = \frac{25}{4}$	$\frac{25}{8}$
4	$\left[\frac{5}{2}, 2\right]$	$t_4 = 3$	$f(3) = 9$	$\frac{9}{2}$

$$\sum_{k=1}^{4} f(t_k)\,\Delta x = \frac{9}{8} + 2 + \frac{25}{2} + \frac{9}{2} = \frac{43}{4}$$

135. Use the **left endpoints** as samples points to estimate the area under the function $f(x) = 1 + 4x^2$ on the interval [2, 8] by dividing into four subintervals of equal with.

Solution

$$\Delta x = \frac{b - a}{n} = \frac{8 - 2}{4} = \frac{6}{4} = \frac{3}{2}$$

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K	Subinterval	t_k	$f(t_k)$	$f(t_k)\,\Delta x$
1	[2 3.5]	$t_1 = 2$	17	25.5
2	[3.5 5]	$t_2 = 3.5$	50	75
3	[5, 6.5]	$t_3 = 5$	101	151.5
4	[6.5 8]	$t_4 = 6.5$	170	255

The area is approximately: $A = 25.5 + 75 + 151.5 + 255 = 507$

Riemann sums and Definite Integrals

In area of region approximation (end point and midpoint) use rectangles of equal with Δx, whose hight are the values of $f(x)$ at the endpoints or midpoints of subintervals.

In **Riemann sum** approximation the rectangles need not have equal width and the height may be any value of $f(x)$ within the subinterval.

To specify a, we choose a partition and a set of sample points:

- **Partition P** of size n:

  Divide: [a, b] into n subintervals

  P: $a = x_0 < x_1 < x_2 < \ldots\ldots < x_n = b$

- **Sample points** $t = \{t_1, t_2 \ldots\ldots t_n\}$:

  t_i belongs to the subinterval

  $[x_{i-1}, x_i]$ for all $i = 1, 2, \ldots\ldots n$

- Width of the rectangle: $\Delta x_i = x_{i-1}$
- The height of the i^{th} rectangle is $f(t_i)$
- The area of the i^{th} rectangle is $f(t_i)\,\Delta x_i$ if $f(t_i) \geq 0$.
- The **Riemann** sum is the sum

$$\sum_{i=1}^{n} f(t_i)\,\Delta x_i = f(t_1)\,\Delta x_1 + f(t_2)\,\Delta x_2 + \ldots\ldots + f(t_n)\,\Delta x_n$$

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Definition of Riemann sum

Let f be defined on the closed interval [a, b] and let Δ be aparition of [a, b] given by

$a = x_0 < x_1 < x_2 < .... < x_{n-1} < x_n = b$

Where Δx_i is the width of the i^{th} subinterval $[x_{i-1}, x_i]$

If t_i is any a sample point in the i^{th} subinterval, then the sum

$$\sum_{i=1}^{n} f(t_i)\Delta x_i \ , \quad x_{i-1} \le t_i \le x_i$$

is called a **Riemann sum of f for petition Δ**

Note: If every subinterval is of equal width, then the partition is regular and is denote by $\Delta x = \frac{b - a}{n}$

Illustrative Example

136. Calculate the Riemann sum for $f(x) = x^2$ on [3, 10]

Where partition $P : x_0 = 3, x_1 = 4, x_2 = 6, x_3 = 9$ and $x_4 = 10$

$t_1 = 3, t_2 = 5, t_3 = 8, t_4 = 9$

Solution

The partion $P = [3, 4], [4, 6] [6, 9], [9, 10]$

$\Rightarrow \Delta x_1 = 1, \Delta x_2 = 2, \Delta x_3 = 3$ and $\Delta x_4 = 1$

i	Subinterval	the width Δx_i	Sample point t_i	Height $f(t_i)$	$f(t_i)\Delta x_i$
1	[3, 4]	$\Delta x_1 = 1$	$t_1 = 3$	9	9
2	[4, 6]	$\Delta x_2 = 2$	$t_2 = 5$	25	50
3	[6, 9]	$\Delta x_3 = 3$	$t_3 = 8$	64	192
4	[9, 10]	$\Delta x_4 = 1$	$t_4 = 9$	81	81

The Riemann sum is the sum

$$\sum f(t_i)\Delta x_i = 9 + 50 + 192 + 81 = 332$$

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The Definite Integral

As the rectangles get thinner and thinner (the partition tends to zero), the number of rectangle n tends to infinity (∞) and they approximate the area under the graph more closely,. This leads to the following definition f is **integrable** over [a, b] if all of the **Riemann sums** approach one and unique number as $\Delta x \to 0$

This unique number is called the **definite integral** of f over [a, b]. The definite integral as the **Riemann sum** of a function, f over the interval [a, b]

$$\sum_{i=1}^{n} f(t_i)\,\Delta x = \int_a^b f(x)\,dx$$

as $n \to \infty$ and $\Delta x \to 0$, and we choice t_i in the i^{th} subinterval of each partition Δx

DEFINITION: The Definite Integral

Let f be a function defined on closed interval [a, b]. Then the definite integral of f from a to b, defined to be

$$\int_a^b f(x)\,dx = \sum_{i=1}^{n} f(t_i)\,\Delta x$$

as $n \to \infty$, and $\Delta x \to 0$

The definite integral is often called the integral of f over [a, b]. The process of computing integrals is called **integration** and $f(x)$ is called the **integrand.**

Illustrative Example

137. Evaluate the definite integral as Riemann sum: $\int_3^2 4x\,dx$

Solution

The function $f(x) = 4x$ is integrable on the interval [-3, 2], because it is continuous on [-3, 2].

To evaluate, we begin by subdividing the interval [-3, 2] into n - subintervals, each of width : $\Delta x = \dfrac{2 - (-3)}{n} = \dfrac{5}{n}$

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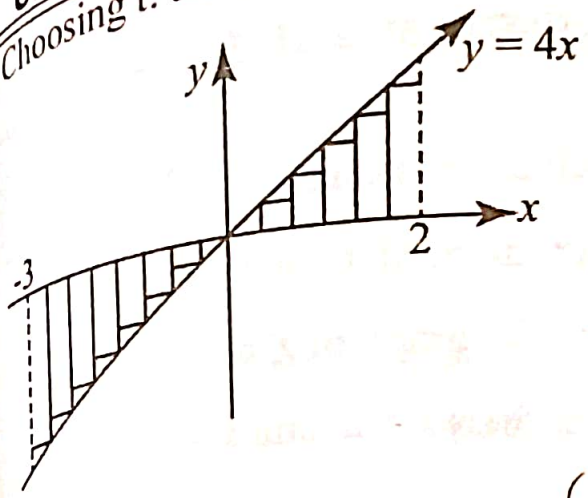
Choosing t_i as the right endpoint of each subinterval produces:



$$t_i = a + i\Delta x = -3 + \frac{5i}{n}$$

So the definite integral is

$$\int_{-3}^{2} 4x\,dx = \sum_{i=1}^{n} f(t_i)\Delta x,$$

$$as\ n \longrightarrow \infty,\ \Delta x \longrightarrow 0$$

$$f(t_i) = 4t_i = 4\left(-3 + \frac{5i}{n}\right)$$

$$\sum_{i=1}^{n} 4\left(-3 + \frac{5i}{n}\right)\left(\frac{5}{n}\right) = \frac{20}{n}\sum_{i=1}^{n}\left(-3 + \frac{5i}{n}\right)$$

$$= \frac{20}{n}\left[-3\sum_{i=1}^{n} 1 + \frac{5}{n}\sum_{i=1}^{n} i\right]$$

$$= \frac{20}{n}\left[-3n + \frac{5}{n}\left(\frac{n(n+1)}{2}\right)\right]$$

$$= -60 + \frac{100}{2n^2}(n^2 + n)$$

$$= -6 + 50\left(1 + \frac{1}{n}\right)$$

$$= -10 + \frac{50}{n},\ as\ n \longrightarrow \infty,\ \frac{50}{n} \longrightarrow 0$$

$$\therefore \int_{-3}^{2} 4x\,dx = -10$$

Definite Integrals as Area

Because the definite integral in example above is negative, it does not represent the area of the region. Definite integrals can be positive, negative, or zero. For definite integrals to be interpreted as an area, the function f must be continuous and non - negative on [a, b].

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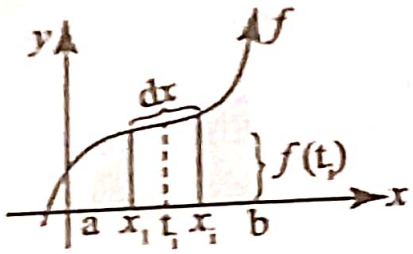
The Definite Integral as the Riemann sun of the area of the rectangle of a region



If f is continuous and non - negative on the closed interval [a, b], then area of the region bounded by the graph of f, the x - axis, and the vertical lines $x = a$ and $x = b$ is

$$\text{Area : } A = \sum_{i=1}^{n} f(t_i)\,\Delta x = \int_a^b f(x)\,dx$$

Illustrative Example

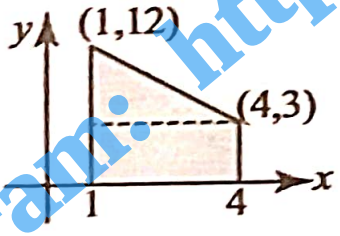
138. Find the area of the region bounded by the graph of $f(x) = 15 - 3x$ The x - axis and the vertical line $x = 1$ and $x = 4$

(Use the right endpoints t_i as sample point)

Solution

We begin by subdividing [1 4] into n subintervals of equal width:

$$\Delta x = \frac{b-a}{n} = \frac{4-1}{n} = \frac{3}{n}$$



Chose ti as the right endpoint of each subinterval

$$t_i = a + i\Delta x = 1 + \frac{3i}{n}$$

Hence $f(t_i) = 15 - 3t_i = 15 - 3\left(1 + \frac{3i}{n}\right) = 12 - \frac{9}{n}i$

The Riemann sunn:

$$\sum_{i=1}^{n} f(t_i)\,\Delta x = \sum_{i=1}^{n}\left(12 - \frac{9i}{n}\right)\left(\frac{3}{n}\right) = \int_1^4 f(x)\,dx$$

$$= \sum_{i=1}^{n}\left(\frac{36}{n} - \frac{27i}{n^2}\right)$$

$$= \frac{36}{n}\sum_{i=1}^{n} 1 - \frac{27}{n^2}\sum_{i=1}^{n} i$$

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$$= \frac{36}{n}(n) - \frac{27}{2n^2}\left(\frac{n(n+1)}{2}\right)$$

$$= 36 - \frac{27}{2n^2}(n^2 + n)$$

$$= 36 - \frac{27}{2}\left(\frac{n^2}{n^2} + \frac{n}{n^2}\right)$$

$$= 36 - \frac{27}{2}\left(1 + \frac{1}{n}\right), \text{ as } n \longrightarrow \infty$$

$$= 36 - \frac{27}{2}(1 + 0)$$

$$= 22.5$$

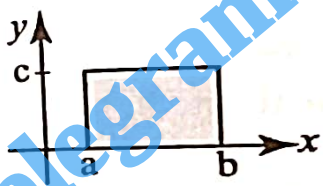
Also $\int_1^4 f(x)\,dx = \int_1^4 (15 - 3x) = 22.5$

Illustrative Example

139. Evaluate the given definite integral

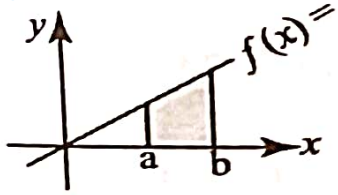
a. $\int_a^b c\,dx$ b. $\int_a^b x\,dx$ c. $\int_a^b 2\,dx$ d. $\int_a^b x\,dx$

Solution: Let $f(x) = C$

a. 

Area = (base) height = (b - a)c

$\therefore$ $A = \int_a^b c\,dx = c\,(b - a)$

b. 

Area $= \frac{h}{2}(b_1 + b_2) = \left(\frac{b-a}{2}\right)(a + b)$

$= \frac{b^2 - a^2}{2}$

$\therefore$ $\int_a^b x\,dx = \frac{b^2 - a^2}{2}$

c. $\int_1^8 2\,dx = 2\,(8 - 1) = 14$

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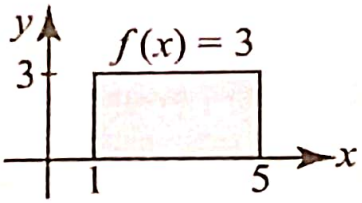
d. $\int_2^6 x\,dx = \frac{6^2 - 2^2}{2} = \frac{36 - 4}{2} = \frac{32}{2} = 16$

140. Evaluate each integral using a geometric formula. (Areas of common geometric figures)

a. $\int_1^5 3\,dx$ b. $\int_0^3 4x\,dx$ c. $\int_0^4 (3x + 2)\,dx$

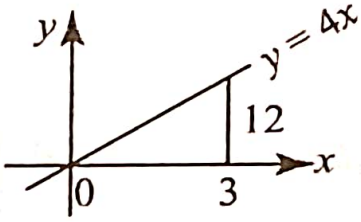
Solution

a. This region is a rectangle of height 3 and width4.



$\therefore \quad \int_1^5 3\,dx$ = Area of rectangle = 3(4) = 12

b. This region is a triangle of height 12 and width 3

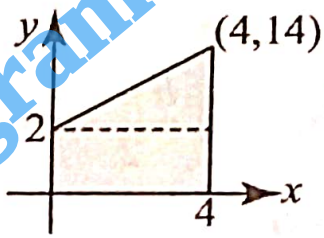


$\therefore \quad \int_0^3 4x\,dx$ = Area of rectangle

$= \frac{1}{2}(12)(3) = 18$

c. This region a trapezium with altitude of 4 and parallel bases of length 2 and 12.

Area of trapezium $= \frac{1}{2} h\,(b_1 + b_2)$



$\therefore \quad \int_0^4 (3x + 2)$ = Area of trapezium

$= \frac{1}{2}(4)(2 + 14) = 34$

141. What is the definite integral of f over a closed interval [a, b]?

Answer : $I = \int_a^b f(x)\,dx$

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11.7 Anti derivative of function

What is an antiderivative of a function? Finding a function from its derivative $f'(x)$ is anti derivative of a function. Anti derivative is the reverse process of derivative. As an example

$$x^3 + 2x \underset{\text{anti - derivative}}{\overset{\text{derivative}}{\rightleftharpoons}} 3x^2 + 2$$

$$x^3 + 2x + 4 \underset{\text{anti - derivative}}{\overset{\text{derivative}}{\rightleftharpoons}} 3x^2 + 2$$

$$x^3 + 2x + c \underset{\text{anti - derivative}}{\overset{\text{derivative}}{\rightleftharpoons}} 3x^2 + 2$$

- The set of all anti derivative of $3x^2 + 2$ is $x^3 + 2x + c$, denoted by
$\int (3x^2 + 2)\, dx = x^3 + 2x + c$

- The set of all anti derivative of $f(x)$ is denoted by
$\int f(x)\, dx = F(x) + c$

Finding Anti derivatives - Indefinite Integrals

DEFINITION

A function $F(x)$ is an anti derivative of a function $f(x)$ if

$$F'(x) = f(x)$$

for all x in the domain of f.

- The set of all anti derivative of f is the **indefinite integral** of f with respect to x, denoted by $\int f(x)\, dx = F(x) + c$

The symbol $\int$ is an **integrand sign**

- The function f is the integrand of the integral and x is the **variable of integration.** C is the constant of integration

Note: The set of all anti derivative of the function f differ from F by a constant.

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Integral formula

Power Rule:

For any real number n # -1

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c$$

If $f(x) = \frac{x^{n+1}}{n+1}$ is an antiderivative of $f(x) = x^n$

$$F'(x) = \frac{d}{dx}\left(\frac{x^{n+1}}{n+1}\right) = \frac{n+1}{n+1}x^n = x^n$$

Here are some examples:

a. $\int x^3 dx = \frac{x^{3+1}}{3+1} + c = \frac{x^4}{4} + c$

b. $\int x^{-6} dx = \frac{x^{-6+1}}{-6+1} + c = -\frac{1}{5}x^{-5} + c$

c. $\int x^{2/3} dx = \frac{x^{\frac{2}{3}+1}}{\frac{2}{3}+1} + c = \frac{3}{5}x^{5/3} + c$

Integration of constant

- $\int k dx = kx + c, k \neq 0$
- $\int o dx = k$
- $\int \pi dx = \pi x + c$
- $\int e dx = ex + c$

Integration of e^x: a. $\int e^x dx = e^x + c$

b. $\int e^x dx = ke^x + c$

Illustrative Example

142. Find the indefinite integral

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a. $\int x^{-7}\,dx$

b. $\int x^{\frac{-4}{3}}\,dx$

c. $\int x^{\frac{1}{5}}\,dx$

d. $\int x^{-2}\,dx$

e. $\int x^{\frac{-2}{5}}\,dx$

Solution

a. $\int x^{-7}\,dx = \dfrac{x^{-7+1}}{-7+1} + C = \dfrac{x^{-6}}{-6} + c$

b. $\int x^{\frac{-4}{3}}\,dx = \dfrac{1}{\frac{-4}{3}+1}\, x^{\frac{-4}{3}+1} + c = -3x^{\frac{-1}{3}} + c$

c. $\int x^{\frac{1}{5}}\,dx = \dfrac{1}{\frac{1}{5}+1}\, x^{\frac{1}{5}+1} = \dfrac{5}{6}x^{6/5} + c$

d. $\int x^{-2}\,dx = \dfrac{1}{-2+1}x^{-2+1} + c = -x^{-1} + c$

e. $\int x^{\frac{-2}{5}}\,dx = \dfrac{x^{\frac{-2}{5}+1}}{\frac{-2}{5}+1} + c = \dfrac{5}{3}x^{3/5} + c$

Sum and difference Rule of Integration

Sum: $\int (f(x) + g(x))\,dx = \int f(x)\,dx + \int g(x)dx.$

$\int (f(x) - g(x)\,dx = \int f(x)\,dx - \int g(x)\,dx$

Illustrative Example

143. Evaluate : $\int (3x^2 - 5x^{2/3} + x^{-3})\,dx$

Solution

$$\int (3x^4 - 5x^{2/3} + x^{-3})\,dx = \int 3x^4 dx - \int 5x^{2/3}dx + \int x^{-3}\,dx$$

$$= \frac{3x^{4+1}}{4+1} - \frac{5x^{\frac{2}{3}+1}}{\frac{2}{3}+1} + \frac{x^{-3+1}}{-3+1} + c = \frac{3}{5}x^5 - 3x^{5/3} - \frac{1}{2}x^{-2} + c$$

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Some Integrand Needs Rewrite Before Integrating

Be sure that one of the most important steps in integrations is rewriting the integrand in a form that first on of the basic integration rule.

Illustrative Example

144. Evaluate

a. $\int \frac{x-2}{\sqrt{x}}\,dx$

b. $\int \frac{x+4}{\sqrt[3]{x}}\,dx$

c. $\int \frac{x^2+3x}{\sqrt{x^3}}\,dx$

d. $\int x\sqrt[4]{x^5}\,dx$

Solution: Rewriting before integrating

a. $\int \frac{x-2}{\sqrt{x}}\,dx = \int \frac{x-2}{x^{\frac{1}{2}}} = \int\left(x^{\frac{1}{2}} - 2x^{\frac{-1}{2}}\right)dx$

$$= \frac{x^{\frac{1}{2}+1}}{\frac{-1}{2}+1} - c\frac{2x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + C$$

$$= \frac{2}{3}x^{\frac{2}{3}} - 4x^{\frac{1}{2}} + c$$

b. $\int \frac{x+4}{\sqrt[3]{x}}\,dx = \int \frac{x+4}{x^{\frac{1}{3}}}\,dx = \int\left(x^{\frac{2}{3}} + 4x^{\frac{-1}{3}}\right)\,dx$

$$= \frac{x^{\frac{2}{3}+1}}{\frac{2}{3}+1} + 4\,\frac{x^{\frac{-1}{3}+1}}{\frac{-1}{3}+1} + c$$

$$= \frac{3}{5}x^{\frac{5}{3}} + 6x^{\frac{2}{3}} + c$$

c. $\int \frac{x^2+3x}{\sqrt{x^3}}\,dx = \int \frac{(x^2+3x)}{x^{\frac{3}{2}}}\,dx = \int\left(x^{\frac{4}{3}} + 3x^{\frac{-1}{3}}\right)dx$

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$$= \frac{x^{\frac{4}{3}+1}}{\frac{4}{3}+1} + \frac{3x^{\frac{-1}{3}+1}}{\frac{-1}{3}+1} + c$$

$$= \frac{3}{7}x^{7/3} + \frac{9}{2}x^{2/3} + c$$

d. $\int x\sqrt[4]{x^5}\,dx = \int x^{9/4}\,dx$

$$= \frac{x^{\frac{9}{4}+1}}{\frac{9}{4}+1} + c$$

$$= \frac{4}{13}x^{13/4} + c$$

145. Evaluate

a. $\int \sqrt{x}\,dx$

b. $\int \frac{1}{\sqrt{x}}\,dx$

c. $\int \frac{1}{\sqrt[3]{x}}\,dx$

d. $\int \frac{4}{\sqrt[3]{x^2}}\,dx$

Solution

a. $\int \sqrt{x}\,dx = \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + c = \frac{2}{3}x^{3/2} + c$

b. $\int \frac{1}{\sqrt{x}}\,dx = \int x^{\frac{-1}{2}}\,dx = \frac{x^{\frac{-1}{2}+1}}{\frac{-1}{2}+1} + c = 2x^{\frac{1}{2}} + c$

c. $\int \frac{1}{\sqrt[3]{x}}dx = \frac{x^{\frac{-1}{3}+1}}{\frac{-1}{3}+1} + c = \frac{3}{2}x^{\frac{2}{3}} + c = \frac{3}{2}\sqrt[3]{x^2} + c$

d. $\int \frac{4}{\sqrt[3]{x^2}}\,dx = \frac{4x^{\frac{-2}{3}+1}}{\frac{-2}{3}+1} + c = 12x^{\frac{1}{3}} + c = 12\sqrt[3]{x} + c$

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146. Evaluate the indefinite integral

a. $\int (2x+1)^2\,dx$ b. $\int \left(x+\frac{1}{x}\right)^2 dx$ c. $\int (3x+1)(3x-1)\,dx$

Solution

First rewrite by expanding binomial.

a. $\int (2x+1)^2\,dx = \int (4x^2+4x+1)\,dx$

$$= \frac{4x^3}{3} + 2x^2 + x + c$$

b. $\int \left(x+\frac{1}{x}\right)^2 dx = \int \left(x^2+\frac{1}{x^2}+2\right)$

$$= \frac{x^3}{3} - \frac{1}{x} + 2x + c$$

c. $\int (3x+1)(3x-1)\,dx = \int (9x^2-1)\,dx$

$$= 3x^3 - x + c$$

Evaluate: $\int \frac{x+1}{x^3}\,dx$

Solution

$$\int \frac{x+1}{x^3}dx = \int \frac{1}{x^2}dx + \int \frac{1}{x^3}dx = \frac{-1}{x} + \frac{-2}{x^2} + c = \frac{-(x+2)}{x^2} + c$$

Conceptual Example

147. Suppose that $F'(x) = f(x)$ and $G'(x) = g(x)$.
Which of the following statement are true? Explain.

a. If $f = g$, then $F = G$

b. If F and G differ by a constant, then $f = g$

c. If f and g differ by a constant, then $F = G$

Solution

a. False, as example,

Let $F(x) = x^2 + 4$, and $G(x) = x^2 + 1$

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$\Rightarrow$ $F'(x) = 2x = f(x)$ and $G(x) = 2x = g(x)$

b. True.

c. False, $F(x) - G(x) = ax + b$.

148. If $\int f(x)\,dx = F(x)$ means

A. $f'(x) = F(x)$

B. $f(x) = F'(x)$

C. $f(x) = F'(x) + c$

D. $f(x) = F(b) - F(a)$

Solution

F is an anti derivative of f if $F'(x) = f(x)$

Answer: B

149. Suppose $h'(x) = g(x)$, then which of the following statements are true?

a. $g(x)$ is an anti derivative of $h(x)$

b. $h(x)$ is an anti derivative of $g(x)$

c. $\int_a^b h(x)\,dx = g(b) - g(a)$

d. $\int_a^b g(x)\,dx = h(b) - h(a)$

Solution

$h'(x) = g(x)$ means h is ant anti derivative of g, and, denoted by

$\int g(x)\,dx = h(x) + c$

a. False b. True c. False d. True

150. Let F be an anti derivative of f. Then which of the following is Not true about F and f?

A. $F'(x) = f(x)$

B. $\frac{d}{dx}\int F(x)\,dx = F(x)$

C. $\int \frac{d}{dx} f(x)\,dx = f(x) + c$

D. $\int F(x)\,dx = f(x) + c$

Solution

$F(x) = \int f(x)\,dx$ and $F'(x) = f(x)$

Answer: D

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151. Can a function have more than one anti derivative ? If so, how are the anti derivative related? Explain

Answer: Yes

The set of all anti derivative of f is $\int f(x)\,dx = F(x) + c$

152. What is an indefinite integral? How do you denoted?

Solution

The set of all derivative of f is the **indefinite integral** of f with respect to x denoted by $\int f(x)\,dx$

153. Suppose that F and G are both anti derivative f on an interval [a, b]. Determine whether each statements are true or false

a. $F(x) = G(x)$

b. $F'(x) = G'(x)$

c. $F(x) = G(x) + c$

d. $G'(x) = \int f(x)\,dx$

Solution

a. False b. True c. True d. False

154. The slope of tangent line to a function (curve) is given by

$$f'(x) = 6x^2 - 4x + 3$$

If the point (-1, 2) is on the curve, find the equation of the curve.

Solution

$$f(x) = \int f'(x)\,dx = \int (6x^2 - 4x + 3)\,dx$$

$2x^3 - 2x^2 + 3x + c$

since $f(-1) = -2 - 2 + 3(-1) + c = 2$

$\therefore \quad c = 2 + 7 = 9$

$\therefore$ The equation of the curve is $f(x) = 2x^3 - 3x + 9$.

The Fundamental Theorem of calculus

Here, we will introduce the two basic relation ships between definite and indefinite integrals together constitute a result called the **fundamental Theorem of calculus (FTC)**

- The First part of FTC will relate the rectangle and anti derivative

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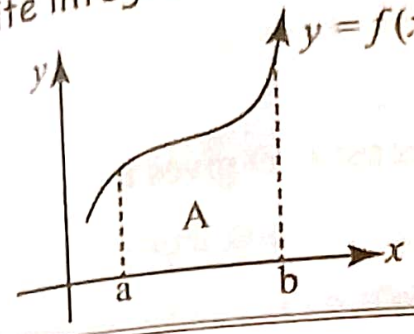
methods for calculating areas

- The second part of "FTC" will provide a powerful method for evaluating definite integrals using anti derivatives.

Let $f(x) \geq 0$ on an interval [a, b], then the area A under the graph of f over the interval [a, b] is represented by the definite integral:

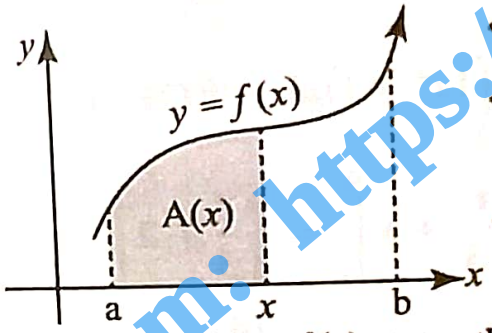


$$A = \int_a^b f(x)\,dx$$

$$= A(b) - A(a)$$

Let A (x) is the area under the graph of f from a to x, then :

$$A(x) = \int_a^x f(x)\,dx$$



- $A'(x) = f(x)$
- A (a) = 0, the area under the curve from a to a.
- A (b) = A the area under the curve from a to b is A

The expression $A'(x) = f(x)$ states that A (x) is an anti derivative of $f(x)$ on [a, b].

Evaluating The Definite Integrals

The definite integrals can be evaluated by find any anti derivative of the integrand and then subtracting the value of this anti derivative at lower limit of integration from its value at the upper limit of integration.

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DEFINITION

If f is continuous on [a, b] and F is any anti derivative of f on on [a, b]

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

Note:

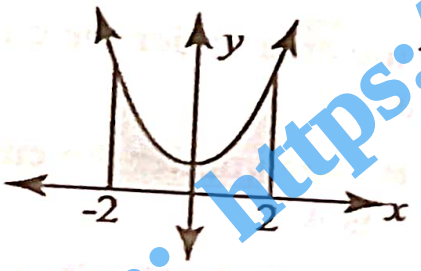
(i) $\int f(x)\,dx = F(x) + c$, we interpreter as: $f(x)$ gives the rate of change of $F(x)$

(ii) $\int_a^b f(x)\,dx$ gives total change of $F(x)$ as x changes from a to b.

Illustrative Example

155. Find the area under the graph of $f(x) = x^2 + 1$ over [-2, 2]

Solution



$f(x) = x^2 + 1$

$$\text{Area: } A = \int_{-2}^{2} (x^2 + 1)\,dx = F(2) - F(2)$$

$$= \left[\frac{x^3}{3} + x\right]_{-2}^{2} = \left[\frac{8}{3} + 2\right] - \left[\frac{-8}{3} - 2\right]$$

$$= \frac{14}{3} + \frac{14}{3} = \frac{28}{3}$$

156. Find the area under the graph of $f(x) = x^3$ over [1 3]

Solution

$$F(x) = \int_1^3 x^3 dx = \left[\frac{x^4}{4}\right]_1^3$$

$$\text{Area} = F(3) - F(1) = \frac{3^4}{4} - \frac{1^4}{4} = \frac{81}{4} - \frac{1}{4} = \frac{80}{4} = 20$$

157. What is the area under the graph of $f(x) = x^2 - 2x + 5$ over [-1, 2] in square unite?

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A. $\frac{8}{3}$ B. 3 C. $\frac{7}{3}$ D. 15

Solution : Since $f(x) > 0$ on $[-1, 2]$

$$\therefore \quad \text{Area} = \int_{-1}^{2} (x^2 - 2x + 5)\, dx$$

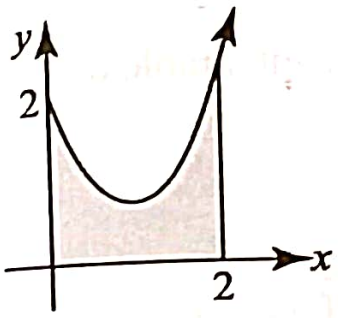
$$= \left[\frac{x^3}{3} - x^2 + 5x\right]_{-1}^{2} = F(2) - F(-1)$$

$$= \left[\frac{8}{3} - 4 + 10\right] - \left[\frac{-1}{3} - 1 - 5\right] = 15$$

Answer: D

158. Find the area under the graph of $f(x) = 2x^2 - 3x + 2$ over $[0, 2]$.

Solution: since $f(x) > 0$ on interval $[0, 2]$.



$$\text{Area} = \int_{0}^{2} (2x^2 - 3x + 2)\, dx$$

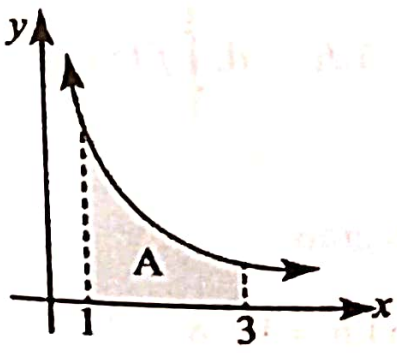
$$= \left[\frac{2x^3}{3} - \frac{3x^2}{2} + 2x\right]_{0}^{2}$$

$$= \left[\frac{16}{3} - 6 + 4\right] - [0 - 0 + 0]$$

$$= \frac{10}{3}$$

159. What is the area under the graph of $f(x) = \frac{1}{x^2}$ over $[1, 3]$ in square units

A. $\frac{2}{3}$ B. $\frac{1}{3}$ C. $\frac{4}{3}$ D. 4



Since $f(x) > 0$ on $[1, 3]$

$$\therefore \quad \text{Area} = \int_{1}^{3} x^{-2}\, dx = \left[\frac{x^{-2+1}}{-2+1}\right]_{1}^{3}$$

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$$= \left[\frac{-1}{x}\right]_1^3 = F(3) - F(1) = \frac{-1}{3} - \left(\frac{-1}{1}\right) = \frac{-1}{3} + 1 = \frac{2}{3}$$

Answer: A

Properties of Definite Integrates

1. If f is defined as $x = a$, then $\int_a^a f(x)\,dx = 0$. $\int_a^a k\,dx = k(b - a)$, for any constant K.

2. If f is integrable on [a, b], then $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$.

3. $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx,\ c \in [a, b]$

4. $\int_a^b k f(x)\,dx = k\int_a^b f(x)\,dx$, for any constant k.

5. If f and g are integrable on [a, b] then $f \pm g$ are integrable on [a, b]

$$\int_a^b [f(x) \pm g(x)]\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$$

6. If $f(x) \geq 0$ on the interval [a, b] then $0 \leq \int_a^b f(x)\,dx$

7. If $f(x) \leq g(x)$ for every x in [a, b] then $\int_a^b f(x)\,dx \leq \int_a^b g(x)\,dx$

Illustrative Example

160. Suppose, $\int_0^2 f(x)\,dx = 1$, $\int_0^3 f(x)\,dx = 6$ and $\int_2^5 f(x)\,dx = 8$, then Evalute:

a. $\int_0^5 f(x)\,dx$ b. $\int_2^3 f(x)\,dx$ c. $\int_3^5 f(x)\,dx$ d. $\int_5^2 f(x)\,dx$

Solution

Using properties of definite integrals:

$$\text{a. } \int_0^5 f(x)\,dx = \int_0^2 f(x)\,dx + \int_2^5 f(x)\,dx = 1 + 8 = 9$$

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b. $\int_2^3 f(x)\,dx = -\int_2^0 f(x)\,dx + \int_0^3 = -1 + 6 = 5$

c. $\int_3^5 f(x)\,dx = -\int_3^0 f(x)\,dx + \int_0^2 f(x)\,dx + \int_2^5 f(x)\,dx$

$= -6 + 1 + 8 = 3$

d. $\int_5^2 f(x)\,dx = -\int_2^5 f(x)\,dx = -8$

161. If $\int_0^4 f(x)\,dx = 6$ and $\int_0^4 g(x)\,dx = 10$ then evalute

a. $\int_0^4 f(x)\,dx + g(x)\,dx$ b. $\int_0^4 \left(2f(x) - \frac{1}{2}g(x)\right)dx$

Solution

a. $\int_0^4 f(x)\,dx + \int_0^4 g(x)\,dx = 6 + 10 = 16$

b. $2\int_0^4 f(x) - \frac{1}{2}\int_0^4 g(x)\,dx = 2(6) - \frac{1}{2}(10) = 7$

162. Suppose $\int_1^3 f(x)\,dx = 6$ and $\int_1^2 f(x)\,dx = 10$, what is the value of $\int_2^3 f(x)\,dx$?

A. 16 B. 4 C. 2 D. -4

Solution

$$\int_2^3 f(x)\,dx = -\int_2^1 f(x)\,dx + \int_1^3 f(x)\,dx$$

$$= -10 + 6 = -4$$

Answer: D

163. Let $f(x) = \begin{cases} 1 - x, & if\ -1 \le x < 0 \\ x^2 & if\ 0 \le x < 2 \\ -1 & if\ 2 \le x \le 3 \end{cases}$ then evalute: $\int_{-1}^3 f(x)\,dx$

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Solution

$$\int_{-1}^{3} f(x)\,dx = \int_{-1}^{0} (1 - x)\,dx + \int_{0}^{2} x^2\,dx + \int_{2}^{3} (-1)\,dx$$

$$= \left[x - \frac{x^2}{2}\right]_{-1}^{0} + \left[\frac{x^3}{3}\right]_{0}^{2} + [-x]_{2}^{3}$$

$$= \frac{3}{2} + \frac{8}{3} - 1 = \frac{19}{6}$$

164. Let $f(x) = \begin{cases} x^{2/3} & \text{if } -8 \le x < 0 \\ -4 & \text{if } 0 \le x \le 3 \end{cases}$ then evalute: $\int_{-8}^{3} f(x)\,dx$

Solution

$$\int_{-8}^{3} f(x)\,dx = \int_{-8}^{0} x^{2/3}\,dx + \int_{0}^{3} -4\,dx = \left[\frac{3}{5}x^{5/3}\right]_{-8}^{0} + [-4x]_{0}^{3}$$

$$= \frac{96}{5} - 12 = 36/5$$

165. Evalute $\int_{0}^{4} f(x)\,dx$ where $f(x)$ is defined by $f(x) = \begin{cases} 3x & \text{if } x \le 3 \\ 2 & \text{if } x > 3 \end{cases}$

Solution

$$\int_{0}^{4} f(x)\,dx = \int_{0}^{3} f(x)\,dx + \int_{3}^{4} f(x)\,dx$$

$$= \int_{0}^{3} 3x - dx + \int_{3}^{4} 2dx = \left[\frac{3x^2}{2}\right]_{0}^{3} + [2x]_{3}^{4}$$

$$= \frac{27}{2} + 8 - 6 = \frac{31}{2}$$

166. Suppose that $F'(x) = f(x)$ and $F(0) = 4$, $F(3) = 9$. What is the area under $y = f(x)$ over $[0, 3]$ if $(x) \ge 0$?

Solution

$F'(x) = f(x)$ means F is an anti derivative of f, denoted by

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$\int_0^1 f(x)\,dx = F(3) - F(0) = 9 - 4 = 5$

167. Suppose that F is anti derivative of f with $F(1) = -11$ and $F(0) = 11$. Then $\int_0^1 f(x)\,dx$ is equal to

A. 11 B. -22 C. 22 D. 0

Solution

$\int_0^1 f(x)\,dx = F(1) - F(0) = -11 - 11 = 22$

Answer: B

168. Evalute $\int_3^8 f'(x)\,dx$, where f is differentiable and $f(3) = f(8) = 5$

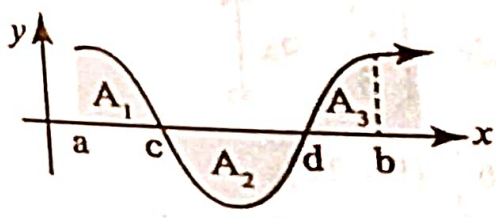
Solution

$\int_3^8 f'(x)\,dx, = f(8) - f(3) = 5 - 5 = 0$

11.7.1 Definite Integrals and Total Area

If $y = f(x)$ has both positive and negative values on an interval [a, b] then the areas for f on [a, b] add the areas of the region that lies above the x - axis to axis to the negative of the region that lies below the x - axis.

The value of the definite integral is the area above the x - axis minus the area below the x - axis.



i. $A_1 = \int_a^c f(x)\,dx$, since $f(x) \geq 0$

ii. $A_2 = -\int_c^d f(x)\,dx$, since $f(x) < 0$

ii. $A_3 = \int_d^b f(x)\,dx$, $f(x) \geq 0$

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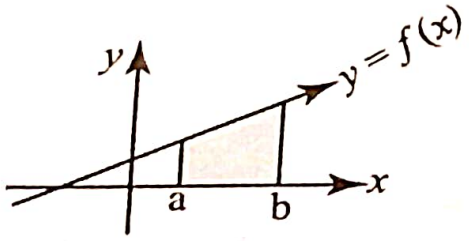
Total area $= A_1 + A_2 + A_3$

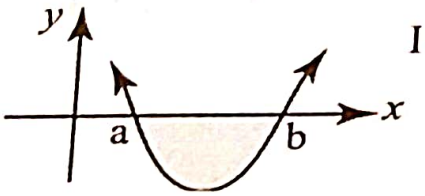
$$A = \int_a^c f(x)\,dx - \int_c^d f(x)\,dx + \int_d^b f(x)\,dx$$

$$\text{Area of a region} = \begin{pmatrix}\text{Area}\\ \text{above } x\text{ - axis}\end{pmatrix} - \begin{pmatrix}\text{Area}\\ \text{below } x\text{-axis}\end{pmatrix}$$



If $f(x) \geq 0$, on [a, b] then the value of the definite integral is area :

$$A = \int_a^b f(x)\,dx$$



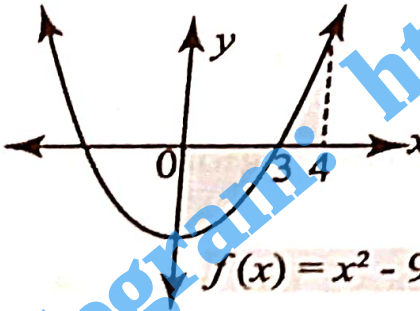
If $f(x) < 0$, on [a, b], then the area of the region is

$$A = -\int_a^b f(x)\,dx \longleftarrow \text{Area below } x\text{-axis}$$

Illustrative Example

169. Find the area (A) of the region between the graph of $f(x) = x^2 - 9$, the x axis and on interval [0, 4]

Solution



Since $f(x) < 0$ on [0, 3], area below the x-axis:

$\therefore$ $$A_1 = -\int_0^3 f(x)\,dx = -\int (x^2 - 9)\,dx = -\left[\frac{x^3}{3} - 9x\right]_0^3$$

$$F(3) - F(0) = -\left[\frac{27}{3} - 27\right] - 0 = -(9 - 27) = 18$$

Since $f(x) \geq 0$ on [3, 4] **area above the x - axis**

$$A_2 = \int_3^4 f(x)\,dx = \int_3^4 (x^2 - 9)\,dx = \left[\frac{x^3}{3} - 9x\right]_3^4$$

$$F_2(4) - F_2(3) = \left(\frac{64}{3} - 36\right) - (9 - 27)$$

$$= \frac{-44}{3} + 18 = \frac{10}{3}$$

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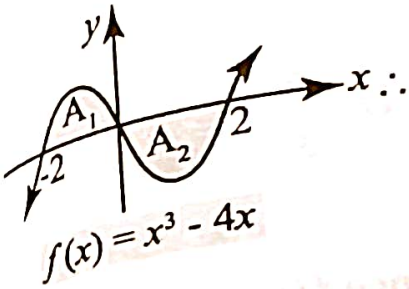
$\therefore$ Area: $A = A_1 + A_2 = 18 + \frac{10}{3} = \frac{64}{3}$

170. Find The area of the region bounded by the graph of $f(x) = x^3 - 4x$ and the x - axis in square units.

Solution: x - intercept $x(x^2 - 4) = 0$ are $x = 0, x = 2$ and $x = -2$

Since, $f(x) < 0$, on $[0, 2]$

$$A_2 = -\int_0^2 f(x)\,dx, \text{ area below } x\text{-axis}$$



$f(x) = x^3 - 4x$

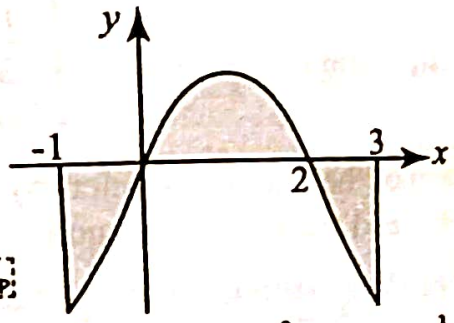
$\therefore$ $A = A_1 + A_2 = \int_{-2}^{0}(x^3 - 4x)\,dx - \int_0^2 (x^3 - 4x)\,dx$

$$A = \left[\frac{x^4}{4} - 2x^2\right]_{-2}^{0} - \left[\frac{x^4}{4} - 2x^2\right]_0^2$$

$$= \left[(0 - 0) - \left(\frac{16}{4} - 8\right)\right] - \left[\frac{16}{4} - 8\right]$$

$\therefore$ $A = 4 + 4 = 8$

171. Find the area of the region between the graph of $(x) = -x^2 + 2x$ the x-axis and on the interval $[-1, 3]$



Solution: [P] [SEP]

Area: $A = \int_{-1}^{3} f(x)\,dx = -\int_{-1}^{0} f(x)\,dx + \int_0^2 f(x)\,dx - \int_2^3 f(x)\,dx$

$$A = \int_{-1}^{3}(-x^2 - 2x)\,dx$$

$$= -\int_{-1}^{0}(-x^2 + 2x)\,dx + \int_0^2 (-x^2 + 2x)\,dx - \int_2^3 (-x^2 + 2x)$$

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$$= -\left[\frac{-x^3}{3} + x^2\right]_{-1}^{0} + \left[\frac{-x^3}{3} + x^2\right]_{0}^{2} - \left[\left(\frac{-x^3}{3} + x^2\right]_{2}^{3}\right.$$

$$= -\left[(0+0) - \left(\frac{1}{3} + 1\right)\right] + \left[\frac{-8}{3} + 4 - 0\right] - \left[-9 + 9 - \left(\frac{-8}{3} + 4\right)\right]$$

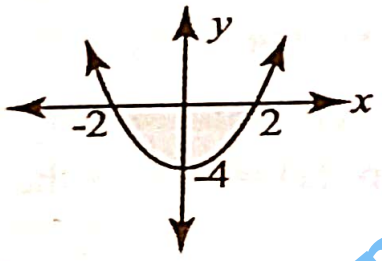
$\therefore$ Area $= \frac{4}{3} + \frac{4}{3} + \frac{4}{3} = 4$ square untis

172. What is the area of the region sounded by the graph of $f(x) = x^2 - 4$ and the x - axis in square units?

A. 32 B. 16 C. $\frac{32}{2}$ D. $\frac{16}{3}$

Solution

The x - intercept of $f(x) = x^2 - 4 = 0$, $x = \pm 2$ since $f(x) <$ on $[-2, 2]$.



$$\text{Area} = -\int_{-2}^{2}(x^2 - 4)\,dx = -\left[\frac{x^3}{3} - 4x\right]_{-2}^{2}$$

$$F(2) - F(-2) = -\left[\left(\frac{8}{3} - 8\right) - \left(\frac{-8}{3} + 8\right)\right]$$

$$= -\left[\frac{-16}{3} - \frac{16}{3}\right] = \frac{32}{3}$$

Answer: C

173. What is the area of the region bounded by the graph of $f(x) = x^2 - 5x + 6$ and the x - axis in square units?

A. $\frac{1}{2}$ B. $\frac{9}{2}$ C. $\frac{14}{3}$ D. $\frac{55}{6}$

Solution: The x - intercept $f(x) = x^2 - 5x + 6$ is $x = 2$ and $x = 3$

Since $f(x) < 0$ on $[2, 3]$

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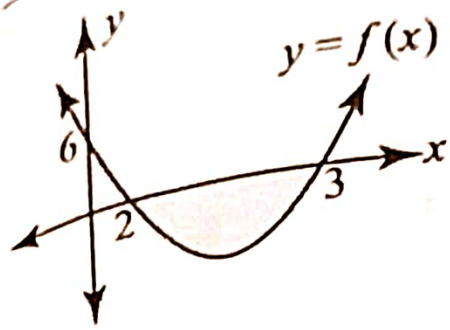


$$\therefore \quad A = -\int_2^3 (x^2 - 5x + 6)\,dx = -\left[\frac{x^3}{3} - \frac{5}{2}x^2 + 6x\right]_2^3$$

$$F(3) - F(2) = -\left[\left(9 - \frac{45}{2} + 18\right) + \left(\frac{8}{3} + 10 + 12\right)\right]$$

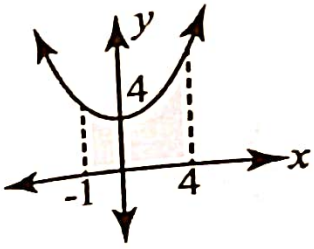
$$-\left(\frac{9}{2} - \frac{14}{3}\right) = -\left(\frac{-1}{2}\right) = \frac{1}{2}$$

Answer: A

174. What is area of the region bounded by the graph of $f(x) = x^2 + 4$, the x - axis and the line $x = -1$ and $x = 4$ in square units?

A. $\frac{101}{3}$ B. $\frac{125}{3}$ C. $\frac{123}{3}$ D. $\frac{109}{3}$

Solution



Since $f(x) \geq 0$ on $[-1, 4]$

$$\therefore \quad A = \int_{-1}^{4} (x^2 + 4)dx = \left[\frac{x^3}{3} + 4x\right]_{-1}^{4}$$

$$F(4) - F(-1) = \left[\left(\frac{64}{3} + 16\right) - \left(\frac{-1}{3} - 4\right)\right]$$

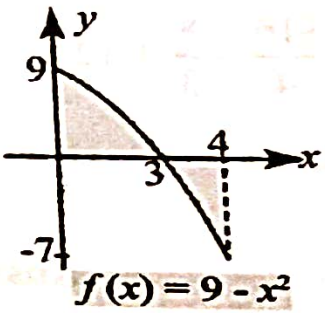
$$= \frac{125}{3} \text{ sq unit}$$

Answer: B

175. What is area of the region bounded by the graph of $f(x) = 9 - x^2$, $0 \leq x \leq 4$, and x - axis

A. $\frac{44}{3}$ B. 18 C. $\frac{10}{3}$ D. $\frac{64}{3}$

Solution: The x - intercept of $f(x)$ divides $[0, 4]$ in to subinterval of $[0, 3]$ and $[3, 4]$:



$$A_1 = \int_0^3 (9 - x^2)\,dx = \left[9x - \frac{x^3}{3}\right]_0^3$$

$$= \left[\left(27 - \frac{27}{3}\right) - (0 - 0)\right] = 18$$

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Since $f(x) < 0$ on $[3, 4]$

$$A_2 = -\int_3^4 (9 - x^2)\,dx = -\left[9x - \frac{x^3}{3}\right]_3^4$$

$$= -\left[\left(36 - \frac{64}{3}\right) - (27 - 9)\right] = -\left(\frac{-10}{3}\right) = \frac{10}{3}$$

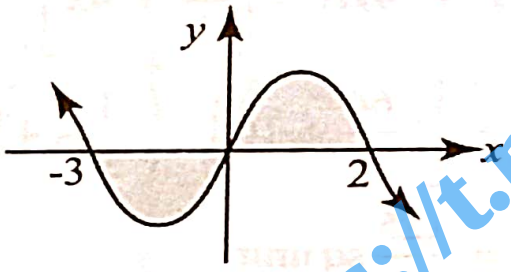
$$\therefore \quad A = A_1 + A_2 = 18 + \frac{10}{3} = \frac{64}{3}$$

176. What is the area of the region bounded by $f(x) = -x(x + 3)(x - 2)$ and x - axis

Answer: D

A. $\frac{253}{12}$ B. $\frac{125}{12}$ C. $\frac{63}{4}$ D. $\frac{16}{3}$

Solution: Since $f(x) < 0$ on $[-3, 0]$



Integral on $[-3, 0] = -\int_3^0 f(x)\,dx$

Area: $A = -\int_{-3}^0 f(x)\,dx + \int_0^2 f(x)\,dx$

To find the indefinite integrals we multiply:

$$\therefore \quad f(x) = -x(x + 3)(x - 2) = -x^3 - x^2 + 6x$$

$$A = -\int_{-3}^0 (-x^3 - x^2 + 6x)\,dx + \int_0^2 (-x^3 - x^2 + 6x)\,dx$$

$$= -\left[\frac{-x^4}{4} - \frac{x^3}{3} + 3x^2\right]_{-3}^0 + \left[\frac{-x^4}{4} - \frac{x^3}{3} + 3x^2\right]_0^2$$

$$= -\left[(0 - 0 + 0) - \left(\frac{-81}{4} + \frac{27}{3} + 27\right)\right] + \left[\frac{-16}{4} - \frac{8}{3} + 12\right]$$

$$= -\left[-\frac{63}{4}\right] + \frac{16}{3} = \frac{63}{4} + \frac{16}{3} = \frac{253}{12}$$

Answer: A

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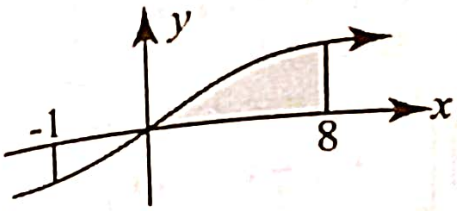
177. What is are area of the region enclosed by the graph of $f(x) = x^{1/3}$, $-1 \le x \le 8$ and x - axis in square unit?

A. 16 B. $\frac{3}{4}$ C. $\frac{51}{4}$ D. $\frac{45}{4}$

Solution



$$\text{Area} = -\int_{-1}^{0} x^{1/3}\,dx + \int_{0}^{8} x^{1/3}\,dx$$

$$= -\left[\frac{3}{4}x^{4/3}\right]_{-1}^{0} + \left[\frac{3}{4}x^{4/3}\right]_{0}^{8}$$

$$= -\left[0 - \frac{3}{4}(-1)^{4/3}\right] + \left[\frac{3}{4}(8)^{4/3} - 0\right]$$

$$= \frac{3}{4} + \frac{3}{4}\left(\sqrt[3]{8}\right)^4 = \frac{3}{4} + 12 = \frac{51}{4}$$

Answer: C

Definite integral involving Absolute value

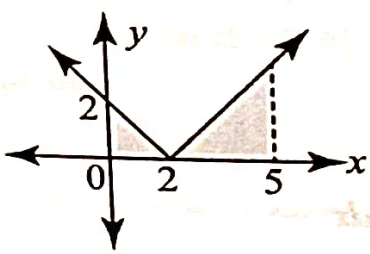
178. Evaluate

a. $\int_0^5 |x - 2|\,dx$ b. $\int_0^2 |3x - 2|\,dx$

Solution

Using the definition of absolute value, you can rewriter the integrand

a. $|x - 2| = \begin{cases} x - 2 & \text{if } x \ge 2 \\ -(x - 2) & \text{if } x < 2 \end{cases}$



Then $\int_0^5 |x - 2|\,dx = \int_0^2 -(x - 2)\,dx + \int_2^5 (x - 2)\,dx$

$$= -\left[\frac{x^2}{2} - 2x\right]_0^2 + \left[\frac{x^2}{2} - 2x\right]_2^5$$

$$= -[2 - 4 - (0 - 0)] + \left[\left(\frac{25}{2} - \frac{10}{2}\right) - (2 - 4)\right]$$

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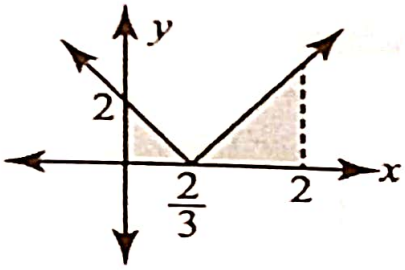
$\therefore$ Area: $A = 2 + 9.5 = 11.5$

b. $|3x - 2| = \begin{cases} 3x - 2 & \text{if } x \geq \frac{2}{3} \\ -(3x - 2) & \text{if } x < \frac{2}{3} \end{cases}$



Then $A = \int_0^2 |3x - 2|\,dx = \int_0^{2/3} -(3x - 2)\,dx + \int_{2/3}^2 (3x - 2)\,dx$

$= \left[-\left(\frac{3}{2}x^2 - 2x\right)\right]_0^{2/3} + \left[\frac{3}{2}x^2 - 2x\right]_{2/3}^2$

$= \frac{2}{3} + \left(2 - \left(-\frac{2}{3}\right)\right) = \frac{2}{3} + 2 + \frac{2}{3} = \frac{10}{3}$

179. Let $f(x) = 6 - 3x$, then

a. Evaluate $\int_0^8 f(x)\,dx$

b. Find the area of the graph $f(x)$ on interval $[0, 8]$ and x - axis.

Solution

a. $\int_0^8 f(x)\,dx = \int_0^8 (6 - 3x)\,dx = \left[6x - \frac{3x^2}{2}\right]_0^8$

$= F(8) = F(0) = 48 - \frac{3(64)}{2} = -48$

$\therefore \quad \int_0^8 f(x)\,dx = -48$

b. To find the area, we use x - intercept: $6 - 3x = 0, x = 2$

Since $f(x) < 0$, on $[2, 8]$ then

Area: $A = -\int_2^8 f(x)\,dx + \int_0^2 f(x)\,dx$

$= -\int_2^8 (6 - 3x)\,dx + \int_0^2 (6 - 3x)\,dx$

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$$= -\left[6x - \frac{3x^2}{2}\right]_2^8 + \left[6x - \frac{3x^2}{2}\right]_0^2$$

$$= -[(48 - 96) - (12 - 6)] + [(12 - 6) - (0 - 1)]$$

$$= 56 + 6 = 62$$

$\therefore$ Area = 62 square unit

180. Find the value of a such that the area bounded by $f(x) = x^2$, the x-axis, $x = -a$ and $x = a$ is 18 square units.

A. 4 B. 3 C. 5 D. 2

Solution

$$A = \int_{-a}^{a} x^2 dx = \left[\frac{x^3}{3}\right]_{-a}^{a} = 18$$

$$\Rightarrow \frac{1}{3}\left(a^3 - (-a^3)\right] = 18, \text{ implies } 2a^3 = 54$$

$\therefore$ $a^3 = 27$, $a = 3$

Answer: B

Critical thinking

181. Let $S = \int_3^8 f(x)dx$, where f is continuous state whether the following are true or false.

a. The values S is the area between the graph and the x-axis over [3, 8]

b. If $f(x) \geq 0$, then S is the area between the graph of $f(x)$ and the x-axis over [3, 8].

c. If $f(x) < 0$, then -S is the area between the graph of $f(x)$ and the x-axis over [3, 8]

Solution

a. False b. True c. True

Conceptual Check

182. Determine where each of the following statement is true of false:

a. The indefinite integral is another term for the family f of all anti

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derivative of a function.

b. The indefinite integral of x^n is $\frac{x^{n+1}}{n+1}$ for all real numbers n.

c. The velocity function is an anti derivative of the acceleration function.

d. The indefinite integral gives the instantaneous rate of change of a function.

e. The definite integral gives an area of a curve.

f. The definite integral of a function is always a positive quantity.

Solution

a. True

b. False , $n \neq -1$

c. True

d. True

e. False

f. False

Net Change as the Integral of Rate of change

The net change in S (t) over an interval (t_1, t_2) is given by

$$\underbrace{\int_{t_1}^{t_2} S'(t)\,dt}_{\text{Integral of areate of changee}} = \underbrace{S(t_2) - S(t_1)}_{\text{Net change over } [t_1, t_2]}$$

Illustrative Example

183. Water flows into an empty reservoir at rate of S (t) = 3000 + 20t litres per hour. What is the quantity of water into the reservoir after 5h?

Solution

quantity of water: $S(t) = \int_0^5 (3000 + 20t)\,dt$

$$S(5) - S(0) = \left[3000t + \frac{20t^2}{2}\right]_0^5 = 15250 \text{ liter}$$

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The Integral of Velocity

For an object in linear motion with velocity V (t), then

(i) Displacement during $[t_1, t_2] = \int_{t_1}^{t_2} v(t)\,dt$

(ii) Distance travelled during $[t_1, t_2]$

$= \int_{t_1}^{t_2} v(t),$ if $V(t) \geq 0$ on $[t_1, t_2]$

$= -\int_{t_1}^{t_2} v(t),$ if $V(t) < 0$ on $[t_1, t_2]$

Relation between position, velocity and acceleration:

$S'(t) = V(t) \Rightarrow S(t) = \int v(t)\,dt$

$V'(t) = a(t) \Rightarrow V(t) = \int a(t)\,dt$

Illustrative Example

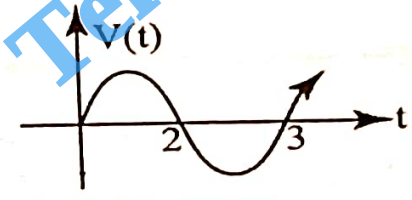
184. A Particle has velocity $V(t) = t^3 - 5t^3 + 6t$ in m/s.

a. Find the displacement over [0, 3]

b. Find the total distance over [0, 3].

Solution

a. The displacement over the time interval [0, 3] is



$V(t) = t^3 - 5t^2 + 6t$

$$\int_0^3 V(t)\,dt = \int_0^3 (t^3 - 5t^2 + 6t)\,dt = \frac{t^4}{4} - \frac{5}{3}t^3 + 3t^2\Big|_0^3$$

$$= \frac{3^4}{4} - \frac{5}{3}(3)^3 + 3(3)^2 = \frac{9}{4}\text{m}$$

b. The distance d is given by the definite integrnal (area):

$$\text{Distance} = \int_0^2 (t^3 - 5t^2 + 6t)\,dt - \int_2^3 (t^3 - 5t^2 + 6t)\,dt$$

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$$= \left[\frac{t^4}{4} - \frac{5t^3}{3} + 3t^2\right]_0^2 - \left[\frac{t^4}{4} - \frac{5t^3}{3} + 3t^2\right]_2^3$$

$$= \frac{8}{3} - \left[\frac{9}{4} - \frac{8}{3}\right] = \frac{8}{3} + \frac{5}{12} = \frac{37}{12}$$

185. Velocity: For a particular object, $a(t) = 6t^2 + 5$ and $V(0) = 8$. Find the velocity: V (t)

Solution

$$a = \frac{dv}{dt},\ dv = a.dt$$

$$\therefore \quad V(t) = \int a.dt = \int (6t^2 + 5)dt = \frac{6t^3}{3} + 5t$$

$$= \frac{6t^3}{3} + 5t + c = 2t^3 + 5t + c$$

$$\therefore \quad V(0) = 0 + 5.0 + c = 8 \Rightarrow c = 8$$

$$\therefore \quad V(t) = 2t^3 + 5t + 8$$

186. Distance: Suppose $V(t) = 9t^2 - 3\sqrt{t}$ and $S(1) = 6$. Find S (t)

Solution

$$V(t) = \frac{dS}{dt} \text{ implies that } dS = V(t)\,dt$$

$$\therefore \quad S(t) = \int V(t)\,dt = \int (9t^2 - 3\sqrt{t})dt$$

$$= \frac{9t^3}{3} - (3)\frac{2}{3}t^{2/3} + c$$

$$S(1) = 3(1)^3 - 2(1) + c = 6,\ c = 6 - 1 = 5$$

$$\therefore \quad S(t) = 3t^3 - 2\sqrt{t^3} + 5$$

187. Distance: Suppose acceleration of an object: $a(t) = 12t + 8$, $V(1) = 10$ and $S(1) = 10$. Find S (t):

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Solution

$$a(t) = \frac{dv}{dt} \text{ and } dv = a(t)\,dt \quad \int dv = \int a(t)\,dt$$

$$V(t) = \int (12t + 8)dt = 6t^2 + 8t + c$$

Since $V(1) = 6.1 + 8.1 + c = 10$, $C = -4$

$$\therefore \quad V(t) = 6t^2 + 8t - 4$$

To find S(t): $V(t) = \frac{dS}{dt}$:

$$dS = V(t)\,dt \Rightarrow \int dS = \int V(t)\,dt$$

$$S(t) = \int (6t^2 + 8t - 4)dt = 2t^3 + 4t^2 - 4t + c$$

Since $S(1) = 2 + 4 - 4 + c = 15$, $C = 13$

$$\therefore \quad S(t) = 2t^3 + 4t^2 - 4t + 13$$

Business and Economics:

Let C (x) = the cost of a manufacture. Then C ' (x) = the marginal cost. The cost of increasing production from a units to b units is the net change C (b) - C (a)

Which is equal to the integral of the marginal cost:

Cost of increasing production from a units to b units

$$C(b) - c(a) = \int_c^b C'(x)\,dx$$

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Illustrative Example

188. The marginal cost of producing x table computers is $C(x) = 60x^2 - 400x$ (ETB per tablet). Find the cost of increasing production from 10 to 15 tablets

Solution

$$C(15) - c(10) = \int_{10}^{15} (60x^2 - 400x)\,dx = \left[20x^3 - 200x^2\right]_{10}^{15}$$

$$= 22,500 \text{ ETB}$$

189. Suppose the revenue from the first pound of soap sold is ETB 4 and the marginal revenue is given by $R'(x) = 4 - 0.02x$ for $1 \leq x \leq 400$.

a. What is the revenue accruing from a sale of 30 pounds of soap?

Solution

In economics the marginal revenue function is by the definition the derivative of the total revenue function R.

$$R(x) = \int R'(x)\,dx = \int (4 - 0.02x)dx = 4x - 0.01x^2$$

Then $R(30) = 4(30) - 0.01(30)^2$

$= 120 - (0.01)\,900$

$= 120 - 9 = 111$ Birr

190. **Profit**: The marginal profit of a small - fast food stand is given by:

$P'(x) = \sqrt[3]{x}$

Where x is the sales of roasted bread.

a. Find the profit function

b. Find the profit from selling 64 roasted breed.

Solution

a. To find $P(x)$, we integrate $P'(x)$

$$P(x) = \int P'(x)\,dx = \int \sqrt[3]{x}\,dx = \frac{3}{4}x^{4/3} + c$$

b. The profit: $P(64) = \int_0^{64} \sqrt[3]{x}\,dx$

$$= \left[\frac{3}{4}x^{4/3}\right]_0^{64} = \frac{3}{4}(64)^{4/3} = 192$$

$\therefore$ From selling of 64 roasted breed, the profit is 192 ETB.

191. The rate of reaction to a drug is given by $r'(t) = 48t^{-2}$ where t is the number of hours since the drug was administered. Find the total reaction to the drug from $t = 1$ to $t = 6$

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Solution : $r(t) = \int r'(t)\,dt$

$$r(6) - r(1) = \int_1^6 48t^{-2}dt = \left[\frac{-48}{t}\right]_1^6$$

$$= \frac{-48}{6} - \left(\frac{-48}{1}\right) = -8 + 48 = 40$$

Review exercise

1. Find the average rate of change of the function over the interval:
   a. $f(x) = x^3 - x$, [1, 2]
   b. $f(x) = x^2 - 2$, [1, 3]
   c. $f(x) = \sqrt{x^2 - 1}$, [1, 4]
   d. $f(x) = \sqrt{3x - 2}$, [1, 2]
2. Find the instantaneous rate of change at given point
   a. $f(x) = x^3 - x$, at $x = 1$
   b. $f(x) = x^2 + 3x$ at $x = -4$
3. Find an equation of the tangent line to the graph of the given function at the indicated values of x.
   a. $f(x) = 2x^3 - 1, x = -1$
   b. $f(x) = -x^3 + 6x^2, x = 1$
4. **Profit:** Suppose that the total profit in hundreds of ETB from selling x - items is given by:: $P(x) = 2x^2 - 5x + 6$
   a. Find the average rate of change of profit between $x = 2$ and $x = 4$
   b. Find the instantaneous rate of change profit to the item produced when $x = 2$.
   c. Find the marginal profit at $x = 4$
5. **Revenue:** The revenue (in thousands of ETB) from producing x - units of an item is $R(x) = 10x - 0.002x^2$
   a. Find the instantaneous rate of change of revenue to the item produced to when $x = 1000$.
   b. Find the marginal revenue at $x = 1500$
6. Find an equation of the normal line to the graph of the given function at the indicated value of x.
   a. $f(x) = -x^2 + 3, x = 2$

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b. $f(x) = \frac{1}{3}x^3 - 2x^2, \quad x = -1$

c. $f(x) = x^4 - x, \quad x = 1$

7. Find an equation of the tangent line to the graph of $f(x) = 2x^4 + 3x^3 - 8x$ at $x = -1$.

8. Find the point on the graph of $f(x) = x^3 - 2x^2 + 4$ at which the slope of the tangent line is -1.

9. Find the point on the graph of $f(x) = 3x^2 + 4x$ at which the slope of the normal line is 2.

10. Find the values of a and b such that the slope of the tangent to the graph of $f(x) = ax^2 + bx$ at (1, 7) is 2.

11. Find derivative of the function at the indicated value of a.

a. $f(x) = (x^2 + 4)^3$, at a = -1

b. $f(x) = \frac{1}{(3x + 1)^2}$, $at = 0$

c. $f(x) = x^2 (x - 1)^3$, at a = 2

12. Suppose $f(1) = 2$, $f'(1) = -3$, $g(1) = 6$, and $g'(1) = 6$

Evaluate the given derivative.

a. $\frac{d}{dx} f(x)\, g(x)\Big|_{x=1}$

b. $\frac{d}{dx} \frac{g(x)}{f(x)}\Big|_{x=1}$

c. $\frac{d}{dx}\left(\frac{1 + 2f(x)}{x - g(x)}\right)\Big|_{x=1}$

d. $\frac{d}{dx}\left(\frac{4g(x)}{x} + f(x)g(x)\right)\Big|_{x=1}$

13. Find the critical values of the given function.

a. $f(x) = x^3 - 12x + 8$

b. $f(x) = 2x^3 - 15x^2 - 36x$

c. $f(x) = \frac{x^2}{x - 1}$

d. $f(x) = (x + 4)^{2/3}$

e. $f(x) = x^{1/3}(x - 4)$

14. Determine the intervals on which the given function f is increasing and the intervals on which f is decreasing.

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a. $f(x) = x^2 + 8x - 1$

b. $f(x) = x^3 - 3x^2$

c. $f(x) = x^4 - 4x - 4x^3 + 9$

15. Water runs into a conical tank at the rate of 12π cm^3/min. The tank stands point down and has a height of 8cm and a base radius of 4cm. How fast is the water level rising when the water is 3cm deep?

16. Heating - plate: when a circular plate of mater is heated in an oven its radius increases at the rate of 0.01 cm/min. At what rate is the plate's area increasing when the radius is 50cm?

17. An ice cub that is 4cm on each side is melting at a rate of 6cm^3 per min. How fast is the length of the side decreasing?

18. Evaluate each definite Integral

a. $\int_2^4 -3dx$

b. $\int_{-4}^{1} \sqrt{2}dx$

c. $\int_1^4 \left(5x\sqrt{x} + 3\sqrt{x}\right)dx$

d. $\int_1^{64} \frac{7\sqrt{x} - 2}{\sqrt[3]{x}}dx$

19. What must be true of F (x) and G (x) if both are anti derivatives of $f(x)$?

20. What would be the anti derivative of velocity function?
What would be the anti derivative of acceleration

21. Evaluate the following integers.

a. $\int 4\,dx$

b. $\int \left(5\sqrt{x} + \sqrt{2}\right)dx$

c. $\int \left(4\sqrt{x} - 2x^{3/5}\right)dx$

d. $\int \left(\frac{\pi^3}{x^2} - \frac{\sqrt{\pi}}{\sqrt{x}}\right)$

e. $\int (x - 2)^2 dx$

f. $\int \frac{\sqrt{x} + 1}{\sqrt[3]{x}}\,dx$

g. $\int \frac{1 - 2\sqrt[3]{x}}{\sqrt[3]{x}}\,dx$

h. $\int \left(x^{1/4} + \pi^{1/4}\right)dx$

22. Find the total area between the region and the x - axis

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a. $f(x) = -x^2 - 2x, \quad -3 \le x \le 2$

b. $f(x) = 3x^2 - 3, \quad -2 \le x \le 2$

c. $f(x) = x^3 - 3x^2 + 2x, \; 0 \le x \le 2$

Answer and Solution

1. a. $Vav = \dfrac{f(2) - f(1)}{2 - 1} = \dfrac{6 - 0}{1} = 6$

b. $Vav = \dfrac{f(3) - f(1)}{3 - 1} = \dfrac{7 - (-1)}{2} = 4$

c. $Vav = \dfrac{f(4) - f(1)}{4 - 1} = \dfrac{\sqrt{15} - 0}{4 - 1} = \dfrac{\sqrt{15}}{3}$

d. $Vav = \dfrac{f(2) - f(1)}{2 - 1} = \dfrac{2 - 1}{1} = 1$

2. a. $r_{inst} = f'(1) = 3(1) - 1 = 2$

b. $r_{inst} = f'(-4) = 2(-4) + 3 = -5$

3. a. $y - f(-1) = f'(-1)(x + 1)$

$y + 3 = -6(x + 1)$

b. $y - f(1) = f'(1)(x - 1)$

$y - 5 = 9(x - 1)$

4. a. Average rate $= \dfrac{f(4) - f(2)}{4 - 2} = \dfrac{18 - 4}{2} = 7$

b. $P'(2) = 8 - 5 = 3$

c. $P'(4) = 16 - 5 = 11$

5. a. $R'(1000) = 10 - 0.004(1000) = 6$

b. $R'(1500) = 10 - 0.004(1500) = 4$

6. a. $y - f(2) = \dfrac{-1}{f'(2)}(x - 2)$

$y + 1 = \dfrac{1}{4}(x - 2)$

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b. $y - f(-1) = \frac{-1}{f'(-1)}(x + 1)$

$y + \frac{7}{3} = -\frac{1}{5}(x + 1)$

c. $y - f(1) = \frac{-1}{f'(1)}(x - 1)$

$y = \frac{1}{5}(x - 1)$

7. $y - f(-1) = \frac{-1}{f'(-1)}(x + 1)$

$y - 7 = \frac{1}{7}(x + 1)$

8. $f'(x) = -1,\ f'(x) = 3x^2 - 4x$

$3x^2 - 4x = -1 \Rightarrow 3x^2 - 4x + 1 = 0$

$\therefore \quad x = \frac{1}{3}$ or $x = 1$

$\therefore \quad (1, 3)$ and $\left(\frac{1}{3}, \frac{103}{27}\right)$

9. $f'(x) = 6x + 4$ and $\frac{-1}{f'(a)} = 2,\ f'(a) = \frac{-1}{2}$

$f'(a) = 6a + 4 = \frac{-1}{2},\ a = \frac{-3}{4}$

$\left(\frac{-3}{4}, \frac{21}{16}\right)$ is point on the graph

10. $\left.\begin{array}{l} f(1) = a + b = 7 \\ f'(1) = 2a + b = 2 \end{array}\right\}\ a = -5,\ b = 12$

11. a. $f'(x) = 3(x^2 + 4)^2 (2x)$

$f'(-1) = 3(1 + 4)^2 (-2) = 150$

b. $f'(x) = -2(3x + 1)^{-3} (3)$

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$f'(0) = -2(1)(3) = -6$

c. $f'(x) = 2x(x-1)^3 + 3x^2(x-1)^2$

$f'(2) = 4 + 12 = 16$

12. a. $f'(x)\,g(x) + f(x)\,g'(x)$

$f'(1)\,g(1) + f(1)\,g'(1) = -3(6) + 2(6) = -6$

b. $\dfrac{g'(x)\,f(x) - f'(x)\,g(x)}{f(x)^2}$

$\dfrac{g'(1)\,f(10 - f'(1)\,g(1)}{f(1)^2} = \dfrac{6(2) - (-3)(6}{2^2} = \dfrac{15}{2}$

c. $\dfrac{(2f'(x)(x - g(x)) - (1 + 2f(x))(1 - g'(x))}{[x - g(x)]^2}$

$= \dfrac{2f'(1)[1 - g(1)] - (1 + 2f(1))(1 - g'(1))}{[x - g(x)]^2}$

$= \dfrac{2(-3)[1 - 6) - [(1 + 2(2))(1 - 6)}{(1 - 6)^2} = \dfrac{11}{5}$

13. a. $f'(c) = 3c^2 - 12 = 0$, $c = -2$ and $c = 2$

b. $f'(c) = 6c^2 - 30c - 36 = 0$, $c = 6$ and $= -1$

c. $f'(c) = 0$ when $c = 0$ and $c = 2$

d. $f'(c) =$ does not exist when $c = -4$

$\therefore$ $c = -4$ is critical number

e. $f'(x) = \dfrac{4}{3}x^{-2/3}$ $(x) = \dfrac{4(x - 1)}{3x^{2/3}}$

$\therefore$ $c = 1$ and $c = 0$ are critical number

14. a. f is increasing on $[-4, \infty)$ and decreasing on $(-\infty, -4]$.

b. f is increasing on $\left(-\sqrt{2}\ \ 0\right)$ and $\left[\sqrt{2},\ \infty\right)$

f is decreasing on $\left(-\infty,\ -\sqrt{2}\right]$ and $\left[0\ \ \sqrt{2}\right]$

c. f is decreasing on $(-\infty, 0] \cup [0, 3)$ f is increasing on $[3, \infty)$.

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15. $\text{Vol} = \frac{1}{3}\pi r^2 h$ and $\frac{r}{h} = \frac{4}{8} = \frac{1}{2}, r = \frac{h}{2}$

$V = \frac{\pi h^3}{12}$, and $\frac{dv}{dt} = \frac{3\pi h^2}{12}\frac{dh}{dt} = \pi h^2 \frac{dh}{dt}$

$12\pi = \pi(3)^2 \frac{dh}{dt}, \frac{dh}{dt} = \frac{12}{9} = \frac{4}{3}\ cm/\text{min}$

16. $A = \pi r^2, \frac{dA}{dt} = 2\pi r \frac{dr}{dt}$

$\therefore \quad \frac{dA}{dt} = 2\pi(50)\ (0.01) = \pi\ cm^2/\text{min}$

17. $V = x^2, \frac{dV}{dt} = 3x^2 \frac{dx}{dt}$

$\frac{dx}{dt} = \frac{6}{3(4)^2} = \frac{1}{8}\ cm/\text{min}$

18. a. $\left[-3x\right]_2^4 = -12 - (-6) = -6$

b. $\left[\sqrt{2}x\right]_{-4}^{1} = \sqrt{2} - \sqrt{2}(-4) = 5\sqrt{2}$

c. $\left[2\sqrt{x^5} + 2\sqrt{x^3}\right]_1^4 = 76$

d. 21

19. F (x) and G (x) differ by constant C

20. Displacement (distance)

21. Velocity of function

22. a. $4x + c$

b. $\frac{10}{3}x^{3/2} + \sqrt{2}x + c$

c. $\frac{8}{3}x^{3/2} + \frac{5}{6}x^{8/5} + c$

d. $\frac{-\pi}{x} + 2\sqrt{\pi x} + c$

e. $\frac{1}{3}x^3 - 2x^2 + 4x + c$

f. $\frac{6}{7}x^{7/6} + \frac{3}{2}x^{2/3} + c$

g. $\frac{3}{2}x^{2/3} - 2x + c$

h. $\frac{4}{5}x^{5/4} + \pi^{1/4}x + c$

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23. a. $A = -\int_{-3}^{-2}(-x^2 - 2x)\,dx + \int_{-2}^{0}(-x^2 - 2x)dx - \int_{0}^{2}(-x^2 - 2x)\,dx$

$= \frac{4}{3} + \frac{4}{3} + \frac{20}{3} = \frac{28}{3}$ sq. unit

b. $A = \int_{-2}^{-1}(3x^2 - 3)dx - \int_{-1}^{1}(3x^{-2} - 3)dx + \int_{1}^{2}(3x^2 - 3)\,dx$

$= 4 + 4 + 4 = 12$ square unite

c. $A = \int_{0}^{1}(x^3 - 3x^2 + 2x)\,dx - \int_{1}^{2}(x^3 - 3x^2 + 2x)\,dx$

$= \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ sq. unite

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UNIT 12

12. Linear Programming

12.1 Inequalities and Linear Programing

What is an inequalities

An **inequality** is statement that indicate that two algebraic expressions are not equal in a specific way, one expression being greater than or less than the other. In this unit we shall see the solution and other graph of the following types of the following types of inequalities:

(i) Linear inequality in one variable
(ii) Linear inequality in two variable
(iii) System of linear inequality in two variable.

12.2 Solving Linear inequality in one Variable

A **linear inequality** in one variable is any relationships form: $ax + b < 0$, $ax + b < 0$, $ax + b \leq 0$, $ax + b > 0$ and $ax + b \geq 0$.

The **solution set** to a linear inequality contains an infinite number of elements and can be expressed in set builder notation, in interval notation, or graph.

How to Solve Linear Inequality

A linear inequality in one variable is solved by collecting the variable on the left side and the constant on the right side. Then the inequality divided by the coefficient of the variable.

Note: If both sides of an inequality are multiplied or divided by a negative number, then the direction of the inequality sign must be reversed.

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Illustrative Example

1. Solve the inequality and express the solution in graph

a. $\frac{x-4}{2} - \frac{3x+4}{10} \geq \frac{-3x}{5}$ b. $\frac{3x-2}{4} - \frac{2-x}{5} \leq \frac{2x-1}{3}$

Solution

a. Multiply both sides by LCM (2, 5, 10) = 10 We get:

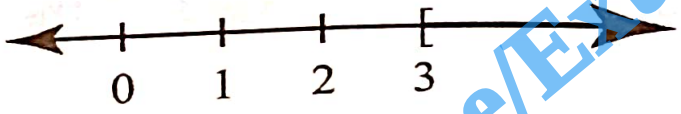
$$10\left(\frac{x-4}{4}\right) - 10\left(\frac{3x+4}{10}\right) \geq 10\left(\frac{-3x}{5}\right)$$

$5x - 20 - 3x - 4 \geq -6x$

$2x + 6x \geq 24$, implies that $8x \geq 24$, $x \geq 3$.

$\therefore$ The solution set in interval notation $[3, \infty)$

- This solution can also shown on the number line graph.



b. Multiplying both sides by LCM (3, 4, 5) = 60

$$60\left(\frac{3x-2}{4}\right) - 60\left(\frac{2-x}{5}\right) \leq 60\left(\frac{2x-1}{3}\right)$$

$15(3x - 2) - 12(2 - x) \leq 20(2x - 1)$

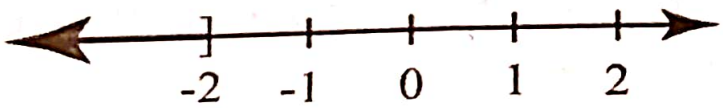
$45x - 30 - 24 + 12x \leq 40x - 20$

$57x - 54 \leq 40x - 20$

$17x \leq -20 + 54$

$x \leq \frac{34}{17}$ this implies that $x \leq 2$.

This solution can also shown on the number line



2. Solve the system of compound inequality in one variable $\begin{cases} 2x - 7 < 3 \\ 3x - 1 > 5 \end{cases}$

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Solution

$2x - 7 < 3$ and $3x - 1 > 5$

$2x < 10$ and $3x > 6$

$x < 5$ and $x > 2$

$(-\infty, 5) \cap (2, \infty) = (2, 5)$

3. Solve the system of compound inequality

a. $5 < -2x + 7 \leq 11$ b. $1 \leq \frac{5x - 4}{2} < 3$

Solution

a. $5 < -2x + 7 < 11$. Subtract (-7) from three inequality parts.

$5 - 7 < -2x + 7 - 7 \leq 11 - 7$

$-2 < 2x \leq 4.$

$\frac{-2}{-2} > \frac{-2x}{-2x} \geq \frac{4}{2}$. Divide all three parts by (-2)

$1 > x \geq -2$ equivalntely $-2 \leq x < 1$.

b. Multiply all three parts by 2

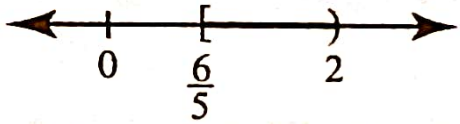
$2(1) \leq 2\left(\frac{5x - 4}{4}\right) < 3(2)$

$2 \leq 5x - 4 < 6$

$2 + 4 \leq 5x - 4 + 4 < 6 + 4$

$6 \leq 5x < 10$

$\frac{6}{5} \leq x < 2$



Word Expression Leading to Linear inequalities

There are variety of ways to express inequalities verbally. A word problems containing phrases like greater than or less than can be described as follows:

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Word Expression	Inequality
x is at least a	$x \geq a$
x is not less than a	$x \geq a$
x is more than a	$x > a$
x is at most a	$x \leq a$
x is not more than a	$x \leq a$
x is less than a	$x < a$

4. To receive Grade A in a course one must obtain an average of 90 marks or more in five examination (each of 100 marks). If Daniya's marks in first four examinations are 95, 94, 87 and 92. Find the minimum mark that Daniya's must obtain in fifth examination to get grade A in the course.

Solution

$$\text{Average marks} = \frac{95 + 94 + 87 + 92 + x}{5} \geq 90$$

$\therefore \quad 368 + x \geq 450$

$\therefore \quad x > 450 - 368, \quad x \geq 82$

$\therefore$ Minimum marks to be obtained is 82.

12.3 Linear Inequalities in Two Variable

A linear inequalities is a linear equation with the equal sign replaced by an inequality.

A linear inequality in two variables has the form:

$ax + by < c$, $ax + by \leq c$, $ax + by > c$ or $ax + by \geq c$.

Where a and b are not both zero.

- A **solution** to an inequality in two variables is an ordered pair that satisfies the inequality.

Illustrative Example

5. Determine whether the ordered pair is solution of $3x + 4y < 12$

a. $(0, 0)$ b. $(5, 1)$ c. $(-2, 2)$ d. $(4, 0)$

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Solution

(x, y)	Substitute	Conclusion
a. (0, 0)	$3(0) + 4(0) = 0 < 12$	(0, 0) is a solution
b. (5, 1)	$3(5) + 4(1) = 19 < 12$	(5, 1) is not a solution
c. (-2, 2)	$3(-2) + 4(2) = 2 < 12$	(-2, 2) is a solution
d. (4, 0)	$3(4) + 4(0) = 12 < 12$	(4, 0) is not a solution

6. Determine whether the ordered pair is a solution of the inequality

a. $y < 2x + 3$, (-3, -1)

b. $y > -3x + 5$, (2, 3)

c. $y < -x + 4$, (3, 1)

d. $y > 2x + 1$, (-1, 1)

Solution

a. It is not: $-1 < -6 + 3$ is false

b. It is: $3 > -1$

c. It is not : $1 < 1$ is false

d. It is: $1 > -1$

12.4 Graphing Linear Inequalities in Two Variables

Every non - vertical line divides the xy plane in to two regions, called **two half plane.**

- One region (half plane) is above the line.
- The other region (half plane) is below the line.

Method of Graphing Inequality

Method 1. Solve the inequality for y, then graph $y = mx + b$.

The graph of the inequality:

1. $y \geq mx + b$ is satisfied above the solid line $y = mx + b$.
2. $y \leq mx + b$ is satisfied below the solid line $y = mx + b$.

3. The Test point Method

The graph of any line $ax + by = c$ separates the xy plane into two regions (two half - plane). Every point on one side of the line satisfies $ax + by < c$: and every point on the other side satisfies

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the inequality $ax + by > c$. This facts to graph an inequality by the test
- point **method:**

1. Graph the corresponding equation
2. Choose any point not on the line

- If the test point satisfies the inequality, then, all points are solution set of the inequality.
- If not, then the solution set is the other region. With this method it is not necessary to solve the inequality for y.

Illustrative Example

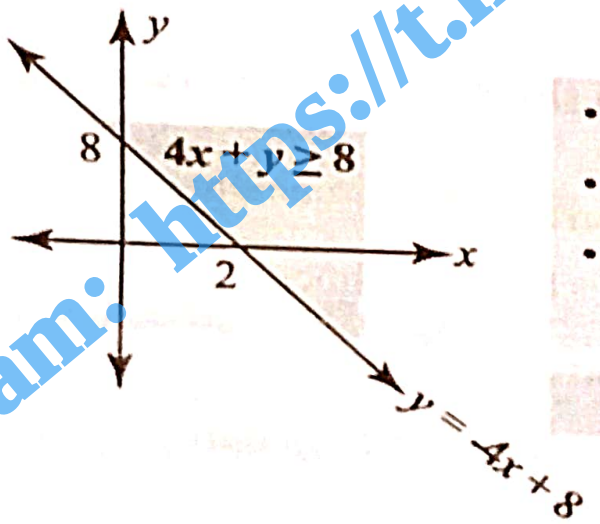
7. By using method of graphing inequality graph the solution set of the inequality:

   a. $4x + y > 8$ b. $2x - 3y < 6$.

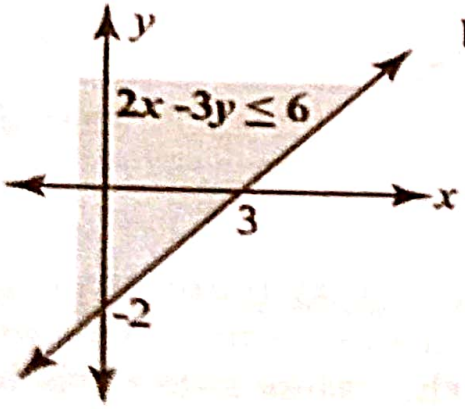
Solution

a. By using method 1.

Solve the inequality for y, then graph the boundary line.



- $4x + y \geq 8$ is equivalent to $y \geq -4x + 8$
- The slope is -5, and y - intercept of 8
- The graph of the inequality: $y \geq -4x + 8$ is the half plane region above the line $y = -4x + 8$



b. By using Test point method :

First graph the equation $2x - 3y = 6$

Using the x - intercept and the y - intercept.

If $x = 0$, then $y = -2$, $(0, -2)$

If $y = 0$, then $x = 3$, and x - intercept is $(3, 0)$

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Select a point on one side of the line, say (0, 0) to test the inequality.

Because $2(0) - 3(0) \leq 6$ is true then the solution set is the region containing the test point and shade the region.

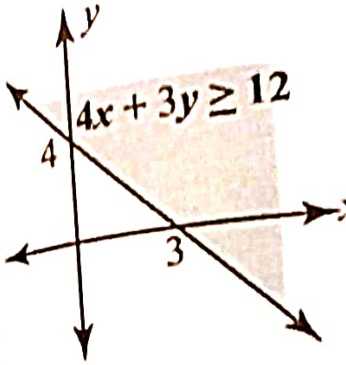
8. Graph the solution set of the following inequalities

a. $4x + 3y \geq 12$

b. $3x - y \geq 6$

Solution

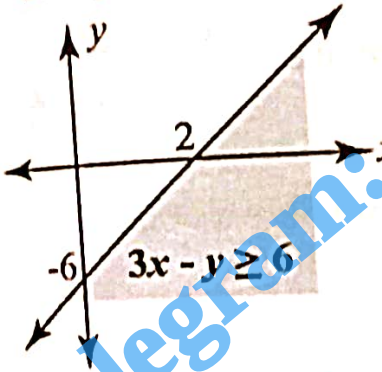
a. By using method 1. Solve the inequality for y, then graph the boundary line



- $4x + 3y \geq 12$ is equivalent to $y \geq -\frac{4}{3}x + 4$

  The y intercept is (0, 4).
- If $y = 0$, then $x = 3$, and the x - intercept is (3, 0).
- The graph of the inequality: $y \geq -\frac{4}{3}x + 4$ is the region above line $y = -\frac{4}{3}x + 4$

b. By using. Test point method. First graph the equation $3x - y = 6$.



- If $x = 0$, then $y = -6$, $(0, -6)$
- If $y = 0$, then $x = 2$, $(2, 0)$

To graph the inequality, select a point on one side of the line, say (0, 0) to test the in equality, Because $3(0) - 0 = 0 \geq 6$ is false, the region on the other side of region the satisfies the inequality.

12.4.1 Solving System of Inequalities in Two Variable

Two or more inequalities in two common variable make up a system of inequalities in two variables. The solution set is the set of order pairs that satisfy each inequality in the system. To graph the solution set to system of inequalities, graph the solution sets to the individual inequalities first. The **solution set to the system of inequalities is the intersection of the graphs of the inequalities.**

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Illustrative Example

9. Determine whether the order pairs is a solution to the system of the inequalities $\begin{cases} 2x - 4y < 9 \\ -3x + y \geq 4 \end{cases}$

a. (5, 3) b. (3, -4) c. (3, 4) d. (-2, 3) e. (0, 0)

Solution

Ordered pair	Satisfies $2x - 4y < 9$	A satisfies $-3x + y \geq 4$	Satisfies both inequality
a. (5, 3)	Yes	No	No
b. (3, -4)	No	No	No
c. (3, 4)	Yes	No	No
d. (-2, 3)	Yes	Yes	Yes
e. (0, 0)	Yes	No	No

The ordered pair (-2, 3) satisfies the system $2x - 4y < 9$ and $-3x + y \geq 4$.

10. Graph the solution set to the system of the inequalities $\begin{cases} x + y \geq 3 \\ 2x + 3y \leq 6 \end{cases}$

Solution

First solve for y, then: $x + y \geq 3$ is equivalent to $y \geq -x + 3$.

$2x - 3y \leq 6$ is equivalent to : $y \geq \frac{2}{3}x - 3$.

The graph of the inequality: $y \geq -x + 3$ is region above the solid line $y = -x + 3$.

$y \geq \frac{2}{3}x - 3$ is the region above the solid line $y \geq \frac{2}{3}x - 3$

The intersection of the shaded region is the solution of the system.

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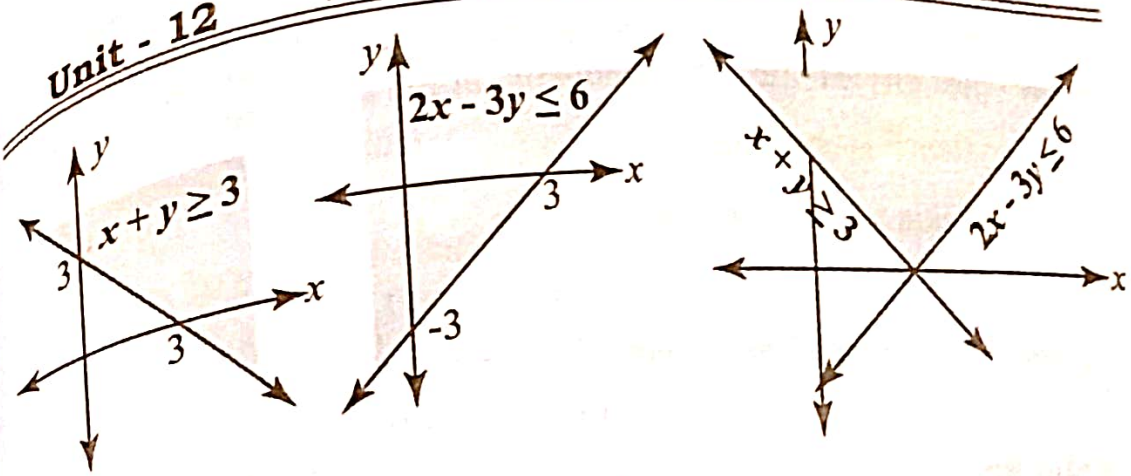


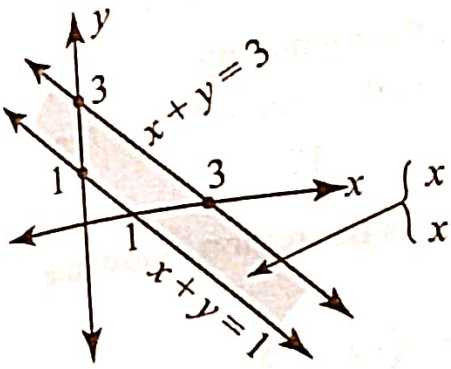
11. Graph the solution of the system $\begin{cases} x + y \le 3 \\ x + y \ge 1 \end{cases}$

Solution: First solve for y:

$x + y \le 3$ equivalent to $y \le -x + 3$

$x + y \ge 1$ equivalent to $y \ge -x + 1$.

Then the graph of inequality:



$\begin{cases} x + y \le 3 \\ x + y \ge 1 \end{cases}$

1. $y \le -x + 3$ is the region below the line $y = -x + 3$
2. $y \ge -x + 1$ is the region above the line $y = -x + 1$.

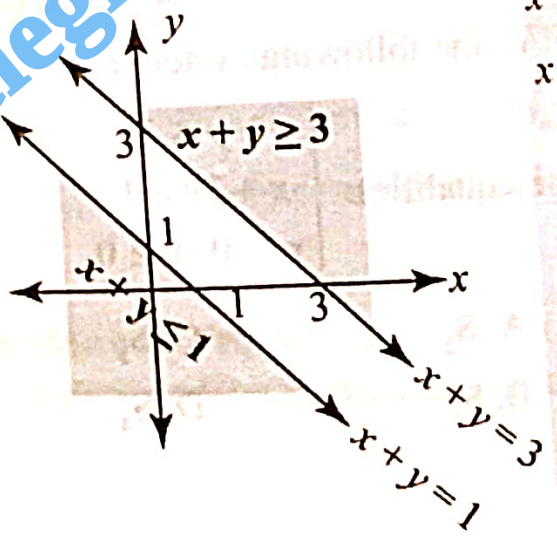
12. Graph the solution of the system $\begin{cases} x + y \ge 3 \\ x + y \le 1 \end{cases}$

Solution: First solve for y:

$x + y \ge 3$ equivalent to $y \ge -x + 3$

$x + y \le 1$ equivalent to $y \le -x + 1$

- The graph of the inequality:
- $y \ge -x + 3$ is the region above the line $y = -x + 3$
- $y \le -x + 1$ is the region below line $y = -x + 1$



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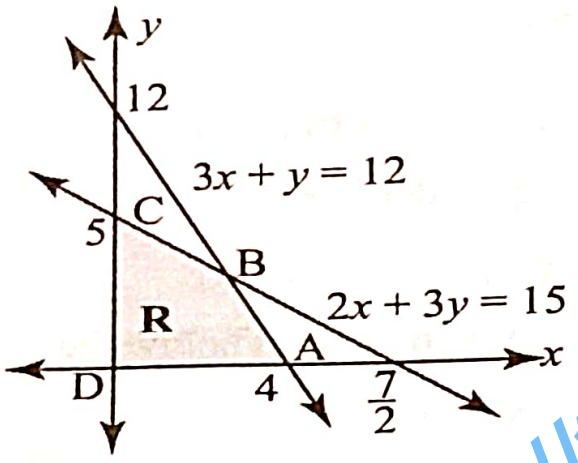
From the figure, because no overlaping region results, there are no points in the xy - plan that satisfies both inequality at the same time

$\therefore$ The system has no solution

$\therefore$ The solution set of system is empty set

13. Graph the solution of the system $\begin{cases} 3x + y \le 12 \\ 2x + 3y \le 15 \\ x \ge 0,\ y > 0. \end{cases}$

Solution



First solve for y:

$3x + y \le 12$, equivalent to $y \le -3x + 12$

$2x + 3y \le 15$ equivalent to $y \le \frac{-2}{3}x + 5$

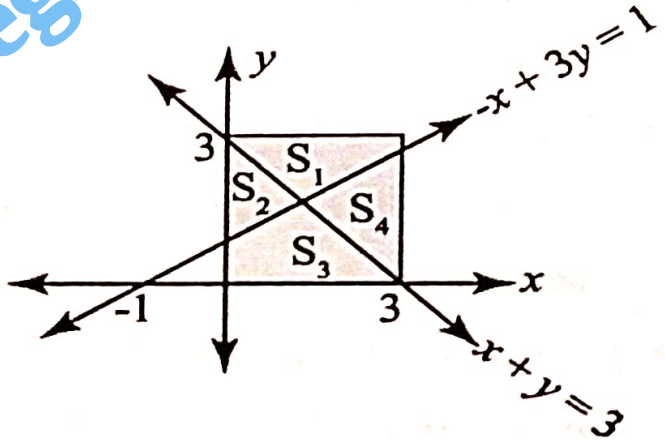
Then the graph the inequality:

- $y \le -3x + 12$ is the region below the solid line $y = -3x + 12$
- $y \le \frac{-2}{3}x + 5$ is the region below the solid line $y = \frac{-2}{3}x + 5$

A region R consisting of the overlapping parts of the inequalities usually called **feasible region** or the region of **feasible solution.**

The point on the boundary of R, A (4, 0), B (3, 3), C (0, 5) and D (0, 0) are called a **corner point** of R

14. Which are the region shaded in the figure below is the solution region of the following system of inequalities: $\begin{cases} x + y \le 3 \\ -x + 3y \ge 1 \\ x \ge 0,\ y \ge 0 \end{cases}$



A. S_2 C. S_3

B. S_1 D. S_4

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Solution

First we solve for y:

$x + y \leq 3$ equivalent $y \leq -x + 3$

$-x + 3y \geq 1$ equivalent to $y \geq \frac{1}{3}x + \frac{1}{3}$

The graph of the inequality:

1. $y \leq -x + 3$ is the region below the line $y = -x + 3$, this region contains (S_2) and S_3.

2. $y \geq \frac{1}{3}x + \frac{1}{3}$ is the region above the line $y = \frac{1}{3}x + \frac{1}{3}$ this region contains (S_2) ans S_1.

$\therefore$ The intersection of the two inequalities ① and ② is S_2

Answer: A

15. Form the above figure, which region is the solution region of the following system of inequalities: $\begin{cases} x + y \leq 3 \\ -x + 3y \leq 1 \\ x \geq 0, y > 0 \end{cases}$

A. S_2 B. S_1 C. S_3 D. S_4

Solution

The graph of the inequality:

1. $y \leq -x + 3$ is the region below the line $y = -x + 3$, this region contains S_2 and (S_3)

2. $y \leq \frac{1}{3}x + \frac{1}{3}$ is the region below the line $y = \frac{1}{3}x + \frac{1}{3}$, this region contains S_4 and (S_3)

$\therefore$ The intersection the inequalities 1 and 2 is S_3

Answer: C

16. Match the feasible region R_1 through R_7 in figure below with appropriate corresponding inequalities. In all cases assume $x \geq 0$, $y \geq 0$.

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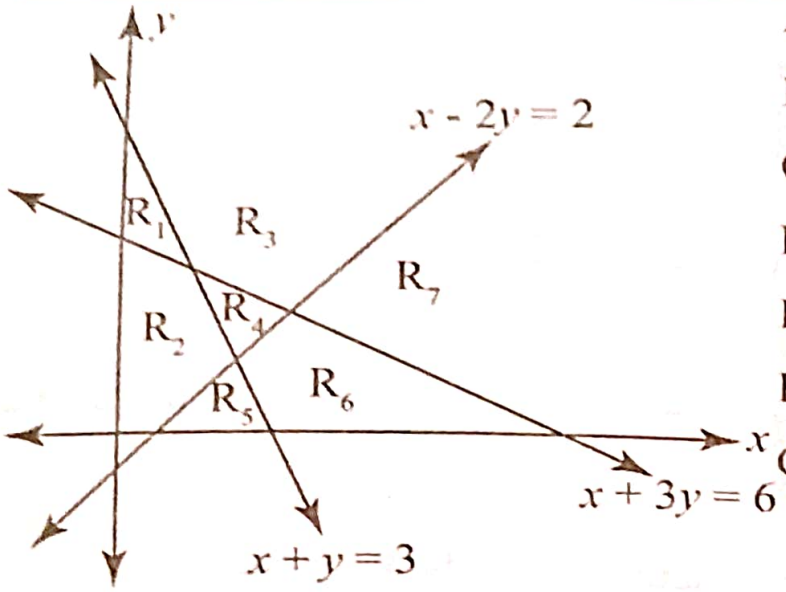

A. $x + y \geq 3, x + 3y \leq 6, x - 2y \leq 2$,

B. $x + y \leq 3, x + 3y \leq 6, x - 2y \leq 2$,

C. $x + 3y \geq 6, x - 2y \geq 2, x + y \geq 3$,

D. $x + y \leq 3, x + 3y \geq 6, x - 2y \leq 2$,

E. $x + y \leq 3, x + 3y \leq 6, x - 2y \geq 2$,

F. $x + y \geq 3, x + 3y \geq 6, x - 2y \leq 2$,

G. $x + y \geq 3, x + 3y \leq 6, x - 2y \geq 2$

Solution

Region R_1 is described by inequalities in D

Region R_2 is described by inequalities in B

Region R_3 is described by inequalities in F

Region R_4 is described by inequalities in A

Region R_5 is described by inequalities in E

Region R_6 is described by inequalities in G

Region R_7 is described by inequalities in C

17. Which one of the following points in the region of the inequalities: $-x + 3y \leq 10, x + y \leq 6, x - y \leq 2, x \geq 0, y \geq 0$.

A. (3, 4) B. (1, 1) C. (3, 0) D. (0, 4)

Solution

(x, y)	$-x + 3y \leq 10$	$x + y \leq 6$	$x - y \leq 2$
A (3, 4)	$-3 + 12 \leq 10 = T$	$3 + 4 < 6 \equiv F$	$3 - 4 \leq 2 = T$
B (1, 1)	$-1 + 3 \leq 10 = \mathbf{T}$	$1 + 21 \leq 6 = \mathbf{T}$	$1 - 1 \leq 2 = T$
C (3, 0)	$-3 + 0 \leq 10 = \mathbf{T}$	$3 + 0 \leq 6 = \mathbf{T}$	$3 - 0 \leq 2 = F$

$\therefore$ The point (1, 1) satisfies all the inequality

Answer: B

18. Which of the following ordered pair is a solution of the system of

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linear inequalities. $\begin{cases} x \geq -1 \\ y < 2 \\ y > 2x + 2 \end{cases}$

A. (0, 0) B. (-1, 1) C. $\left(\frac{-1}{2}, 1\right)$ D. (-1, 0)

Answer: B

12.5 Minimum and Maximum Value in Linear Programming Problem

What is a linear programming?

Many real life problems involve a process called **optimization**, which means finding the minimum or maximum value of some quantity. Here you will study one type of optimization process called **linear programming**. Linear programing is the process used to **maximize** or **minimize** a linear function subject to linear constraints.

- The function to be optimized (maximize or minimized) in linear programming is called **objective function.**
- The objective function often has two. Independent variable: $Z = f(x, y) = ax + by$, the variable Z is dependent on x and y.
- In general, every linear programming problem has two components:

1. A linear objective function: $Z = ax + by$ that is maximize or minimized.
2. A collection of linear inequalities they are called **constraints.**

A **feasible region** is a region consisting of the intersection parts of two or more graph of inequalities in a system. Every point in the feasible region satisfies all the constraint.

A **corner point** is a point in the feasible region where the boundary line of two constraints cross.

A feasible region is **bounded** if the region is enclosed by boundary lines on all sides. Other wise the feasible regions is **unbounded.**

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How to find the maximum or minimum value or the object quantity ($Z = ax + by$)?

The steps to solve a linear programming problem

Step - 1 : write an object function $Z = f(x, y)$

Step - 2 : Graph the feasible region of a given linear inequalities (constants)

Step - 3 : Identify the vertices (corner points).

Step - 4 : Evaluate the object function at each vertex of the feasible region.

- Then the least of value is the minimum of the feasible region. The greatest of these value is the maximum of the feasible region.

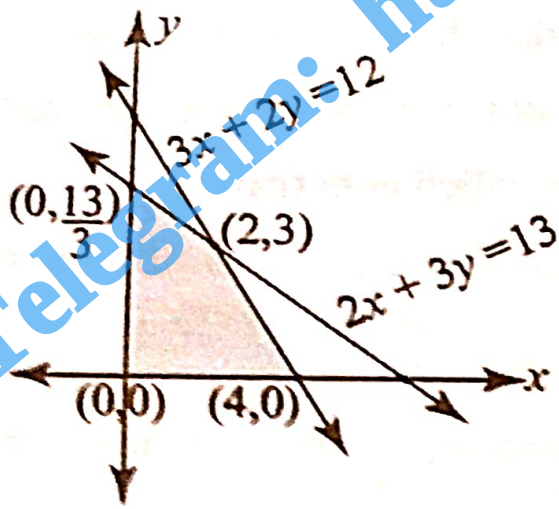
Illustrative Example

19. Solve a linear programming problem. Find the minimum and maximum value of objective function. $Z = 5x + 7y$

Subject to the constraints: $\begin{cases} 2x + 3y \leq 13 \\ 3x + 2y \leq 12 \\ x \geq 0,\ y \geq 0. \end{cases}$

Solution

First solve for y, in each inequalities



$2x + 3y \leq 13$ equivalent to $y \leq \frac{-2}{3}x + \frac{13}{3}$

$3x + 2y \leq 12$ equivalent to $y \leq \frac{-3}{2}x + 6$

Then the graph of the inequality:

1. $y \leq \frac{-2}{3}x + \frac{13}{3}$ is the region below the line $2x + 3y = 13$

2. $y \leq \frac{-3}{2}x + 6$ is the region below line $3x + 2y = 12$

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A corner point in a feasible region are $(0, 0)$, $(4, 0)$, $(2, 3)$ and $\left(0, \frac{13}{2}\right)$

Then we substitute each of the four corner point into the objective function: $Z = 5x + 7y$.

Corner point	Value of $Z = 5x + 7y$
At $(0, 0)$	$Z = 5(0) + 7(0) = 0$ **minimum**
At $(4, 0)$	$Z = 5(4) + 7(0) = 20$
At $(2, 3)$	$Z = 5(2) + 7(3) = 31$ **maximum**
At $\left(0, \frac{13}{3}\right)$	$Z = 5(0) + 7\left(\frac{13}{3}\right) = \frac{91}{3}$

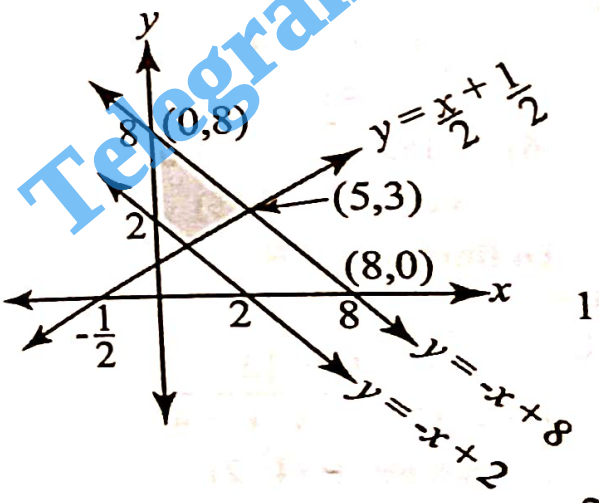
$\therefore$ The minimum value of Z is 0, when $x = 0$ and $y = 0$

$\therefore$ The maximum value of Z is 31 when $x = 2$ and $y = 3$.

20. Find the maximum and minimum value of the objective function $Z = 3x + 8y$. subject to $\begin{cases} x - 2y \leq -1 \\ x + y \geq 2 \\ x + y \leq 8 \end{cases}$

Solution

To make drawing the graph of the inequality easy, first solve for y



$x - 2y \leq -1$ equivalent to $y \geq \frac{1}{2}x + \frac{1}{2}$

$x + y \geq 8$ equivalent to $y \geq -x + 2$

$x + y \leq 8$ equivalent to $y \leq -x + 8$

The graph of the inequality:

1. $y \geq \frac{1}{2}x + \frac{1}{2}$ is the region above the boundary line $y = \frac{1}{2}x + \frac{1}{2}$

2. $y \geq -x + 2$ is the region above the boundary line $y = -x + 2$

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3. $y \leq -x + 8$ is the region below the boundary line $y = -x + 8$

The four vertices (corner point) of the feasible region are (1, 1), (0, 2), (0, 8) and (5, 3).

The value of the objective function $Z = 3x + 8y$ at each of the four corner points:

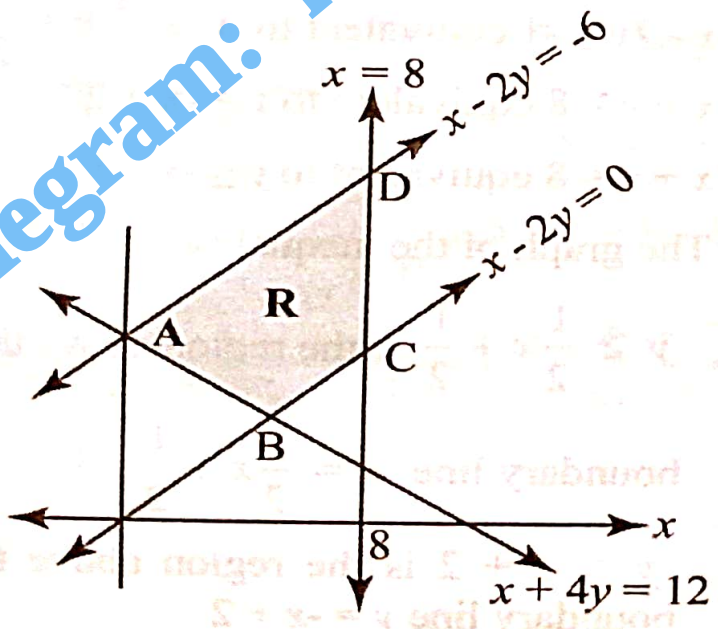
Corner point	Value of $Z = 3x + 8y$
At (1, 1)	$Z = 3(1) + 8(1) = 11$ minimum
At (0, 2)	$Z = 3(0) + 8(1) = 16$
At (0, 8)	$Z = 3(0) + 8(8) = 64$ maximum
At (5, 3)	$Z = 5(3) + 8(3) = 39$

The feasible region is bounded so, the minimum value of objective function Z is 11, it occurs when $x = 1$ and $y = 1$. The maximum value of Z is 64, it occurs when $x = 0$ and $x = 8$.

21. Find the maximum and minimum values of the objective function $Z = 2x + 5y$ subject to the constraints:
$x + 4y \geq 12,\ x - 2y \geq -6,\ x - 2y \leq 0, x \leq 8$

Solution

On the same axes, graph each inequality by graphing the boundary line. Also, we have $y \geq \frac{-1}{4}x + 3$, $y \leq \frac{1}{2}x + 3$ and $y \geq \frac{1}{2}x$



To find vertex A

$$-\begin{cases} x - 2y = -6 \\ x + 4y = 12 \end{cases}$$

$-6y = -18,\ y = 3,\ x = 0$

$\therefore$ Vertex A (0, 3)

To find vertex B

$$-\begin{cases} x - 2y = 0 \\ x + 4y = 12 \end{cases}$$

$-6y = -12,\ y = 3,\ x = 4$

$\therefore$ Vertex B (4, 2)

vertex C (8, 4)

vertex D (8, 7)

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Value of the objective function at corner point of feasible region.

Corner point	Value of $Z = 2x + 5y$
At (0, 3)	$Z = 2(0) + 5(3) = 15$ minimum
At (4, 2)	$Z = 2(4) + 5(2) = 18$
At (8, 4)	$Z = 2(4) + 4(5) = 36$
At (8, 7)	$Z = 2(8) + 5(7) = 51$ maximum

∴ The minimum value of $Z = 2x + 5y$ is 15 at the corner point (4, 2).

The maximum value of $Z = 2x + 5y$ is 51 at the corner point (8, 7).

22. Determiner whether each region is bounded or unbounded for each system of inequalities.

a. $\begin{cases} x + 3y \le 6 \\ 2x + 4y \ge 7 \\ x \ge 0, y \ge 0 \end{cases}$

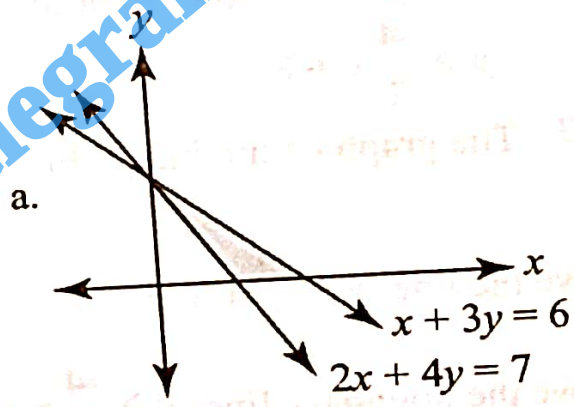
b. $\begin{cases} x + y \le 7 \\ x - y \le -4 \\ x \ge 0, y \ge 0 \end{cases}$

c. $\begin{cases} 2x - y \ge -4 \\ x + y \ge 2 \\ x \ge 0, y \ge 0 \end{cases}$

d. $\begin{cases} 3x + 4y \ge 12 \\ 2x - 3y \le 6 \\ 0 \le y \le 2 \\ x \ge 0 \end{cases}$

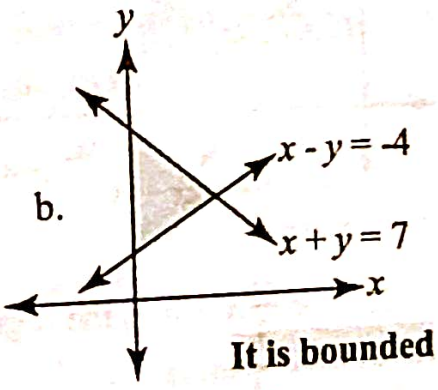
Solution

a.



It is bounded

b.



It is bounded

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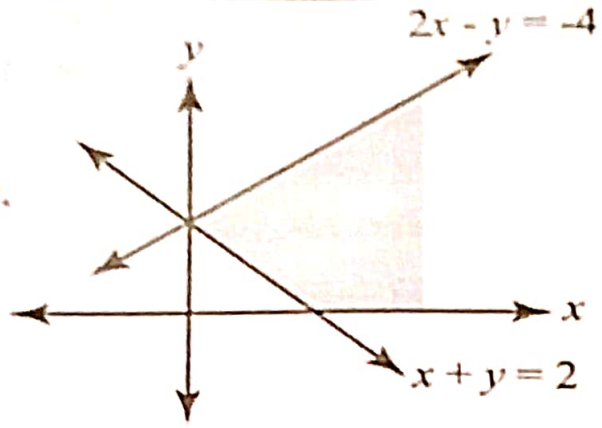
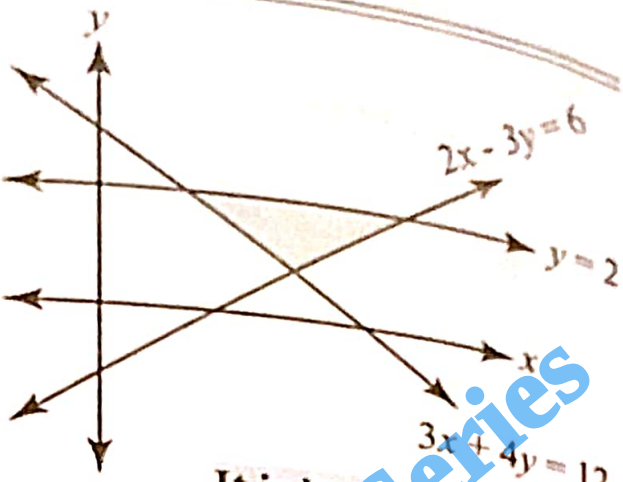
c.



It is unbounded

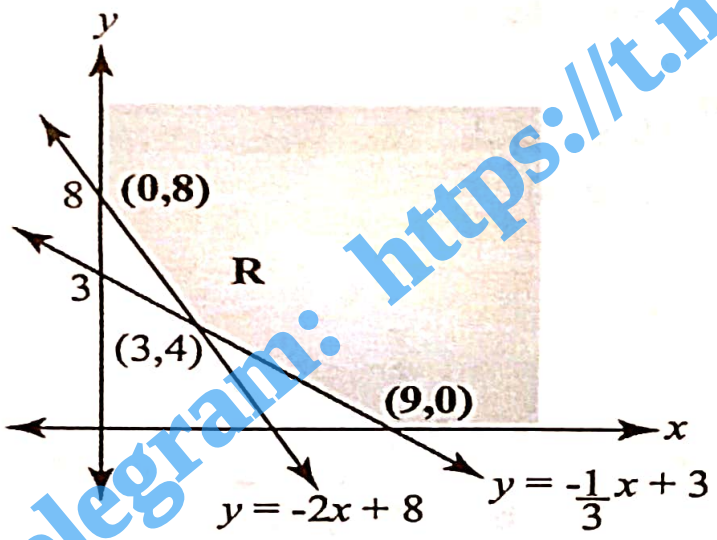
d.



It is bounded

23. Find the values of x and y which minimize and maximize the objective function $Z = 6x + 5y$ subject $\begin{cases} x + 3y \geq 15 \\ 4x + 3y \geq 24 \\ x \geq 0,\ y \geq 0 \end{cases}$

Solution



To make drawing easy, first solve for y:

$x + 3y \geq 15$ equivalent to

$y \geq \frac{-1}{3}x + 5$

$4x + 3y \geq 24$ equivalent to

$y \geq \frac{-4}{3}x + 8$

The graph of the inequality:

1. $y \geq \frac{-1}{3}x + 5$ is the region above the line $y \geq \frac{-1}{3}x + 5$
2. $y \geq \frac{-4}{3}x + 8$ is the region above the boundary line $y \geq \frac{-4}{3}x + 8$.

The three vertices (corner points) are (0, 8), (3, 4), and (15, 0). The value of object function $Z = 6x + 5y$ at each corner point:

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At (0, 8) $Z = 6(0) + 5(8) = 40$

At (3, 4), $Z = 6(3) + 5(4) = 38$

At (15, 0), $Z = 6(15) + 5(0) = 90$

∴ The minimum value of Z is 38 it occurs when $x = 3$ and $y = 4$.

No - maximum value, since the graph of constraints the feasible region is unbounded.

Note: In linear programing problem:

- As long as the feasible region is not empty, and with bounded regions always have solution.
- On the other hand, the feasible region in above example is unbounded, and no Solutions will maximize the values of the objective function be cause Z can be made as large as you like.

24. Find values of x and y which minimize or maximize the value of the objective function $Z = 4x + 12y$ subject to: $\begin{cases} 2x + y \geq 8 \\ x + 3y \geq 9 \\ x > 0,\ y \geq 0 \end{cases}$

Solution

First solve for y:

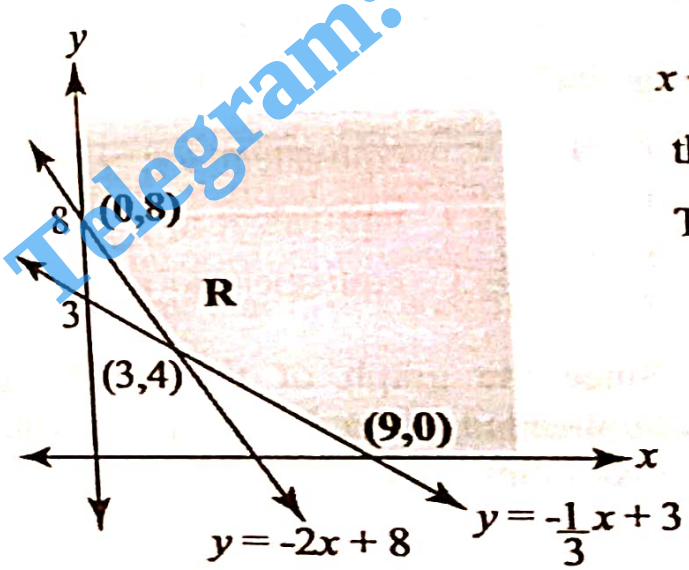
$2x + y \geq 8$ equivalent to $y \geq -2x + 8$

$x + 3y \geq 9$ equivalent to $y \geq \frac{-1}{3}x + 3$

the feasible region:

The graph of the inequality:

1. $y \geq -2x + 8$ is the region above the line $y = -2x + 8$
2. $y \geq \frac{-1}{3}x + 3$ is the region above line $y = \frac{-1}{3}x + 3$



The corner points are (9, 0), (3, 2) and (0, 8).

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Value of the objective function at corner point

Corner point	Value of $Z = 4x + 12y$
At (9, 0)	$Z = 4(9) + 12(0) = 36$
At (3, 2)	$Z = 4(3) + 12(2) = 36$
At (0, 8)	$Z = 4(0) + 12(8) = 96$

Thus both (9, 0) and (3, 2) as well as all the point on the boundary line segment between the two points (9, 0) and (3, 2) gives the same optimal solution of $Z = 36$. Hence, we conclude that there are infinitely many equal values of x and y that will give the same minimum value of the objective function $Z = 4x + 12y$. This minimum value is $Z = 36$.

As example, $\left(4, \frac{5}{4}\right)$, $\left(5, \frac{4}{3}\right)$, (6, 1), $\left(7, \frac{2}{3}\right)$, $\left(8, \frac{1}{3}\right)$, ... (9, 0)

are gives the same optimal solution $Z = 36$. No - maximum solution.

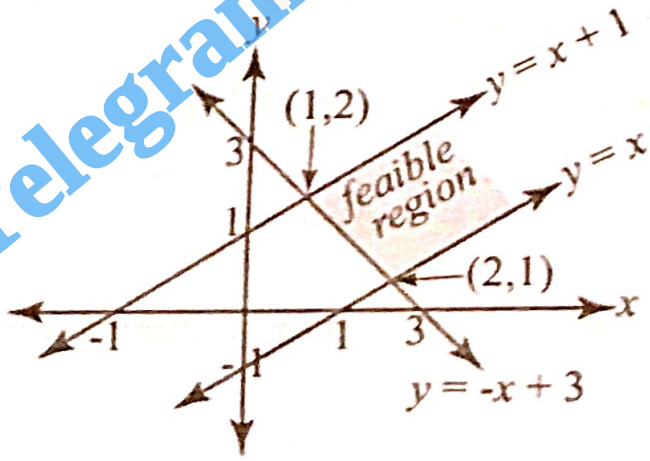
25. What is the maximum value of $Z = -3x + 2y$

subject $\begin{cases} -x + y \leq 1 \\ x - y \leq 1 \\ x + y \geq 3 \end{cases}$ ..... UEE

A. 1 B. -4 C. 6 D. It has no maximum

Solution

The graph of a feasible region is as follows:



$\begin{cases} -x + y \leq 1 & \text{equivalent } y \leq x + 1 \\ x - y \leq 1 & \text{equivalent to } y \geq x - 1 \\ x + y \geq 3 & \text{equivalent to } y \geq -x + 3 \end{cases}$

Since the graph of the feasible is unbounded, then it has no specific maximum value.

But it has minimum value at $x = 2$ and $y = 1$, with value $Z = -4$.

Answer: D

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Conceptual skill

26. Where does the maximum or minimum value of liner function subject to linear constraints occur?

Explanation

It occurs at vertex (corner) of feasible region.

27. What is objective function?

Explanation

The function to be optimised in a linear programming is objective function.

28. What is the feasible region?

Explanation

The region in the plane that represents the solution set to a system of constraints.

29. Is there necessarily a uniqe point in the feasible region where the maximum or minimum occur.

Answer: No, it may not be uniqe point. Refer, example # _____

30. Determine whether each of the following statements is true or false, and explain why?

a. The feasible solution that gives an optimal solution to linear programing problem is always uniqe, if it exist.

b. Every linear programing problem with a feasible region that is not empty either has solution or is unbounded.

c. The optimal solution to a linear programing problem can occur at a point that is not a corner point.

Solution

a. False, refer, example # 24.

b. True

c. True, refer example # 24.

31. Which of the following ordered pair is in the feasible region of the linear programming problem: objective function $Z = 7x + 4y$

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Subject to: $\begin{cases} 2x + y \leq 15 \\ 3x + 8y \leq 30 \\ x \geq 0,\ y \geq 0 \end{cases}$

A. (4, 7) B. (7, 1) C. (5, 5) D. (3, 3)

Solution

The point (7, 1) is in the feasible region, it satisfies all the inequalities.

Answer: B

32. In a linear programing problem (L.P.P) the objective function: maximize $Z = 7x + 4y$

Subject to: $\begin{cases} 3x + 8y \leq 30 \\ 4x + 2y \leq 15 \\ x \geq 0,\ y \geq 0 \end{cases}$ then which of the following is true?

A. The point (2, 3) is a corner point of the L.P.P.

B. The point (2, 3) is in the feasible region of the L.P.P

C. The optimal solution to occurs at point (2, 3)

D. The graph of the feasible region unbounded.

Solution

A. False B. True C. False D. False

Answer: B

33. Which point maximizes $Z = 13x + 2y$ and lies with in the feasible region of the constraints $\begin{cases} -3x + 2y \leq 8 \\ -8x + y \geq -48 \\ x \geq 0,\ y \geq 0 \end{cases}$

A. (8, 16) C. (4, 3)

B. (6, 0) D. (8, 4)

Solution

The maximum value of $Z = 13x + 2y$ is 136 occurs when $x = 8$, and $y = 16$

Answer: A

34. Which point minimizes $Z = 3x + 4y$ and lies in the feasible region of

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the constrains: $\begin{cases} x + 5y \geq 16 \\ -x + 4y \leq 20 \\ 2x + y \geq 5 \\ 2x + y \leq 14 \end{cases}$

A. (4, 6) C. (1, 3)

B. (2, 6) D. (2, 4)

Solution

The minimum value of $Z = 3x + 4y$ is 15 occurs when $x = 1$ and $y = 3$

Answer: C

35. **Critical thinking:** Is there necessarily a unique point in the feasible region where the maximum or minimum occurs? Why or why not

Answer: No

When the optimal solution occurs at two consecutive corner points every point is also an optimal solution

Example: $Z = 2x + 4y$, subject to $x + 2y \geq 10, 3x + y \geq 10, x \geq 0, y \geq 0$.

The point (2, 4) (10, 0), (8, 1), (6, 2) are all points on the boundary line, gives the same minimum value of objective function $Z = 2x + 4y$. This minimum value is 20.

Applications of Linear Programming

We can use a line programming models to analze cost, and to maximize profit

Illustrative Example

36. A farmer needs to raises only goats and cows. He wants to raise no more than 16 animals, including no more than 10goats. He spends Birr 2500 to raise a goat and Birr 7500 to raise a cow, and he Birr 90,000 for this project. Each goat produces Birr 1200 in profit and each cow Birr 4000 in profit. How many goats and how many cows should he raises to maximize profit?

Solution

First set up a table that shows information given in the problem

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	Goat	Cows	Total
Number Raised	x	y	≤ 16
Goat limit	x		≤ 10
Cost to Raise	2500	7500	$\leq 90{,}000$
Each	1200	4000	

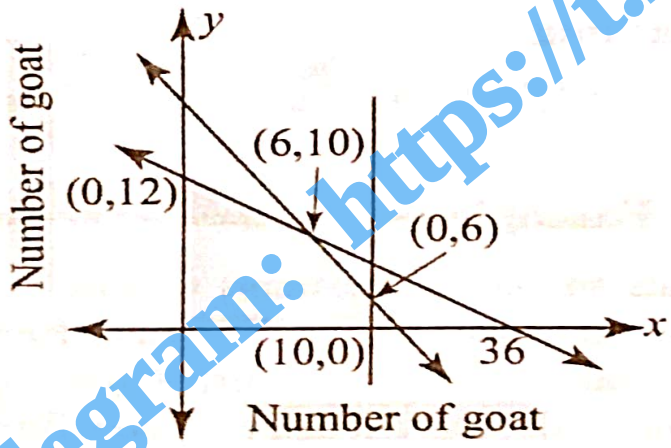
Since total number of animals can not exceed 16, so that $x + y \leq 16$.

No more than 10 goats means: $x \leq 10$

The cost to raise x - goats at Birr 2500 per goat is $2500x$ Birr. The cost to raise y - cows at Birr 7500 each is $7500y$ - Birr. Since only Birr 9000 available. $2500x + 7500y \leq 90000$ Divide both sides by 2500, gives $x + 3y \leq 36$. The number of goats and cows can not negative: $x \geq 0, y \geq 0$

The farmers wants to know how many goats and cows to raise in order to produce maximum profit. Each goat Birr 1200 profit and each cow Birr 4000 profit. If Z represents total profit, then $Z = 1200x + 4000y$.

Maximize: $Z = 1200x + 4000y$.



Subject to: $\begin{cases} x + y \leq 16 \\ x + 3y \leq 36 \\ x \leq 10, x \leq 0, y \leq 0. \end{cases}$

Value of $Z = 1200x + 4000y$

At (10, 0) $Z = 12000$

At (0, 6), $Z = 24000$

At (0, 12), $Z = 48000$

At (6, 10), $Z = 7200 + 40000 + 47200$

$\therefore$ The maximum profit is Birr 48000 occurs as $x = 0$ (No goats) and $y = 12$ cows will produce maximum profit.

37. A school wants rent buses to transport at least 3600 students to exam hall center. Large buses hold 60 people and small buses hold 45 people. Further more, the number of available bus driver is at most 75. Each large bus costs Birr 1200 to rent and each small buses that will

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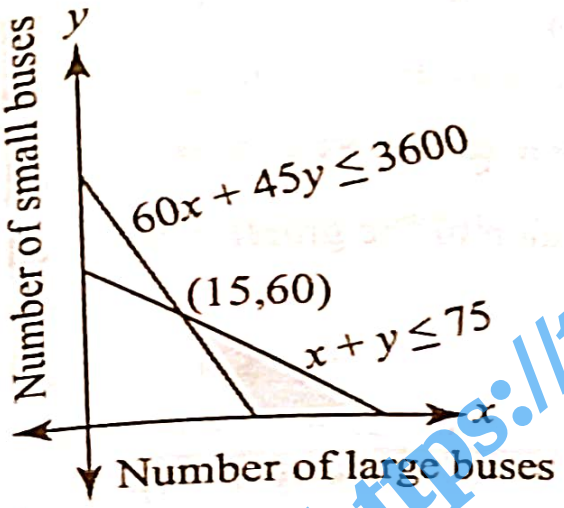
minimize cost. Find the optimal number of large and small buses that will minimize cost.

Solution

First set up a table that shows information given in the problem.

	Large bus	Small bus	
Rent Bus	x	y	≤ 75
Capacity	60	45	≤ 3600
Cost	1200	800	

Total number of buses can not exceeds 75, so that, write as $x + y \leq 75$



Total number of capacity of buses: $60x + 45y \leq 3600$

Dividing both sides by 60, gives equivalent in quality: x 60. $75y \leq 60$. number large buses and small buses can not negative: $x \geq 0, y \geq 0$. The corner points are (60, 0), (75, 0) and (15, 60).

The objective function (cost function: $Z = 1200x + 800$).

Corner point	$Z = 1200x - 800y$
At (60,0)	$Z = 1200\ (60) + 800(0) = 72{,}000$
At (75,0)	$Z = 1200\ (75) + 800(0) = 90000$
At (15,60)	$Z = 1200\ (15) + 800\ (60) = 66000$

The cost would be minimized if the school rents 15 large buses and 60 small buses. The minimum cost is Birr 66000.

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Review Exercise 12

1. What is a linear programing
2. What is the constraint
3. What is a feasible region?
4. How is the objective function used in a linear programing problem?
5. How is the system of constraint inequality used?
6. Decide whether the ordered pair is in the feasible region of the system of linear inequalities $\begin{cases} x + y \leq 6 \\ x + 2y \leq 8 \\ x + 4y \leq 14 \\ x \geq 0,\ y \geq 0 \end{cases}$

   a. (2, 4) b. (4, 2) c. (1, 5) d. (5, 1)

7. Determine whether the feasible region of the graph is bounded or unbounded.

   a. $\begin{cases} 2x + y \leq 4 \\ x + 2y \geq 4 \\ x \geq 0,\ y \geq 0 \end{cases}$ c. $\begin{cases} x + y \leq 4 \\ 2x + y \geq 6 \\ x \geq 0,\ y \geq 0 \end{cases}$

   b. $\begin{cases} 3x + y \geq 6 \\ 4x + 3y \geq 12 \\ x \geq 0,\ y \geq 0 \end{cases}$ d. $\begin{cases} x + y \geq 2 \\ 2x + 3y \leq 12 \\ 3x + y \leq 12 \end{cases}$

8. Find values of $x \geq 0$ and $y \geq 0$ that maximize $Z = 5x + 6y$ subject to each set of constraints

   a. $\begin{cases} x + y \leq 20 \\ x + 3y \leq 24 \end{cases}$ b. $\begin{cases} 3x + y \leq 15 \\ x + 2y \leq 24 \end{cases}$

9. Find values of $x \geq 0$ and $y \geq 0$ that minimize $Z = 5x + 8y$ subject to each set of constraints

   a. $\begin{cases} 4x + 3y \geq 30 \\ x + 3y \geq 21 \end{cases}$ b. $\begin{cases} x + y \geq 8 \\ x + 5y \leq 20 \end{cases}$

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10. Find the minimum and maximum value of the objective function

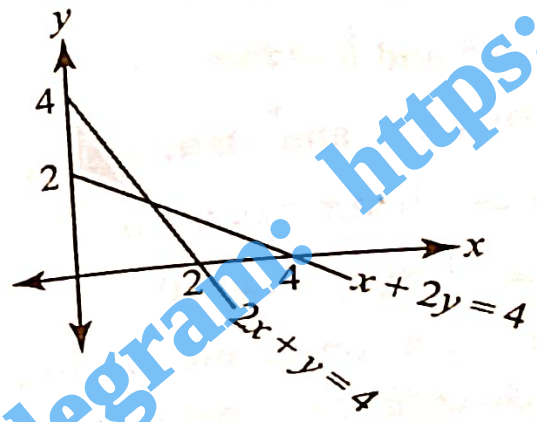
$Z = 2x + 4y$. Subject to: $\begin{cases} 3x + 2y \le 12 \\ 5x + y \ge 5 \\ x \ge 0,\ y \ge 0 \end{cases}$

11. Find the minimum and maximum value of the objective function

$Z = 4x + 2y$. Subject to: $\begin{cases} x + y \le 50 \\ 2x + y \ge 20 \\ x + 2y \ge 30 \\ x \ge 0,\ y \ge 0 \end{cases}$

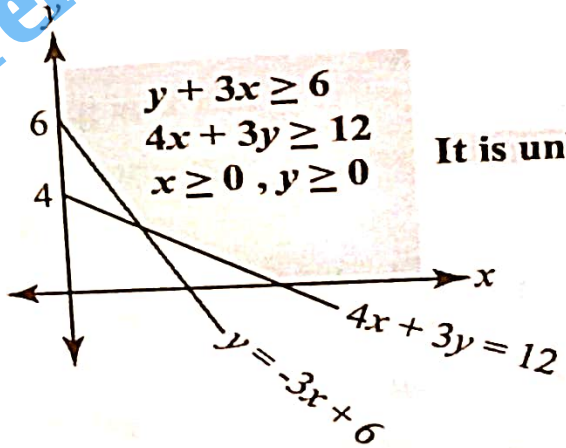
Answer to Review Exercise 12

3. The **feasible region,** containts all points that satisfy all the constraints.
4. Objective function is evaluated at the vertices formed by the constrains.
5. System of constraints define the region of possible solution.
6. a. It is not b. It is c. It is not d. It is
7. a.



It is bounded, the corner point are (0, 4), (0, 2) and $\left(\frac{4}{3}, \frac{4}{3}\right)$

b.



$y + 3x \ge 6$
$4x + 3y \ge 12$
$x \ge 0, y \ge 0$

It is unbounded

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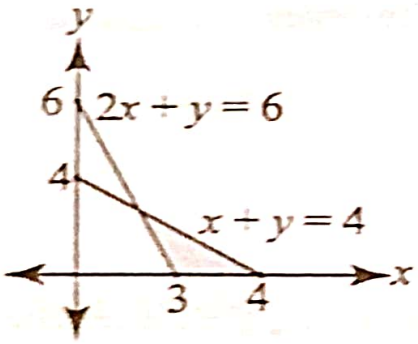
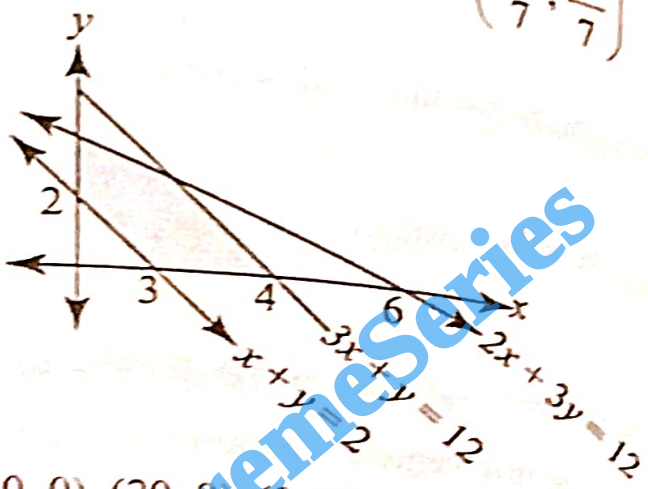
c. It is bounded with corner point(2, 2), (3, 0) and (4, 0)



d. It is bounded with corner points (0, 2), (0, 4), (2, 0), (4, 0) and $\left(\frac{24}{7}, \frac{12}{7}\right)$



8. a. The four corner points are (0, 0), (20, 0), (0, 8) and (18, 2). The maximum value is 102 when $x = 18$ and $y = 2$

   b. The maximum values is $\frac{354}{5}$ when $x = \frac{12}{5}$ and $y = \frac{49}{5}$

9. a. The minimum value is 63 when $x = 3$, and $y = 6$

   b. The minimum value is 49 when $x = 5$ and $y = 3$

10. The four corner points are (1, 0), (4, 0), (0, 5) and (0, 6)

   - Maximum value of $Z = 2x + 4y$ is 24 when $x = 0, y = 6$
   - Minimum value of $Z = 2x + 4y$ is 2 when $x = 1, y = 0$

11. Minimum of 40 at any point on the segment connecting (0, 20) and $\left(\frac{10}{3}, \frac{40}{3}\right)$, No. maximum value

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UNIT 13

13. Mathematics of Business and Finance

Buying a house, car usually requires both some saving for a down payment and a loan for the balance. Every one uses money. Some times you work for your money other times your money works for you. Here, in this units we going to see the principles of saving, investing and borrowing money.

13.1 What is a Ratio ?

A ratio, expresses a comparison between two quantities - it is a relation between two like numbers for two like values.

The ratio of a to b is written:

(i) with the colon a : b

(ii) as a fraction $\frac{a}{b}$

(iii) as a division a ÷ b.

The quantity that are being compared should be expressed in the same units

Illustrative Example

1. Find the ratio :

a. 12kg and 16kg

b. 9 months and 24 years

c. 40 minutes and one hour

d. 800 gm and 1 killogram

Solution

a. Ratio $= \frac{12\,kg}{16\ kg} = \frac{3}{4} = 3 : 4$

b. Ratio $= \frac{9\ months}{2\ years} = \frac{9\ months}{2 \times 12\ months} = \frac{3}{8} = 3 : 8$

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c. Ratio $= \frac{40 \text{ minutes}}{\text{one hours}} = \frac{40 \text{ minutes}}{60 \text{ minutes}} = \frac{2}{3} = 2:3$

d. Ratio $= \frac{800 \text{ gm}}{1 \text{ kgm}} = \frac{800 \text{ gm}}{1000 \text{ gm}} = \frac{4}{5} = 4:5$

2. A birr 50,000 is divided between two people in the ratio 3: 7. How much does each person get?

Solution

Let the number be $3x$ and $7x$

Total share: $3x + 7x = 50{,}000$, $x = 5000$.

$\therefore$ share of $3x = 3\ (5000) = 15{,}000$

share of $7x = 7(5000) = 35{,}000$.

Therefore, each of them get birr 15,000 and birr 35,000.

3. Three numbers are in the ratio 2:3:4. What are the number if their sum is 270.

Solution

Let the umber be $2x$, $3x$ and $4x$

Then: $2x + 3x + 4x = 270$ implies $9x = 270$

$\therefore \quad x = \frac{270}{9} = 30$

$\therefore$ The numbers are 60, 90 and 120.

4. The ratio of the number of boys and girls a school is 2:5. If there are 350 students in the school find the number of girls in the school.

Solution

$$\frac{5}{2+5} \times 350 = \frac{5}{7} \times 350 = 250 \text{ girls}$$

5. The ratio between two numbers is 3:5. If each number is increased by 4, the ratio become 2:3. Find the number.

Solution

Let the numbers be $3x$ and $5x$

Then $\frac{3x+4}{5x+4} = \frac{2}{3}$ implies $10x + 8 = 9x + 12$

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$x = 4$, then the numbers 12 and 20.

6. If Birr 2,600 is divided among Hanan, Rahel and Abebe in the ratio 11: 18: 23. Then, find the share of a person.

Solution

Let $11x$, $18x$, and $23x$, be the share

$11x + 18x + 23x = 2600$ implies $x = \frac{2600}{52} = 50$.

$\therefore$ share of : Hanan $= 11 \times 50 = 550$

share of Rahel $= 18 \times 50 = 900$

share of Abebe $= 23 \times 50 = 1{,}150$

7. A sum of money is divided among A, B and C in the ratio 3:5:7. If the share of B is Birr 1,020. What is the total amount of money shared?

A. Birr 3060 B. Birr 1428 C. Birr 5000 D. Birr 600

Solution

Let $3x$, $5x$ and $7x$ be the share

Then $5x = 1020$, so that $= x = \frac{1020}{5} = 204$.

Therefore $3(204) + 5(204) + 7(204) = 3060$

Answer: A

Continued Ratio

Three quantities a, b and c are said to be in **continued ratio** if the ratio between the first and the second quantities is same as the ratio between the second and the third quantities.

That is a : b : c is equal to a:b = b : c.

8. If A: B = 7 : 9 and B : C = 8 : 11, then A : B : C is equal to

A. 72: 56: 99 B. 56: 72: 99 C. 99: 56: 72 D. 56: 99: 72

Solution

We have: A : B = 7:9 = 7 × 8: 9 × 8 = 56 : 72

B : C = 8: 11 = 8 × 11 = 11 × 9 = 72 : 99

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$\therefore$ A : B : C = 56 : 72 : 99

Answer: B

9. The ratio of the number of boys and girls in a school is 4:5. If the percentage increase in the number of boys and girls is 15% and 22% respectively, what will be the new ratio

A. 6 : 7 B. 6 : 11 C. 46 : 61 D. 41 : 90

Solution

Let the boys be $4x$ and girls be $5x$

$$\frac{4x + 15\% \text{ of } 4x}{5x + 22\% \text{ of } 5x} = \frac{4x(1 + 0.15)}{5x(1 + 0.22)} = \frac{4.6}{6.1} = \frac{46}{61}$$

10. If Birr 782 be divided into three parts in the ratio $\frac{1}{2} : \frac{2}{3} : \frac{3}{4}$ then the first part is

A. Birr 182 B. Birr 190 C. Birr 196 D. Birr 204

Solution

$$\frac{1}{2} : \frac{2}{3} : \frac{3}{4} = \frac{1}{2} \times 12 : \frac{2}{3} \times 12 : \frac{3}{4} \times 12 = 6 : 8 : 9$$

$$\text{First part: } \frac{6}{6 + 8 + 9} \times 782 = 6 \times 34 = 204$$

Answer: D

13.2 What is Rate

Rate is the ratio of two measurements having different units of measure For example: a speed of 40km per hour.

- A rate that has a denominator of 1 is called a **unit rate.**
- A rate that identifies the cost of an item per units is called the **unit percent.**

Illustrative example

11. A car has a fuel consumption of 7 km per liter. How much fuel will be needed for a journey of 105 km?

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Solution

$$\frac{7kg}{1\ell t} = \frac{105\ km}{x}\ ,\ \text{then}\ x = \frac{105\ km}{7\ km} \times 1\ell t = 15\ell t$$

A potatoes are sold at Birr 330 per 55 kg. Calculate the cost per kg.

Solution

$$\frac{330\ Birr}{55\ kg} = \frac{x}{1\ kg}\ ,\ \text{then}\ x = \frac{330\ km}{55}\ Birr\ = 0$$

$\therefore$ 6 Birr per kg

12. If the average price of an item increased from Birr 24 to Birr 32 in the last four years. Find the rate of increase?

Solution

$$\text{Rate of increase} = \frac{\text{change of increase}}{\text{original price}}$$

$\therefore$ The rate of increase is $0.333 \times 100\% = 33.3\%$

13. Which has the better unit price 60kg Teff for Birr 2,990 or 120kg of Teff for Birr 4,500?

Solution

$$\text{First Rate per kg} = \frac{2990}{60} = Birr\ 49.83/kg$$

$$\text{Second rate per kg} = \frac{4500}{120} = Birr\ 37.5/kg$$

Therefore, 120 kg of Teff for Birr 4,500 is better unit price.

13.3 What is a Proportion

A Proportion is an equation expressing the equality between two ratio. The equality of the ratios $\frac{a}{b}$ and $\frac{c}{d}$ may be written: $\frac{a}{b} = \frac{c}{d}$

The end terms **a, d** are called the **extremes** of the proportion. The middle terms **b, c** are called the **means** of the proportion.

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Illustrative Example

14. A blue Print paper for a house that 6cm equals 3m. If the length of a wall is 3.6m, how long is the wall in the blue print paper?

Solution

$$\frac{6\text{ cm}}{3\text{m}} = \frac{x}{3.6\text{m}}, \text{ this implies } x = \frac{6\text{cm} \times 3.6\text{m}}{3\text{m}} = 7.2\text{cm}.$$

15. A 10-m length of string weights 350g. How long is a string that weighs 875 g.

Solution

$$\text{Using proportion: } \frac{10}{350} = \frac{x}{875}$$

$$\therefore \quad x = \frac{10}{350} \times 875 = 25\text{m}$$

16. Selam worked 5 hours and was paid Birr 187.50. How much is she paid for working 8 hours?

Solution

$$\text{Using proportion: } \frac{5}{187.50} = \frac{8}{x}$$

$$\therefore \quad x = \left(\frac{187.50}{5}\right)(8) = \text{Birr } 300.$$

Compound Proportion

When two or more quantities are related to each other and:

(i) If the first becomes greater when the second also become greater.

(ii) or, if the first becomes smaller when the second also become smaller then, their relationship is known as **direct proportion.**

DEFINITION: If y is directly proportional to x, then

$\frac{y}{x} = k$ or $y = kx$, or $\frac{y}{x} = k$ where $k \neq 0$ and k is called the **constant of proportionality.**

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y is directly proportional to x, means y varies directly as x.

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- As x gets increase, y gets increase.
- As x gets decrease, y gets decrease.

On the other hand: When two quantities are related to each other and the first become grater but the second become smaller, or greater when the first become smaller their relationship is know as an **inverse proportion.**

DEFINITION: Inverse proportion

If y is inversely proportional to x, then $xy = k$ or $y = \frac{k}{x}$, $k \neq 0$

Where k is a constant of proportionality

- y is said to vary inversely as x,

Illustrative Example

17. Suppose y varies directly as x and $y = 2$, when $x = 3$.

Solution

a. Find y when $x = 24$

b. Find y when $x = 18$

c. Find x when $y = 24$

d. Find x when $y = 18$

Solution

Since y varies directly as x, we can write this proportion.

a. $\frac{2}{3} = \frac{y}{24}$, this implies $y = \frac{2}{3}(24) = 16$

b. $\frac{2}{3} = \frac{y}{18}$, this implies $y = \frac{2}{3}(18) = 12$

c. $\frac{2}{3} = \frac{24}{x}$, this implies $x = \frac{3}{2}(24) = 36$

d. $\frac{2}{3} = \frac{18}{x}$, this implies $x = \frac{3}{2}(18) = 27$

18. Suppose y varies inversely as x. If $x = 3$, when $y = 8$, what is the value of when $x = 6$

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Solution

Since y varies inversely as x, there is a k such that $x_1 y_1 = k$ and $x_2 y_2 = k$

$$(3)\ (8) = 6x \text{ implies } x = \frac{24}{6} = 4$$

$\therefore$ when $x = 6$, $y = 4$.

19. The pressure acting on $10m^3$ of air is raised from 1 atmosphere to 2 atmosphere. The temperature is kept constant. What is the volume of the air after the pressure is applied?

Solution

From Boyle's Law: $P_1V_1 = P_2V_2$

$(10m^3)\ (1\ atm) = (2\ atm)\ V_2$

$V_2 = 5m^3$.

20. Suppose $y = 4$ when $x = 3$, find an equation that relates x and y such that y and x are inversely proportion?

A. $y = \frac{3}{4}x$ B. $xy = 12$ C. $xy = \frac{3}{4}$ D. $y = \frac{4}{3}x$

Solution

If y is inversely proportional to x then xy = constant, so $xy = 12$

Answer: B

21. Suppose $y = 5$ when $x = 3$, find an equation that relates x and y such that x and y vary directly:

A. $xy = 15$ B. $y = \frac{5}{3}x$ C. $y = \frac{3}{5}x$ D. $xy = \frac{5}{3}$

Solution

Direct Proportion: $\frac{y}{x} = \frac{5}{3}$, $y = \frac{5}{3}x$

Answer: B

22. Suppose Z is directly proportional to x and inversely proportional to y.

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If $Z = 0.3$ when the ratio of y to x is 2 : 3 how much is Z when $x = 5$ and $y = 4$.

A. $\frac{5}{4}$ B. $\frac{4}{5}$ C. $\frac{3}{8}$ D. $\frac{1}{4}$

Solution

$Z \alpha x$ and $Z \alpha \frac{1}{y}$

$Z = \frac{kx}{y}$, in this case $0.3 = K\left(\frac{3}{2}\right)$

$\therefore \quad K = \frac{(0.3)\ (2)}{3} = \frac{1}{5}$

$\therefore \quad Z = \frac{kx}{y} = \frac{\left(\frac{1}{5}\right)(5)}{4} = \frac{1}{4}$

Answer: D

23. The amount of sales in a business enterprise is directly proportional to the number of items sold and inversely to the amount of tax paid which is 5% of the amount sold. If the sales amount to Birr 15000 and the number of items sold is 3000, what is the constant of proportionality in this situation?

A. 3750 B. 3650 C. 7500 D. 5700

Solution

Let A = Sales of amount sold = 15000

x = Number of items sold = 3000

y = Amount of tax paid = 5% of A = $\frac{5}{100} \cdot (1500) = 750$

$A \alpha \frac{x}{y}$ implies $A = \frac{kx}{y}$ and $15000 = \frac{3000K}{750}$

$\therefore \quad K = \frac{15000 \times 750}{3000\,K} = 3750$

Answer: A

24. Five men a building a wall take 24 days to complete it. How long would

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it take 15 men to complete it

A. 8 days B. 18 days C. 12 days D. 6 days

Solution

The more men, the less day, and the more day the less men. The more men, the more work.

∴ Number of men is inverse proportion to days, so the product of two relation is constant

$$(5)(24) = (x)(15) \text{ implies } x = \frac{5 \times 24}{15} = 8$$

Answer: A

25. A 15 men decided to do a work in 24 days. How many men can do the same work in

a. 6 days less time b. 6 days more time

Solution

The number of men and number of days are inverse proportion then
(15) (24) = 360 = K = constant

a. 6 days less time means 24 - 6 = 18 days

$$\therefore \quad (18)(x) = 360 \text{ implies } x = \frac{360}{18} = 20$$

∴ 20 men can do the work in 18 days

b. 6 days more means 24 + 6 = 30 days

$$\therefore \quad (30)(x) = 360 \text{ implies } x = \frac{360}{30} = 12 \text{ men}$$

∴ 12men can do the work in 30 days .

Scale Drawing Using Proportion

A scale drawing is used to present something that is too large or to small to be drawn to actual size

$$\text{Scale map} = \frac{\text{distance on the map}}{\text{actual distance}}$$

- The scales on the map is the ratio of the distance on the map to the actual distance.

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Illustrative Example

2. The scale a map is 0.75 cm to 1km. If the distance on the map between two tons is 60cm, then find the actual distance

Solution

$$\frac{0.75 \text{ cm}}{1 \text{ km}} = \frac{60 \text{ cm}}{x}$$

$$x = \frac{60\text{cm} \times 1\text{km}}{0.75 \text{ cm}} = 80 \text{ km}$$

13.4 Percentage, Base, Rate

Percent is fraction with denominator 100. A percent is a ratio that compares a number to 100, dented by %.

Example : $\frac{9}{100} = 9\%$

$$\frac{3}{20} = \frac{3 \times 5}{20 \times 5} = \frac{15}{100} = 15\%$$

$$\frac{3}{4} = \frac{3 \times 25}{4 \times 25} = \frac{75}{100} = 75\%$$

- The Base, rate, and percentage as a percent proportion. A percent proportion compares part of a quantity to the whole quantity, called the base.
- A **base** is the whole quantity.
- A **rate** is the percent of the quantity.
- A **percentage** is the part (portion) of the whole quantity.

DEFINITION: The based (B), rate (R), and percent (P) relation can express in percent proportion as

$$\frac{\text{Percentage (P)}}{\text{base (B)}} = \text{Rate}$$

$$\frac{\text{Percentage}}{\text{base}} = \text{percent}$$

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(i) Finding the percentage: What is 15% of 60?

Percentage (P) = 15% of 16 $= \frac{15}{100}(60) = 9$

So, 15% of 60 is 9.

(ii) Finding the base (B) : 9 is 15% of what ?

$\frac{P}{Base}$ = Rate

Here, P = 9, rate (R) = 15% = 0.15, B = ?

$\frac{P}{B} = 0.15$ implies Base (B) $= \frac{9}{0.15} = 60$

So, 9 is 15% of 60.

(iii) Finding the rate (percent): What percent of 60 is 9

$\frac{9}{60} = \frac{Rate}{100}$, Rate (R) $= \frac{9}{60} \times 100 = 15\%$

Illustrative Example

27. Find each number:

a. What percent of 200 is 80

b. What number is 12% of 72

c. 75 is 20% of what number

d. 24% of what number is 13.2

Solution

We use the percent proportion

a. $\frac{Rate}{100} = \frac{percentage}{Base}$, Rate (R) $= \frac{80}{200} \times 100\% = 40\%$

So, 40% of 200 is 80

b. $\frac{Rate}{100} = \frac{Percentage}{Base}$,

$\frac{12}{100} = \frac{P}{72}$ implies P = (72) (0.12) = 8.64

$\therefore$ 8.64 is 12% of 72.

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c. $\frac{\text{Rate}}{100} = \frac{\text{Percentage}}{\text{Base}}$,

$75 = 20\%\ of\ \ x$

$\therefore \quad x = \frac{75}{0.2} = 375$

$\therefore$ 75 is 20% of 375.

d. 24% of $x = 13.2$

$x = \frac{13.2}{0.24} = 55$

$\therefore$ 24% of 55 *is* 13.2.

28. Mr. Tesfay's salary is Birr 4800 and Mr. Zelalem has 20% more than Mr. Tesfaye's salary.

a. Find the salary of Mr. Zelalem

b. By what percent Tesfay's salary less than Mr. Zelalem salary.

Solution

a. Mr. Tesfay's salary = Birr 4800

$\therefore$ Mr. Zelalem's salary = 4800 + 20% of 4800

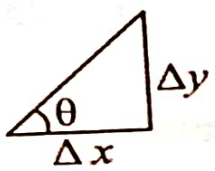
= 4800 + 960 = 5,760

b. Mr. Tesfay's salary less than Birr 960

$\therefore$ In percent $= \frac{960}{5,760}\ 100\% = 16.70\%$

Percentage and Slope

Slope is the change in y with respect to the change in x. Slope is a number that tells the steepness and direction of a line.



$$m = \frac{\text{vertical change}}{\text{horizontal change}}$$

$$m = \frac{\Delta y}{\Delta x} \text{ and } \tan\theta = \frac{\Delta y}{\Delta x}$$

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Illustrative Example

29. What does indicates:

a. The slope is 10% in road

b. The slope 40% in road

Solution

a. $10\ \% = \frac{10}{100} = \frac{1}{10}$, for every 1m horizontal distance change the road up 1 m vertically.

And at an angle of $\theta = \tan^{-1}\left(\frac{1}{10}\right) = 5.7^0$

b. $40\ \% = \frac{40}{100} = \frac{2}{5}$, For every 5m horizontal distance the road up 2m vertically, and angle of $\theta = \tan^{-1}\left(\frac{2}{5}\right) = 21.8^0$

30. You happened to notice that the 8% sales taxes that your paid on the new Television set come to Birr 3,800. How much did the television set cost him.

A. Birr 30,600

B. Birr 35,400

C. Birr 45,800

D. Birr 47,500

Solution

Sale tax: 8% of price television = 3800

$$\therefore \quad \text{Price of television} = \frac{3800}{0.08} = 47,500$$

Answer: D

31. Voter turnout in the local school election declined 40% from last year. This year only 3600 eligible voters went to the polls. How many people voted in last year's school board election?

A. 4500 B. 5000 C. 5500 D. 6000

Solution

Let x = voted last year and y = voted this year

$y = x - 40\%\ x = 3600$ and $x\ (1 - 0.4) = 3600$

$$\therefore \quad x = \frac{3600}{0.6} = 6000$$

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Answer: D

32. If the price of item is increased by 15% and then its sale decreased by 10%. What is the percentage effect on the income?

A. 5% B. 3.5% C. 2.5% D. 25%

Solution

Let x = The price, then

$\text{Effect}_1 = x + 15\% \text{ of } x = x + 0.15x = 1.15x$

$\text{Effect}_2 = 1.15x - 10\% \text{ of } (1.15x) = 1.035x$

$$\therefore \quad \text{Net effect} = \frac{(\text{final}) - \text{orginal}}{\text{orginal}}$$

$$= \left(\frac{1.035x - x}{x}\right) \times 100\% = 3.5\%$$

Answer: B

13.5 Mark up and Discount

What is Mark up?

To make a profit, stores changes more for merchandise than they pay for it. The amount of increase in price is called the **mark up.**

- Mark up is added to the cost of merchandise to a rive at the selling price.
- The percent of increase is the **percent of mark up.**
- Mark up = (Selling price) - (cost price)
- Selling price is equal to the cost price plus the mark up.

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Illustrative Example

33. A shopkeeper buys an article for Birr 560 and sell it for 630. What is the mark up on article?

Solution

Mark up = (selling price) - (cost price) = 630 - 560 = Birr 70

34. An article is sold at Birr 2400 with 35% mark up on selling price. Calculate the cost price and the percent mark up on cost

Solution

The percent of mark up w.r.t selling price is :

$$\frac{\text{mark up}}{\text{Selling price}} = 0.35 \text{ and mark up} = (0.35)\ \text{S.P}$$

Let x = selling price

Mark up = S.P - C.P and C.P = S.P - (mark up)

cost price (C.P) = $x - 0.35x = 0.65x$

$\therefore$ C.P = (0.65) (2400) = Birr 1560

Percent mark up on cost

$$\frac{\text{mark up}}{\text{Cost price}} \times 100\% = \frac{(0.35)\ 2400}{1560} \times 100\% = \frac{840}{1560} \times 100\% = 53.8\%$$

35. Calculate the percent of mark up on cost, if the mark up of the selling price is 25%

Solution

The percent of mark up with selling price

$$\frac{\text{Mark up}}{\text{selling price}} = 0.25$$

Let x = the selling price

mark up = (0.25) selling price = $0.25x$

Mark up (selling price) - (cost price)

$0.25x = x$ - cost, then cost = $x - 0.25x$

The percent of mark up with cost price is:

$$\frac{\text{Mark up}}{\text{Cost price}} \times 100\% = \frac{0.25}{0.75} \times 100 = \frac{1}{3} \times 100\% = 33.3\%$$

36. Calculate the percent of mark up on cost if the mark up on retail is 42%.

Solution

The percent of mark up with respect to selling price is:

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$$\frac{\text{Mark up}}{\text{selling price}} = 0.42$$

So, the mark up = (0.42) selling price.

Mark up = (Selling price) - (cost price)

Cost price = selling price - mark up

Let x = selling price

cost price = $x - 0.42x = 0.58x$

$\therefore$ The percent mark up with respect to cost price is:

$$\frac{\text{Mark up}}{\text{Cose price}} \times 100\% = \frac{0.42x}{0.58x} \times 100\% = \frac{42}{58} \times 100\% = 72.4\%$$

37. One special perfume costs a cosmetic shope Birr 2100. If the shop desires a 25% mark up rate based on the selling price, then which one of the following should be set as a selling price for this perfume **(UEE)**

A. Birr 2300 B. Birr 2400 C. Birr 2600 D. Birr 2800

Solution

Mark up = (Selling price) - (cot price)

Mark up = S.P - C.P = S.P - 2100

And percent of mark up = $\frac{\text{mark up}}{\text{S.P}}$

$\therefore$ mark up = 0.25 (S. P)

0.25 S.P = S.P - 2100 implies S. P - 0.25 (S.P) = 2100

$$\text{S.P} = \frac{2100}{0.75} = 2800$$

Answer: D

Discount

What is Discount

In order to increase the sale or to dispose off old or damged goods, shopkeepers sell their articles at a price lower than their regular price (R.P). This reduction is called **discount.**

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Discount is the amount by which the regular price is reduced.

- The sale price is the regular price minus the discount.
- Selling price = (Regular price) - (Discount)
- The percent of the discount is a percent of decrease
- Percent of discount $= \frac{\text{Discount}}{\text{regular price}} \times 100\%$

Illustrative Example

38. The regular price of a table is Birr 750 and it is available at a discount of 20%. Find the discount and the selling price of the table.

Solution

$$\text{Discount} = 20\% \text{ of Birr } 750 = \frac{20}{100}(750) = \text{Birr } 150$$

Selling price = (regular price) - (Discount)

$$= 750 - 150 = \text{Birr } 600.$$

39. The regular price of an article is Birr 1,850 and is sold at Birr 1,628. Find the discount and the rate of discount.

Solution

Discount = (Regular price) - (Selling price)

$$= 1850 - 1628 = \text{Birr } 222$$

$$\text{Rate of discount} = \frac{\text{Discount}}{\text{Regular price}} \times 100\%$$

$$= \frac{222}{1850} \times 100\% = 12\%$$

40. During a sale a clothing shop the price of T - shirt is 30% discount of Birr 240. Find the regular and selling price

Solution

30% of the regular price = Birr 240

$$\therefore \quad \text{Regular price} = \frac{240}{0.3} = \text{Birr } 800$$

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Selling price = (Regular price) - (Discount)

= 800 - 240 = Birr 560

41. A shopkeeper advertise to sell his goods at 15% discount. Find the regular price of an article which is sold for Birr 1,360?

Solution:

Let x = the regular price

Selling price = (regular price) - (discount)

x - 15% of x = 1360

$x(1 - 0.15) = 1360$ Birr

$\therefore$ Regular price = $x = \frac{1360}{0.85} = 1600$ Birr

42. A dealer sold car at discounted price Birr 2,400,000 with a discount of Birr 800,000. Calculate his percentage of discount

Solution

Selling price = (regular price) - (discount price)

2,400,000 = regular price - 800,000

$\therefore$ Regular price = 2,400,000 + 800,000 = 3,200,000

$\therefore$ Rate of discount = $\frac{\text{Discount}}{\text{Regular price}} \times 100\%$

$= \frac{800,000}{3,200,000} \times 100\% = 25\%$

43. An item is sold for Birr 120 after 20% discount from the regular price. How much was the regular price of the item?

A. Birr 135 B. Birr 150 C. Birr 140 D. Birr 145

Solution: Let x = regular price

$$\text{percent of discount} = \frac{\text{Discount}}{\text{regular price}} = \frac{D}{x}$$

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$$0.2 = \frac{\text{Discount}}{\text{regular price}} = \frac{D}{x}$$

Discount : D = (0.2) (regular price) = $0.2x$

$\therefore$ Selling price = (regular price) - Discount

$$120 = x - 0.2x = 0.8x$$

$$\therefore \quad x = \frac{120}{0.8} = \text{Birr } 150$$

44. An item was marked at Birr 250. After discount of 16% was allowed, the item was sold at a profit of 5% on the original cost. **Answer: B**

Which of the following is not True?

A. The original cost of the item was Birr 235.50

B. The discount was Birr 40

C. The profit was Birr 10.50

D. The selling price of the item Birr 205.

Solution

Marked price is a regular price = Birr 250

Selling price = (regular price) - 16% of regular price

$$= 250 - \frac{16}{100}(250) = \text{Birr } 210$$

Original cost price is C.P = (regular price) - (5%) of regular price

$$= 250 - \left(\frac{5}{100}\right)(250) = \text{Birr } 237.50$$

- Discount = regular price - selling price = 250 - 210 = Birr 40
- After discount cost price = S.P - 5% S.P = $210 - \frac{5}{100}(210) = 199.5$
- Profit: P = (Selling price) - (cost price) = 210 - 199.5 = Birr 10.50

Answer: D

45. The selling price of an item in a shop is Birr 150 with mark up of

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Birr 40. How much profit does the shop owner earn if he sells this item with a discount of 20% of its selling price?

A. Birr 10 B. Birr 20 C. Birr 30 D. Birr 40.

Solution

C.P = S.P - mark = 150 - 40 = 100

New selling price = 150 - 20% (1500 = 120

Profit = (New selling price) - (cost price) = 120 - 100 = Birr 10

Answer; A

46. Calculate the regular price of bed assuming that 20% discount would have been worth Birr 2400.

Solution: Discount rate $= \frac{\text{Discount}}{\text{regluar price}}$

Discount = 20% of regular pricent.

2400 = (0.2) regular price.

$\therefore$ regular pricen $= \frac{2400}{0.2} = 12{,}000.$

$\therefore$ Selling price = R.P - discount = 12000 - 2400 = 9600

47. A shop sales a pack of 20 articles for Birr 550. He planned to clear the articles during holiday. If the shop sells each for Birr 220, With percent discount has it made?

A. 18% B. 15% C. 22% D. 20%

Solution

Price of each articles: $\frac{5500}{220} = 275$

Discount: 270 - 220 = Birr 55

percent discount $= \frac{55}{275} \times 100\% = 20\%$

Answer: D

48. coffee table with regular price of Birr 750 is first discounted by 20% and then by 15% .What is the final selling price of the a coffee table after the discount?

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A. Birr 510 B. Birr 262.5 C. Birr 690 D. Birr 488

Solution

Regular price = 750

First discount = 20% (750) = Birr 150

Net price after 1st discount = 750 - 150 = 600.

Now, second discount = 15% (600) = 90.

∴ The selling price = 600 - 90 = Birr 510.

Answer: A

13.6 Time Value of Money

Every body uses money. Sometimes you work for your money and other time your money works for you.

For example, unless you are attending University on a full scholarship, it is very likely that you and your family have either saved money or borrowed money, or both to pay for your education. When we borrow money, we have to pay interest for that privilege. When we save money, for future purchase or retirement, we are lending money to a bank and we expect to earn interest on your investment. The interest to earn is either **simple or compound.**

13.7 What is the a Simple Interest ?

Interest is money paid for the use of money.

Principal (P) is the money intialy invested

The money that borrows (takes) from the money leader.

If the principal remains the same for the entire loan period, the interest is called **simple interest**

Simple Interest is paid only on the intial principle (P).

Simple Interest is calculated by Using the formula:

$$I = Prt$$

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The accumulated amount (A) of an investment (debt) is the principal plus interest.

$$A = P + I = P + Prt$$

This amount is called the future value of simple interest.

Future Value for Simple Interest

The future value of A of the principal (P) at simple interest rate (r) for t years is :

$$A = P(1 + rt)$$

Illustrative Example

49. Abebe has borrowed Birr 2500 from Dawit. Abebe agrees to pay back Birr 2650 in 4 months at simple interest rate

a. How much interest did he pay back?

b. What is the annual interest rate?

Solution: I PRt

a. Interest: $I = 2650 - 2500 = \text{Birr } 150$

b. Interest rate: $R = \frac{I}{PT} = \frac{150}{(2500)\left(\frac{4}{12}\right)} \times 100\% = 18\%$

50. Daniya borrowed Birr 200000 from a bank inorder to buy a car. She had to repay Birr 10,000 per moth for 36 months in simple interest.

Find the interest rate charged.

Solution

Total amount paid is $36 \times 10{,}000 = 360{,}000$

$\therefore$ the interest: $I = 360{,}000 - 200{,}000 = \text{Birr } 160000$

Using $I = PRT$, then $R = \frac{I}{PT} \times 100\%$

$$R = \frac{160{,}000}{(200000)\left(\frac{36}{12}\right)} \times 100\% = 26.7\%$$

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51. Birr 180000 is invested at rate 12% per year in simple interest. How long will it take for the investment to reach Birr 315000 ?

Solution: r = 12% = 0.12

$$\text{Interest} = 315000 - 180000 = 135{,}000$$

$$\text{Since } I = Prt \text{ and } t = \frac{I}{Pr} = t = \frac{135000}{(180000)(0.12)} = 6.25 \text{ years}$$

52. How long will take for which Birr 5400 will earn Birr 2160 at 8% per year in simple interest?

Solution

$$I = 2160, \text{ rate: } r = 8\% \text{ and } P = 5400$$

$$\therefore \quad t = \frac{I}{Pr} = \frac{2160}{(5400)(0.08)} = \frac{2160}{432} = 5 \text{ years}$$

53. Find the future value of loan of Birr 2500 to be repaid in 8 months with interest of 6% in simple interest.

Solution : $r = 6\% = 0.06$, $t = \frac{8}{12} = \frac{2}{3}$ year. The future value is:

$$A = (P)(1 + rt) = 2500\left[1 + 0.06\left(\frac{2}{3}\right)\right] = \text{Birr } 2600$$

54. Find the future value of simple interest a loan of Birr 13,000 for 72 days at 12% interest.

Solution : $P = 13{,}000$, $r = 12\% = 0.12$, $t = \frac{72}{360}$ .The future value is:

$$A = P(1 + rt) = 13000\left[1 + 0.12\left(\frac{72}{360}\right)\right] = 13{,}312$$

55. To buy furniture for a new apartment, Daneil. Tekst borrowed Birr 45,000 at 8% simple interest for 10 months.

How much interest will he pay?

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Solution : $P = 45{,}000 \quad r = 8\% \quad t = \frac{10}{12}$ years

We use: $I = Prt = (45000)\,(0.08)\left(\frac{10}{12}\right) = 3000.$

56. Find the balance if Birr 10000 is invested at annual simple interest rate of 13% for 4 years

Solution

$I = Prt = 10000\,(0.13)\,(4) = 52{,}000$

$\therefore \quad A = P + I = 10000 + 52{,}000 = 152000.$

57. If you want to earn an annual rate of 10% on your investment, how much should you pay for a note that will be worth Birr 5000 in 9 months.

A. Birr 6,163.10 C. Bir 4651.16

B. Birr 5,261.21 D. Bir 3,983.25

Solution : A 5000, $r = 10\% = 0.1$, $t = \frac{9}{12} = 0.75$

$A = P\,(1 + rt)$ substitute the value:

$5000 = P\,[1 + (0.1)\,(0.75)] = P\,(1.075)$

$\therefore \quad P = \frac{5000}{1.075} = \text{Birr } 4651.16$

Answer: C

58. If Birr 2000 with simple interest in four years amounts to Birr 2400, in what time will Birr 2500 at the same interest rate amount to Birr 3000

A. 4 years B. 3 years C. 5 years D. 2.5 years

Solution

Interest: $I_1 = 2400 - 2000 = \text{Birr } 400$

$I_1 = P_1\ RT_1 \Rightarrow R = \frac{I_1}{P_1\ T_1} = \frac{400}{(2000)(4)} \times 100\% = 5\%$

Interest $I_2 = 3000 - 2500 = \text{Birr } 500$

$I_2 = P_2\ R\ T_2$ implies $R = \frac{I_2}{P_2\ T_2}$. But the rate is the same:

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$0.05 = \frac{500}{(2500)\,T_2}$ this implies $T_2 = \frac{(500)}{(2500)(0.05)} = 4$ years

Answer: A

59. A sum of money doubles itself in 10 years at simple interest. In how many years would it triple itself?

A. 10 years B. 15 years C. 20 years D. 25 years

Solution: t = 10

Since A = P(1 + rt), and A = 2P

2P = P(1 + 10r) implies that 10r = 2 - 1 = 1 then : $r = \frac{1}{10}$

if A = 3P, then $3P = P\left(1 + \frac{1}{10}t\right)$

$1 + \frac{t}{10}$ in this case. t = 10(2) = 20

Answer: C

60. At what rate in simple interest will a sum of money double itself in 8 years?

A. 12.5% B. 10% C. 12% D. 8%

Solution

A = P (1 + rt) and A = 2P, t = 8 years

2P = P (1 + 8r), then 1 + 8r = 2

$\therefore \quad r = \frac{1}{8} \times 100\% = 12.5\%$

Answer: A

13.8 Compound Interest

In case of "simple interest" the principal (P) remains constant through the term of payment (period) for which the money (principal) is borrowed.

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What is Compound interest?

When the principal does not remain the same for the entire period due to addition of interest to the principal after a certain period (one year, half - year, quarter year e.t.c): to from the principal for the next period, the interest is called **compound interest.**

Illustrative Example

61. Suppose Birr 4000 is deposited for 4 years in saving account earning 10% per year compounded annually

a. Find the amount of compound interest earned

b. Find the balance on the account at the end of 4 years.

Solution

$$\text{Year 1: } I_1 = P_1\ r\ t = (4000)\left(\frac{10}{100}\right)(1) = 400$$

$$\text{Amount: } A_1 = P_1 + I_1 = 4000 + 400 = 4400$$

$$\text{Year 2: } I_2 = P_2\ r\ t = (4400)\left(\frac{10}{100}\right)(1) = 440$$

$$\text{Amount : } A_2 = P_2\ r + t = 4400 + 440 = 4840$$

$$\text{Year 3: } I_3 = P_3\ r\ t = 4840\left(\frac{10}{100}\right)(1) = 484$$

$$\text{Amount: } A_3 = P_3 + I_3 = 4840 + 484 = 5{,}324$$

$$\text{year 4: } P_4\ r\ t = (5324)\left(\frac{10}{100}\right)(1) = 532.40$$

$$\text{Amount: } A_4 = P_4 + I_4 = 5324 + 532.4 = 5856.40$$

a. At the end of fourth year, the compound interest is :

C. I = A - P = 5856.40 - 4000 = Birr 1856.40

b. The balance on the account at the end of 4th year is A = Birr 5856.40. This method of finding is of course quite lengthy. It can be done using a the compound interest formula.

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Amount in compound Interest

Suppose the intial deposit (P) is invested at annual interest rate (r), duration of deposit (n): Then the value of the account at time n is : $A_n = P(1 + r)^n$

Compound Interest: $C.I = A - P = P(1 + r)^n - P = P[(1 + r)^n - 1]$

From the above example, we have $P = 4000$, $r = 10\% = 0.1$ and $n = 4$

$A = P(1 + r)^n = 4000(1 + 0.1)^4 = 4000(1.4641) = 5856.40$

Illustrative Example

62. Suppose Birr 6000 is deposited for 5 years in saving account paying 8% per year compounded annually.

    a. What will be the balance on the account at the end of that time?

    b. What is the compound interest of the period ?

    Solution: $A = 6000$, $n = 5$ years, $r = 8\%$

    a. The balance (the compound amount) is

    $A = P(1 + r)^n = 6000(1 + 0.08)^5 =$ Birr 8815.96

    b. To find the compound interest is subtract the initial deposit from the compound amount.

    $\therefore$ $C.I = A - P = 8815.96 - 6000 =$ Birr 2815

63. Find the principal that will amount to Birr 5832 in 2 years at 8% per year.

    Solution : $A = 5832$, $r = 8\%$ and $n = 2$

    $\therefore$ $A = P(1 + r)^n$ implies $5832 = P\left(1 + \frac{8}{100}\right)^2$

    $P(1.08)^2 = 5832$

    $(1.1664)P = 5832$ implies: $P = \frac{5832}{1.1664} = 5000$

    $\therefore$ The required principal: $P =$ Birr 5000

64. Find the principal on which the compound interest in 3 years amounts to Birr 1324 at 10% per year.

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Solution

We have given compound interest: C.I = Birr 1324, r = 10% and n = 3.

$$C.I = P\left[(1 + r)^n - 1\right]$$

$$1324 = P\left[\left(1 + \frac{10}{100}\right)^3 - 1\right] = P\left(\frac{1331}{1000} - 1\right)$$

$$P\left(\frac{1331 - 1000}{1000}\right) = 1324, \text{ and } P = \frac{1324 \times 1000}{331}$$

$\therefore$ The principal required: P = Birr 4000.

13.8.1 Compounding Frequencies

Interest can be compounded more than once per year. Common compounding periods include:

- Annually means once a year m = 1
- Semi - annually means twice per year : m = 2
- Quarterly means four times per year: m = 4
- Monthly means 12 times per year: m = 12
- Daily means 365 times per year: m = 365.

If r is an annual interest rate and m is the number of compounding period per year, then:

- The interest rate each period is: $i = \frac{r}{m}$
- The number of compounding period after t year is **mt**.

Thus, the future value of compound interest formula is express as follows:

Future Value of compound Interest

The principal (P) invested at annual interest rate (**r**) and compounded m times per year for t years, produces the balance: $A = P\left(1 + \frac{r}{m}\right)^{mt}$

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Important:

The future value A when the interest rate is compounded:

i. Annually: $m = 1 : A = P\left(1 + \frac{r}{1}\right)^{t}$

ii. Semi - annually: $m = 2: A = P\left(1 + \frac{r}{2}\right)^{2t}$

iii. Quarterly: $m = 4: A = P\left(1 + \frac{r}{4}\right)^{4t}$

iv. monthly: $m = 12: A = P\left(1 + \frac{r}{12}\right)^{12t}$

Illustrative Example

65. If Birr 3000 is deposited at an annual interest rate of 10% for 4 years, Find the balance and the amount of interest earned if the interest is compounded:

a. annually
b. Semi - annually
c. quarterly
d. Monthly

Solution

$P = 3000$, rate: $r = 10\% = 0.10$ and $t = 4$ years.

a. Compounded annually, $m = 1$, then the balance at the end of 4th year:

$$\therefore \quad A = P\left(1 + \frac{r}{m}\right)^{mt} = 3000\left(1 + \frac{0.10}{1}\right)^{4} = 3000(1.0)^{4} = \text{Birr } 4392.3$$

$$\therefore \quad \text{Interest: C.I} = A - P = 4392.30 - 3000 = \text{Birr } 392.3$$

b. Compounded semi:- annually means $m = 2$.

$\therefore$ The balance at the end of 4th year is

$$A = P\left(1 + \frac{r}{m}\right)^{mt} = 3000\left(1 + \frac{0.10}{1}\right)^{2(4)} = 3000(1.05)^{8} = \text{Birr } 4432.37$$

$$\therefore \quad \text{Interest C. I} = A - P = 4432.37 - 3000 = \text{Birr } 432.37$$

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c. Since the interest is compounded quarterly, means m = 4, then the balance at the end of 4th year:

$$A = P\left(1 + \frac{r}{m}\right)^{mt} = 3000\left(1 + \frac{0.10}{4}\right)^{4(4)} = 3000(1.025)^{16} = \text{Birr } 4453.51$$

$\therefore$ Interest: C.I = A - P = 4453.51 - 3000 = Birr 453.51

d. Compounded monthly, m = 12, then the balance at the end of 4th year:

$$A = P\left(1 + \frac{r}{m}\right)^{mt} = 3000\left(1 + \frac{0.10}{12}\right)^{48} = 3000(1.4893) = \text{Birr } 4468.06$$

$\therefore$ Interest C.I = A - P = 4468.06 - 3000 = Birr 468.06

66. If Birr 2000 is deposited in bank with interest rate 10% per annually, compounded semi annually, then how much birr will it be by the end of one year?

A. 2100 B. 2200 C. 2205 D. 2210

Solution

$$\text{Here, } A = P\left(1 + \frac{r}{m}\right)^{mt} = 2000\ (2000)\left(1 + \frac{0.10}{2}\right)^{2}$$

$$\therefore \quad A = 2000\left(1 + \frac{1}{20}\right)^{2} = 2000\left(\frac{21}{20}\right)^{2} = \frac{2000 \times 21 \times 21}{400} = 2205$$

Answer: C

67. A sum of money is invested at compound interest calculated annually which amounts to Birr 21,632 at the end of the second year and to Birr 22,497.28 at the end of the third year. Which one of the following pairs of value gives the rate of interest and the principal respectively?

A. 0.05 and Birr 21,000 C. 0.03 and Birr 18,550

B. 0.06 and Birr 21550 D. 0.04 and Birr 20,000

Solution: $A_2 = 21,632$ and $A_3 = 22,497.28$

$A_2 = P(1 + r)^2$ and $A_3 = P(1 + r)^3 = P(1 + r)^2 (1 + r)$.

Then $A_3 = A_2(1 + r)$

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$$1 + r = \frac{A_3}{A_2} = \frac{22497.28}{21,632} = 1.04$$

$$\therefore \quad r = 1.04 - 1 = 0.04$$

To find the principal: $P = \frac{A_2}{(1+r)^2}$

$$\therefore \quad P = \frac{21632}{(1+0.04)^2} = \frac{21632}{1.0816} = \text{Birr } 20,000$$

Answer: D

68. An investment of Birr 3000 was made over 3 years at an interest rate of 5% with interest compounding annually. What are the principal at the start of the second year and the interest earned during the second year (in birr), respectively. **(UEE)**

A. 3150 and 152.5
C. 3150 and 157.5
B. 3100.9 and 157.5
D. 3307.5 and 157.5

Solution

$A_1 = P(1 + r)^1 = 3000\ (1 + 0.05) = 3150$

$A_2 = P(1 + r)^2 = 3000\ (1 + 0.05)^2 = 3,307.5$

Interest = 3307.5 - 3150 = 157.5

Principal in 2nd year : 3,307.5 - 157.5 = 3150

Answer: C

69. An investor has an opportunity to purchase two different bonds. Bond M pays 15% compounded monthly and Bond - S pays 15.2% compounded semi - annually. Assuming all others are equal, which is the better investment?

Use: $(1.0125)^{12} = 1.160755$, $(1.076)^{12} = 2.408503$

$(1.0425)^2 = 1.025156$, and $(1.076)^2 = 1.157776$.

A. Bond - S is by far better than Bond M

B. Bond - S is lightly better than Bond M

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C. Bond- S is the preferred investment than Bond S

D. Both of them have the same value at the end of the year.

Solution

Amount of bond : $M = P\left(1 + \frac{0.15}{12}\right)^{12(1)} = P(1.0125)^{12} = 1.16077P$

Amount of bond : $S = P\left(1 + \frac{0.152}{2}\right)^{2} = P(1.076)^{2} = 1.157776P$

Since the amount of bond M > S

Answer: C

Finding the time (t)

Illustrative Example

70. In how many years will sum of Birr 4000 in a 10% saving account compounded annually become Birr 5324?

Solution

Compounded annually means compounded once per year,

$m = 1,\ i = \frac{r}{m} = 0.1$ and $mt = (1)t = t$

$A = P(1 + i)^{mt}$ this implies $5324 = 4000\ (1 + 0.1)^{t}$

$\therefore \quad (1.1)^{t} = \frac{5324}{4000} = \frac{1331}{1000}$ and $\left(\frac{11}{10}\right)^{t} = \frac{1331}{1000}$

$\therefore \quad t = 3$ years

71. In what time will the compound interest on Birr 1000 at the rate of 10% per year will be 210.

Solution

$A = P(1 + r)^{n}$, where $A = P + C.I = 1000 + 210 = 1210,\ r = 10\%$

$1210 = 1000\ (1 + 0.1)^{n} = 1000\ (1.1)^{n}$

$\therefore \quad (1.1)^{n} = \frac{1210}{1000} = 1.21$

$\therefore \quad n = 2$ years

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72. In how many years a sum of Birr 10000, compounded quarterly at the rate of 12% per annually will grows to Birr 10609

Solution

Compounded quarterly means $m = 4$

$i = \frac{12\%}{4} = 3\% = 0.03$, and $mt = 4t$ we need to find t.

Here, $A = P(1 + i)^{mt}$

$10609 = 10000(1 + 0.03)^{4t} = 10000(1.03)^{4t}$

$\left(\frac{103}{100}\right)^{4t} = \frac{10609}{10000}$, this implies $4t = 2$

$\therefore \quad t = \frac{2}{4} = \frac{1}{2}$ year

73. Find the time takes for money invested at an annual rate of 8% to double if the money is compounded quarterly.

Solution : $r = 8\% = 0.08$, $m = 4$ and $A = 2P$

$A = P\left(1 + \frac{r}{m}\right)^{mt}$ this implies $2P = P\left(1 + \frac{0.08}{4}\right)^{4t}$

$(1.02)^{4t} = 2$ implies $4t = log^{2}_{1.02} = \frac{log}{log\ 1.02} = \frac{0.3010}{0.0086} = 35$

$\therefore \quad t = \frac{35}{4} = 8.75$ years

74. If a sum of money at compound interest will grows to three times itself in 4 yeas, then in how many years will be 9 times itself?

A. 12 years B. 9 years C. 8 years D. 6 years

Solution

$A = P(1 + r)^t$, since $A = 3P$

$3 = (1 + r)^4$, this implies $1 + r = 3^{1/4}$.

Then $9 = (1 + r)^t$ implies $1 + r = 9^{1/t}$

$9^{1/t} = 3^{1/4}$ implies $3^{2/t} = 3^{1/4}$

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$\therefore \quad \frac{2}{t} = \frac{1}{4}$, it gives $t = 8$

Answer: C

75. In what time will Birr 800 amount to Birr 882 at 5% per year compounded annually ?

A. 3 years B. 2 years C. 4 years D. 2.5 years

Solution: $A = 882, P = 800, r = 5\%, n = ?$

$A = P(1 + r)^n$ and $882 = 800(1.05)^n$

$\frac{882}{800} = (1.05)^n$ and $(1.05)^n = 1.1025$

$\therefore \quad n = 2$ years,

Answer: B

76. If some amount of money is deposited at the rate of 8% compounded quarterly, how long does it take to 4 times the initial deposit in years. (use $\log 2 = 0.3010$, and $\log (1.02) = 0.0086$)

A. 70 years B. 35 years C. 6 years D. 17.5 years

Solution : $m = 4, i = \frac{r}{m} = \frac{8\%}{4} = 0.02, \quad mt = 4t$ P and $A = 4P$

Here, $A = P(1 + i)^{mt}$ this implies $4P = P(1 + 0.02)^{4t}$

$(1.02)^{4t} = 4$ is equivalent to $\log_{1.02}^{4} = 4t$

$\therefore \quad 4t = \frac{\log^4}{\log(1.02)} = \frac{2\log^2}{\log(1.02)} = \frac{2(0.3010)}{0.0086} = 70$

$t = \frac{70}{4} = 17.5$ years

Answer: D

How to Find the Rate of Compound Interest ?

77. At what rate compounded yearly will Birr 6400 amount grows to Birr 10000 in 2 year

Solution

$$A = P\left(1 + \frac{r}{100}\right)^2 \text{ this implies } 10000 = 6400\left(1 + \frac{r}{100}\right)^2$$

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$$\left(1+\frac{r}{100}\right)^2=\frac{10000}{6400}=\frac{100}{64}$$

$$1+\frac{r}{100}=\frac{10}{8}=\frac{5}{4}$$

$$\therefore \quad r=\frac{5}{4}-1=\frac{1}{4}\times 100\%=25\%$$

13.8.2 Effective Interest Rate

In financial institution, the **effective annual rate of interest** can be used to compare future amounts of a given principal under different interest rate and compounding periods.

By definition, the effective annual rate of interest for a given transaction is the simple interest rate that will yield the same amount in 1 years,

- The **effective interest rate** (r_E) or the equivalent annual rate (EAR) is called the annual yield.
- The annual rate (r) of compound interest is called the **nominal interest rate**, i.e (Nominal rate is the original interest rate)

$$\text{Thus, } A=P\left(1+\frac{r}{m}\right)^m=P\left(1+r_E\right) \text{ in this case}$$

$$P\left(1+\frac{r}{m}\right)^m=P\left(1+r_E\right)$$

$$\therefore r_E=\left(1+\frac{r}{m}\right)^m-1$$

The **effective annual interest rate (r_E)** corresponding to the nominal rate **(r)** compounding **m** times per year is given by:

$$r_E=\left(1+\frac{r}{m}\right)^m-1$$

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Illustrative Example

78. Find the effective interest rate corresponding to a nominal rate of 8% compounded quarterly.

Here, $r_E = \left(1 + \frac{r}{m}\right)^m - 1$

$\therefore \quad r_E = \left(1 + \frac{0.08}{4}\right)^4 - 1 = (1.0824) - 1 = 0.0824$

$\therefore \quad r_E = 0.0824 \times 100\% = 8.24\%$

79. Find the effective interest rate equivalent to 10.5% compounded semi - annually :

Solution:

Here, $r_E = \left(1 + \frac{r}{m}\right)^m - 1 = \left(1 + \frac{0.105}{2}\right)^2 - 1 = 1.1077 - 1 = 0.1077$

$\therefore \quad r_E = 0.1077 \times 100\% = 10.77\%$

80. Oluma needs to borrow money from a bank in the following bank charge

Bank	Rate	Compounding period
Bank : X	8.6%	Semi - annually
Bank : Y	8.5%	monthly

which bank will Oluma the lower amount of the interest?

Solution: the effective rates.

Bank X : $r_E = \left(1 + \frac{0.086}{2}\right)^2 - 1 = 0.0878 = 8.78\%$

Bank Y : $r_E = \left(1 + \frac{0.085}{12}\right)^{12} - 1 = 0.0883 = 8.83\%$

Bank X has the lower effective rate, although it has a higher nominal rate.

81. A saving and a loan wants to offer a certificate of deposit with a monthly compounding rate that has an effective rate of 4%. What annual nominal rate compounded monthly should they use?

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Use: $(1.04)^{1/12} = 1.00373$ and $(1.04)^{12} = 1.601032$

A. 3.9 % B. 4.5 % C. 5.2 % D. 7.2 %

Solution

$$r_E = \left(1 + \frac{r}{m}\right)^m - 1 \text{ this implies } \left(1 + \frac{r}{m}\right)^m = r_E + 1$$

$$\left(1 + \frac{r}{12}\right)^{12} = r_E + 1 = 0.04 + 1 = 1.04$$

$$1 + \frac{r}{12} = (1.04)^{1/12} = 1.00373$$

$$\therefore \quad \frac{r}{12} = 1.00373 - 1 = 0.00373$$

$$\therefore \quad r = 0.04476 \times 100\% = 4.5\%$$

Answer: B

82. How much interest would a bank pay on Birr 1500 deposit for 240 days if the effective interest rate at 12%

Solution

Here, $A = P(1 + r_E) = 1500(1 + 0.12) = \text{Birr } 1680$

Interest: $I = 1680 - 1500 = 180$ Birr per year.

$$\text{Interest per day} = \frac{180 \text{ Birr}}{365 \text{ days}} = 0.493$$

$\therefore$ Interest for 240 days is $I = 0.493 \times 240 = 1180.35$

83. Find the future value of Birr 240,000 at a 18% effective rat for 8 mouths.

Solution

$A = P(1 + r_E) = 240{,}000(1 + 0.18) = 28{,}3200$ Birr at the end of one year

$\therefore$ The value in 8 mouths is $A = 283200 \times \frac{8}{12}$

13.9 Annuity

You studied saving plans that involved a single deposit. Most people, however, prefer to make several deposit over a long period of times.

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- If you deposit Birr 2000 at the end of each month for the paying 8% interest compounded monthly how much will be in your account at the end of this period

What is an Annuity

Annuity: is a sequence of equal payments (deposit) made at regular interval. Two common uses of annuities are to accumulate funds for some goal or to withdraw funds from an account.

Fore example , an annuity may be used to save money for large purchase, such as : an automobile, houses, university eduction, or a down payment on condominium ... etc.

An annuity also may be used to provide monthly payments for retirement, usually the so called "ጡረታ"

Timing of the payment

The time interval for each payment made in an annuity is called the **payment period.** There are two types of annuities based on timing payments.

(i) An ordinary Annuities. (payment made at the end)

(ii) An annuities Due: is annuities in which payments are made at the beginning of each time period.

What is an ordinary Annuities?

If the payments are made at the end of each payment period, the annuity is called an **ordinary annuity.** In this case, the compounding period is the same as the payment period. The time from the beginning of the first payment period to the end of the last period is called the **term of annuity.**

13.9.1 Future Value of an Ordinary Annuities

The future value of the annuities, the final sum on deposit, is defined as the sum of the compounded amounts of all the payments, compounded to the end of the terms. Calculating future value of an Annuities suppose R birr is deposited at the end of each year for the next years in saving account paying i% per year compounded annualy.

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To find the future value of the annuity

Year	Deposit	The future value
1	R	R .The value at the end of 1st years
2	R	$R(1+i)^1$ .The value at the end of 2nd year
3	R	$R(1+i)^2$ .The value at the end of 3rd year
4	R	$R(1+i)^3$ .The value at the end of 4th year
5	R	$R(1+i)^4$ .The value at the end of 5th year

Since the money is deposited at the end of the first year and earns interest for only 4 years. The sum of the values of all the deposit is called the **future value of annuity:**

$$S = R + R(1+i) + R(1+i)^2 + R(1+i)^3 + R(1+i)^4$$

This is the sum of the first 5 terms of payment made a geometric sequence, with

$G_1 = R$, $r = 1 + i$, and $n = 5$

Thus the sum : $S = G_1\left[\frac{r^n - 1}{r - 1}\right] = R\left[\frac{(1+i)^5 - 1}{i}\right]$

To generalize, for regular payments of R is deposited into an account at the end of each period for n- period at rate of interest i per period. If S denote the future value (or sum) of the annuity, then

$$S = R + R(1+i) + R(1+i)^2 .... + R(1+i)^{n-1}$$

This is the sum of the first n - terms of the geometric sequence with first term $G_1 = R$ and common ratio $r = 1 + i$ is

$$S = G_1\left[\frac{r^n - 1}{r - 1}\right] = R\left[\frac{(1+i)^n - 1}{i}\right]$$

The expression in the bracket given as : $\frac{(1+i)^n - 1}{i}$

is called the **annuity factor** denoted by $s\overline{n}|i$ (read "S - angle - n at i).

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Future Value of Ordinary Annuity

Suppose a payment Birr R is made at the end of each payment period and the interest rate per period is i. Then the future value of the annuity after n - periods is

$$S = R\left[\frac{(1+i)^n - 1}{i}\right] = R \cdot S\overline{n|}i$$

Where:

S is the future value, . i is the interest rate per period

R is the periodic payment . n is the number of period.

Illustrative Example

84. Suppose Birr 2000 is deposited at the end of each year for the next 8 years in an account paying 6% per year compounded annually.

a. Find the future value of annuity

b. Find the annuity factor

c. Find the total interest

Solution

Here, R = 2000, n = 8, r = 6% = 0.06

a. $S = R\left[\frac{(1+i)^n - 1}{i}\right] = 2000\left[\frac{(1+0.06)^8 - 1}{0.06}\right] = 2000\,[9.897]$

$= \text{Birr } 19794.9$

The future value after 8 years is **Birr 19794.90**

b. The annuity factor : $S\overline{n|}i = \frac{(1+i)^n - 1}{i}$

$$= \frac{(1+0.06)^8}{0.06} = 9.897$$

c. Interest : $S - R \times n = 19794.90 - 2000 \times 8 = \text{Birr } 3794$

85. If Birr 2500 is deposited at the end of each month into a saving account earning 8% annual interest compounded monthly for the next 10 years

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a. Find the future value of annuity in 10 years.

b. Find the annuity factor

c. Find the total interest

Solution: R = 2500, n = 12 × 10 = 120

$$\text{a. } S = R\left[\frac{(1+i)^n - 1}{i}\right] = 2500\left[\frac{(1+0.0067)^{120} - 1}{0.0067}\right]$$

$$= 2500(182.03) = \text{Birr } 4550089.64$$

b. The annuity factor: $S\overline{n}|i = \frac{(1+i)^n - 1}{i}$

$$= \frac{(1+0.0067)^{120} - 1}{0.0067} = 182.03$$

c. Interest : S - R × n = 455089 - 2500 × 120 = **Birr 155089**

86. Retirement Planning: A 45 - year old man puts Birr 2500 in a retirment account at the end of each quarter until he reaches the age of 70, then makes no further deposits. If the account pays 6% interest compounded quarterly, how much will be in the account when the man retires at age 70.

Solution

$$R = 2500,\ i\ \frac{r}{m} = \frac{0.06}{4} = 0.015,\ n = (70 - 45) \times 4 = 100$$

$$\therefore \quad S = R\left[\frac{(1+i)^n - 1}{i}\right] = 2500\left[\frac{(1+0.015)^{100} - 1}{0.015}\right]$$

$$= 2500(228.80) = \textbf{Birr 572007.}$$

87. Find the annuity factor of the following cases.

a. n = 10 and i = 0.015 b. n = 120 and i = 0.004

Solution

$$\text{Annuity factor : } S\overline{n}|i = \frac{(1+i)^n - 1}{i}$$

$$\text{a. } S\overline{n}|i = S\overline{10}|0.015 = \frac{(1\ 0.015)^{10} - 1}{0.015} = \frac{0.160}{0.015} = 10.70$$

$$\text{b. } S\overline{n}|i = \frac{(1+0.004)^{120} - 1}{0.004} = 153.63$$

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88. **Comparing Two saving plans:** Which would give you a higher balance in 10 years:

A. Monthly deposit of Birr 500 at 10% interest compounded monthly

B. Monthly deposit of Birr 600 at 6% interest compounded monthly.

Solution

$$\text{A. } S = 500\left[\frac{(1+0.0083)^{120}-1}{0.0083}\right] = 205.67(500) = 102833$$

$$\text{B. } S = 600\left[\frac{(1+0.005)^{120}-1}{0.005}\right] = 600\,(163.90) = 98327$$

$\therefore$ Monthly deposit of Birr 500 at 10% interest compounded monthly better.

89. How much must be deposited each month into an ordinary annuity paying 7.5% compounded monthly, to attain a balance of Birr 2500 after five years?

Solution

$$n = 5 \times 12 = 60,\ S = 25000 \text{ and } i = \frac{0.075}{12} = 0.00625\ R = ?$$

$$S = R\left[\frac{(1+i)^n - 1}{i}\right] \text{ implies } 25000 = R.\left[\frac{(1+0.00625)^{60}-1}{0.00625}\right]$$

$$2500 = \left[\frac{0.453}{0.00625}\right], \text{ then } R = \frac{25000}{72.53} = \text{Birr } 344.70$$

13.9.2 Annuity Due

What is Annuity Due?

Annuities due is annuities in which payment are made at the beginning of each time period.

To find the future value of annuity due, treat each payment as if it were made at the end of the preceding period. That is find $S_{n\rceil i}$ for **one additional period** to compensate for this, subtract the amount of one payment

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Future value of Annuity Due

The future value of **annuity due** of n - payments of R Birr each at the beginning of consecutive interests periods, with interest compounded at the rate if i per period is :

$$S = R\left[\frac{(1+i)^{n+1} - 1}{i}\right] - R = R\,S\overline{n|}\,i - R$$

Illustrative Example

90. A father deposits Birr 800 into saving account for his daughter on the day she was born at the beginning of each month with annual interest rate of 7.2% compounded monthly for the next 10 years.

a. Calculate the future value of annuity due.

b. Calculate the total interest will she have after 10 years.

Solution

Here, $i = \frac{0.072}{12} = 0.006$, $n = 10(12) = 120$

a. $S = R\left[\frac{(1+i)^n - 1}{i}\right] = 800\left[\frac{(1+0.006)^{120} - 1}{0.006}\right]$

$= 800\,(175) =$ **Birr 140002.40**

b. Interest: $I = S - R \times$ (12 month) (10 year)

$= 14002 - 800 \times 12 \times 10 =$ Birr 44002

13.9.3 Present Value of an Annuity

The **present value of an annuity (P)** is the amount that would have to be deposited to day to produce the same amount as the annuity at the end of a specified time.

That is $P(1+i)^n = R\left[\frac{(1+i)^n - 1}{i}\right]$

To solve this equation for P, multiply both side by $(1+i)^{-n}$

$$P = R\,(1+i)^{-n}\left[\frac{(1+i)^n - 1}{i}\right] = R\left[\frac{(1+i)^n - 1}{i}\right]$$

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Present Value (P) of an Annuity of n - payments of R Birr each at the end of consecutive interest period with interest compounded at the rate of interest i per period is :

$$P = R\left[\frac{\left(1-(1+i)^{n}\right)}{i}\right]$$

The present annuity factor is $\overline{a_{n}|}i = \frac{\left(1-(1+i)^{n}\right)}{i}$

Illustrative Example

91. Find the present value of each payment of Birr 50000 quarterly for 10 years at 8% compounded quarterly.

Solution: Here: $R = 50000$, $n = 4(10) = 40$

and $i = \frac{0.08}{4} = 0.02$

$$P = R\left[\frac{1-(1+i)^{-n}}{i}\right] = 50000\left[\frac{1-(1+0.02)^{-40}}{0.02}\right]$$

$= 50000(27.36) =$ Birr 1367773.96

92. Find the present value annuity factor for five year annuity with monthly payments at 7.2% compounded monthly

Solution : $n = 12 \times 5 = 60$, $i = \frac{0.072}{12} = 0.006$

∴ The peresent annuity factor:

$$\overline{a_{n}|}i = \frac{1-(1+i)^{-n}}{i} = \frac{1-(1+0.006)^{-60}}{0.006} = \textbf{Birr 50.2}$$

93. Mr. Kassahun wants to buy a car. He borrowed Birr 600000 is to be paid off by monthely payments over 15 years at 8.4% annul interest How much is his monthly payment?

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Solution

Here, $P = 600000$, $n = 12(15) = 1800$, $i = \frac{0.084}{12} = 0.007$

The present value of annuity factor:

$$\frac{1 - (1 + i)^{-n}}{i} = \frac{1 - (1 + 0.007)}{0.007} = 102.16$$

$$P = R(102.16) \text{ this implies } R = \frac{P}{102.16}$$

$$\therefore \quad R = \frac{600000}{102.16} = \text{Birr } 5873.14$$

Hence, 180 monthly payments Birr 5873.14 will pay for the car.

13.9.4 Amortization

A present debt can be amortized (killed off) by a series of regular payment

- A loan is **amortized** if both the principal and interest are paid by a sequence of equal periodic payments.
- The periodic payments (R) needed to a loan may be found by solving present value (P) for R.

Amortization Payment

A loan of Birr (P) at interest rate i per period may be amortized in n equal periodic payments of R made at the end of each period is given by

$$R = P\left[\frac{i}{1 - (1 + i)^{-n}}\right] \text{ or } R = \frac{P}{a_{\overline{n}|i}}$$

Illustrative Example

94. A loan of Birr 100000 is to be paid off by monthly payments over 8 years at 12% annual interest. What is the monthly payment?

Solution

$$\text{Here } P = 100000,\ n = 12(8) = 96,\ i = \frac{0.12}{12} = 0.01$$

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Then the monthly payment is

$$R = P\left[\frac{i}{1-(1+i)^{-n}}\right] = 100000\left[\frac{0.01}{1-(1+0.01)^{-96}}\right]$$

$$= 100000\ [0.016] = \textbf{Birr 1612.90}$$

95. Dawit borrows Birr 2500 for 1 year at 12% annual interest compounded monthly.

a. What is the monthly loan payment

b. In amortization schedules table how much of the fourth payment is interest

c. The principal at the end of 4th payment period

d. How much interest is paid in the first 4 months of the loan

e. The principal

Solution

Here, P = 2500, n = (10 (12) = 12, and $i = \frac{0.12}{12} = 0.01$

a. The monthly payment is $R = \left[\frac{i}{1-(1+i)^{-n}}\right]$

$$\therefore \quad R = 2500\left[\frac{0.01}{1\ (1+0.01)^{-12}}\right] = 2500(0.088.85) = \textbf{Birr 221.30}$$

To make, the amortization table: since the annual interest rate is 12% compounded monthly, so the interest rate per month is $\frac{12\%}{12} = 1\% = 0.01$

1. When 1st payment is made, 1st interest is 2500 (0.01) = 25, payment to the principal is 221.30 - 25 = 196.30

   Hence, the principal at the end of first payment period is

   2500 - 196.30 = 2303.70

2. When 2nd payment is made 1st month interest on (2303.70) (0.01) = 23.037

   - Payment to the principal is 221.30 - 23.037 = 198.26

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∴ The principal at the end of payment 2 is

2303.70 - 198.26 = Birr 2105.44

3. When 3rd payment is made, 2nd month interest is (2105.44) (0.01)

= 21.05

- Payment to the principal: 221.30 - 21.05 = 200.25

∴ The principal at the end of payment 3 period is

2105.44 - 200.25 = 1905.2

4. When 4th payment is made, the interest is (1905.2) (0.01) = 19.05.

- Payment to the principal: 221.30 - 19.05 = Birr 202.25

∴ The principal at the end of payment 4 period is

1905.20 - 20.25 = Birr 1702.95

Amortization of Birr 2500 with 12 % interest rate compounded monthly.

Month	Principal	Payment	Interest	Payment to principal
0	2500			
1	2303.70	221.30	25.00	196.30
2	2105.44	221.30	23.04	198.26
3	1905.20	221.30	21.05	200.25
4	1702.95	221.30	19.05	202.25

b. The 4th payment interest is 19.05

c. The principal (remaining balance) during 4th payment period is 1702.95. That is the present value of 8th payment annuity:

$$P = R\left[\frac{1-(1+i)^{-n}}{i}\right] = 221.30\left[\frac{1-(1+0.01)^{-8}}{0.01}\right] \approx 1702.95$$

d. Interest paid is 25 + 23.04 + 21.05 + 19.05 = 88.14

96. What present value can be amortized by monthly payment of Birr 500 for 4 years at an annual interest rate of 15%

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Solution

Here, $R = 500$, $i = \frac{0.15}{12} = 0.0125$, $n = 4(12) = 48$

$$P = R\left[\frac{1-(1+i)^{-n}}{i}\right] = 500\left[\frac{1-(1+0.0125)^{-48}}{0.0125}\right]$$

$$= 500\,(35.93) = 17{,}965.74$$

97. If a loan of Birr 21000 with 12% annual interest compounded monthly amortized over 10 years, how much is the monthly payment?

Use: $(1.02)^{-120} = 0.024$, $(1.01)^{-120} = 0.3$, $(1.01)^{120} = 3.3$

A. Birr 280 B. Birr 300 C. Birr 333 D. Birr 350

Solution : $n = (10)(12) = 120$, $i = \frac{0.12}{12} = 0.01$

Monthly payment is $R = P\left[\frac{i}{1-(1+i)^{-n}}\right]$

$$\therefore \quad R = 21000\left[\frac{0.01}{1-(1+0.01)^{-120}}\right] = 21000(0.0143) = 300$$

Answer: B

Depreciation

When the value of an article decrease with time (age) this reduction in its called **depreciation.** In order to calculated the depreciated (decreased) value of the article. We use! $A = P(1-r)^n$

Illustrative Example

98. The cost of a computer is Birr 35,000. Its value depreciates at the annual rate of 8%.

a. What will be the value of a computer over 4 years.

b. Find the total depreciation in its value in four year.

Solution

Rate of depreciation: $r = 8\% = 0.08$, $n = 4$

a. $A = 35000\,(1-0.08)^4 =$ Birr 25073.75

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b. Total depreciation: 35000 - 25073.75 = Birr 9926.25

99. Ebrahim purchases a TV for Birr 16000. If the cost of TV after 2 year depreciates to Birr 14440. Find the rate of depreciation

Solution

Present cost of TV: P = 16000, cost of TV after 2 year: A = 14440.

$$A = P\left(1 - \frac{r}{100}\right)^n$$

$$14440 = 16000\left(1 - \frac{r}{100}\right)^2 \text{ implies } \left(1 - \frac{r}{100}\right)^2 = \frac{14440}{16000}$$

$$\left(1 - \frac{r}{100}\right)^2 = \frac{361}{400}, \left(1 - \frac{r}{100}\right)^2 = \frac{361}{400} = \left(\frac{19}{20}\right)^2$$

$$1 - \frac{r}{100} = \frac{19}{20} \text{ this implies } \frac{r}{100} = 1 - \frac{19}{20} = \frac{1}{20}$$

$$\therefore \quad r = \frac{1}{20} \times 100\% = 5\%$$

Straight Line Depreciation

Straight line deprecation method is vary commonly used method for calculation depreciation by taking the price depreciated is the same amount of Birr but not the same percent each year. Now, we assume that:

At the end of its useful life the item has some salvage value called **salvage value or its residual value.** The **straight - line depreciation** value is calculated as ;

$$\text{Depreciation} = \frac{\text{(Intial value) - (Residual value)}}{\text{useful life n year}}$$

Illustrative Example

100. A machine which costs Birr 5500 when new. If the cost of machine after 8 years worth Birr 3300.

Solution

$$\text{Depreciation} = \frac{\text{(original value) - (Reidual value)}}{\text{useful life}}$$

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$$= \frac{5500 - 3300}{8} = \frac{2200}{8} = 275 \text{ Birr per year}$$

The depreciate value in percent is

$$\frac{275}{5500} \times 100\% = 5\ \% \text{ per year.}$$

101. The cost of a TV was Birr 12000 when new. The useful life is 10 years, with the salvage value of Birr 1800. Find the amount of straight line depreciation per year.

Solution : n = 10

$$\text{Straight line depreciation} = \frac{\text{original value - salvage value}}{\cap}$$

$$= \frac{12000 - 1800}{10} = \text{Birr 1020 per year}$$

102. A publishing company has purchased a new machine at cost of Birr 12000. The machine has useful life of 8 years. If the salvage value at the end of 8 years is Birr 2000. What is its book value at the end of its 5th year.

A. Birr 6820 B. Birr 5685 C. Birr 5750 D. Birr 3925

Solution

$$\text{Depreciation charge} = \frac{(\text{cost value}) - (\text{salvage})}{\text{life time}}$$

$$= \frac{12000 - 2000}{8} = \frac{10000}{8} = 1250 \text{ Birr per year}$$

Its value at the end of 5^{th} year is 12000 - 5(1250) = 5750

Answer: C

13.10 Taxation and Its Calculation

State Governments need money to provide different types of facilities such as schools, health construction of road, police.

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The objectives of taxation are:

- Promoting economic growth
- Ensuring economic growth
- Ensuring people's security

In order to meet the expense on such different thing, the state government impose taxes of different types such as **Direct tax and indirect tax.**

Direct Taxes are income based taxes usually known as income tax. Employment in come tax, business income tax, and land use fee are examples of income tax.

Taxable income is the amount of income subject to tax after deduction of all expenses.

Calculation of Income tax

Income Tax = (Taxable income) × (tax rate) - Deduction

103. The following table shows the income rate with deduction

Taxable monthly income	Tax rate	Deduction
Less than 600	0%	0
600 - 1650	10%	60
1650 - 3200	15%	142.5
3200 - 5250	20%	302.5
5250 - 1090	25%	565
7800 - 10900	30%	955
Over 10900	35%	1500

a. Mr. Afework earns basic salary of Birr 15200 per month. What will be the net pay for Mr. Afework?

b. Mr. Tefera basic salary increase from Birr 8000 to Birr 20,000.

(i) What percentage change in income tax?

(ii) What is the percentage change on net income?

(iii) What percentage change

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Solution

a. Income tax = (15200) (0.35) - 1500

= 5320 - 1500 = 3820

Net pay : 15200 - 3820 = 11380.

b. i. Income tax on Birr 8000 is

(8000) (0.30) - 955 = 2400 - 1445

Income tax on Birr 20000 will be

(20000) (0.35) - 1500 = 7000 - 1500 = 5500.

∴ Percent change in income tax is

$$\frac{5500 - 1445}{1445} \times 100\% = 280.62\%$$

ii. Net income on 8000 will be 8000 - 1445 = 6555

Net income on 20000 will be 20000 - 5500 = 14500

∴ Percentage change in net income is

$$\frac{14500 - 6555}{6555} \times 100\% = \frac{7945}{6555} \times 100\% = 121.20\%$$

iii. Percentage change in basic salary is

$$\frac{20000 - 8000}{8000} \times 100\% = 150\%$$

104. In leasing a building, tax is required from rental income. The following is the tax rate to a total amount of rent per year in Birr.

Amount of tax income	Tax rate	Deducted amount
Upto 1800	0	0
1807 - 7800	10%	180
7801 - 1680	15%	570
16801 - 28200	20%	410

If a shop is rented at Birr 2350 per month for one year, what amount of tax is payable in Birr

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A. 4230 B. 5460 C. 2420 D. 3750

Solution

Total amount of rent per year = 2350 × 12 = 28200

Income tax = (taxable income) - (Deduction)

= (28200) (0.20) - 1410 = 4230

Answer: A

Indirect Tax

Indirect taxes are basically taxes that can be passed onto another entity or individual. Indirect taxes usually imposed on a manufacture or supplier who then passes on the tax to the consumer.

The major types of indirect taxes in Ethiopia are value added tax (VAT), tuner over tax (TOT), excise tax, customs tax, and stamp duty tax.

1. **Value added tax (VAT)** is a tax on goods and services which are purchased. The standard rate of VAT in Ethiopia is 15%. Some items exempted from VAT are books.
2. **Turnover tax (TOT)** is an equalization tax imposed on persons not registered for value added tax (VAT) to fulfil their obligations.

Illustrative Example

105. A man buys a coffee table for Birr 1380, the price includes VAT at 15%.

a. What is the price of the table exclusive of VAT?

b. What is the amounts of value added tax

Solution

a. Let x = price exclusive of VAT

$x + 15\%\, x = 1380$ this implies $x + 0.15x = 1380$

$x\,(1 + 0.15) = 1380$, then $x\,\frac{1380}{1.15} = 1200$

$\therefore$ The price of the table with out VAT is Birr 1200

b. The amount of VAT is 15% of 1200

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$= (0.15)\,(1200) =$ Birr 180.

106. A woman buys a washing machine whose price exclusive of VAT is Birr 24000. If VAT is charged at 15%, how much will the woman pay for the machine?

Solution

2400 + 15% of 2400 = 2400 + 360 = **Birr 2760.**

∴ The woman pay Birr 2760.

107. A dealer purchased an items from producer ar Birr 6000 including 15% VAT. If the dealer sold the item for Birr 8000. Find the amount of VAT liable to the dealer

Solution

Purchasing price $= x + 15\%\, x = 6000$

$1.15x = 6000$ implies $x = \dfrac{6000}{1.15} = 5217.40$

(i) The amount of VAT on purchasing is $6000 - 5217.40 = 782.60$

(ii) Selling price, is $x + 15\% = 8000$

∴ $x\,(1.15) = 8000$ implies $\dfrac{8000}{1.15} = 6956.50$

The VAT on selling price is $8000 - 6956.50 = 1043$

Hence, the amount of VAT liable to the dealer is

1043.47 - 782.60 = **Birr 260.87**

Review Exercise

1. A sum of money Birr 16500 shared into three parts in the ratio 2:4:5. Find the share of each.
2. The sides of a triangle are in the ratio $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$ and its perimeter is 156. Find the length of largest side.
3. If 40 articles cost Birr 350, how much do 55 articles cost?
4. A machine makes 15 articles in 2 hours. How many articles will it make in 6 hours.
5. A farmer employs 15 women to pick his fruit it takes them 3 days. How

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long would it take 10 women to pick the fruit?

6. The salary of a worker Purchasing power first increased by 10% and then decreased by 20%. What is the percentage effect in this salary?
7. When sum of money is decreased by 20% it become Birr 3200. What was the original sum.
8. A TV was sold for Birr 2100 has a mark up of 20% on the selling price. What is the cost of TV?
9. Find the future value and simple interest :
   a. Birr 18000 at 4% for 9 months.
   b. Birr 1200 at 8% for 5 years.
10. Find the compound amount for each deposit and the amount of interest earned
    a. Birr 2000 at 6% compounded annually for 8 years.
    b. Birr 3000 at 8% compounded semi - annually for 6 years.
    c. Birr 15000 at 6% compounded monthly for 10 years.
11. Find the interest rate for each deposit and compound amount.
    a. Birr 8000 accumulating to Birr 12000 compounding quarterly for 5 years.
    b. Birr 4500 accumulating to Birr 6000, compounding monthly for 5 years.
12. Find the effective rate corresponding to each nominal rate:
    a. 6% compounded quarterly
    b. 4% compounded semi- annually.
13. Find the present value of
    a. Birr 12000 at 4.8% compounded annually for 6 years.
    b. Birr 8800 at 5% compounded quarterly for 5 years.
14. Find the future value of each ordinary annuity. If regular payments of,
    a. Birr 9200 deposited at 10% interest compounded semi - annually for 7 years
    b. Birr 12500 deposited at 8.4% inters quarterly for 10 years.

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15. How much must be deposited each month into an ordinary annuity paying 8.4% compounded monthly, to attain a balance of Birr 40000 after five year?

16. Find the present value of each ordinarity annuity.

a. Payments of Birr 900 each year for 16 year at 6% compounded annually.

b. Payments of Birr 5000 quarterly for 10 years at 8% compounded quarterly.

17. A machine costing Birr 12000 is amortized by monthly payments at 12% annual interest rate what are the payments if the term of the annuity is 3 year.

Answer to Review Ex. 13

1. Let x shared value: $2x + 4x + 5x = 16500$, $x = 1500$.

   Then each of shared value 3000, 6000 and 7500.

2. $\frac{1}{2} : \frac{1}{3} : \frac{1}{4} = \frac{1}{2}(12) : \frac{1}{4}(12) = 6 : 4 : 3.$

   $\therefore$ largest perimeter: $\frac{6}{13}(156) = 72$ unit

3. $\frac{350}{40} = \frac{x}{55}$ implies $x = \frac{350}{40}(55) = 481.25$

4. $\frac{15}{2} = \frac{x}{6}$ implies $x = \frac{(15)(6)}{2} = 45$ articles

5. $(15)(3) = (x)(10)$ then $x = \frac{45}{10} = 4.5$ days.

6. $(x + 10\%x)$ then $(x + 10\%x) - 20\%$ of $(x + 10\%$ of $x)$

   $= 1.1x - 0.2(1.1x) = 0.88x.$

   $\therefore$ Net effect $= \left(\frac{x - 0.88x}{x}\right) \times 100\% = 12\%$

7. $x - 20\%x = 3200$ implies $x = \frac{3200}{0.8} =$ Birr 4000.

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8. Mark up percent $= \frac{\text{mark up}}{\text{selling price}} \times 100 = 25$

$\therefore$ Mark up = (selling price) = 0.25(21000) = **Birr 525**

9. a. interest: $I = PRT = (18000)(0.04)\left(\frac{9}{12}\right)$ = **Birr 540**

Future value : A = P + PRT = 18000 + 540 = **18540**

b. I = PRT = 1200 (0.08)5 = **Birr 480**

A = P + PRT = 1200 + 480 = **Birr 1680**

10. a. A = P (1 + r)n = 2000 (1 + 0.06)8 = **Birr 3187.70**

I = A - P = 3187.70 - 2000 = **Birr 1187.70**

b. $P(1+r)^n = 3000\left(1+\frac{0.08}{2}\right)^{6(2)}$ = 3000 (.01)12 = **Birr 4803.**

I = A - P = 4803 - 3000 = 1803.

c. $A = P(1+r)n = 15000\left(1+\frac{0.06}{12}\right)^{10(12)} = 27290.95$

I = A - P = 27290.95 - 15.00 = 12290.95

11. a. $A = P(1+r)^n$ implies. $(1+r)^n = \frac{A}{P} = \frac{12000}{8000} = \frac{3}{2}$

$\left(1+\frac{r}{4}\right)^{32} = \frac{3}{2}$, then $1+\frac{r}{4} = \sqrt[32]{\frac{3}{2}} = 1.013$

$\frac{r}{4} = 0.013$, in this case, r = 5.1%

b. $1 + r = \sqrt[n]{\frac{A}{P}}$ implies $r = \sqrt[n]{\frac{A}{P}} - 1$

$\frac{r}{12} = \left(\frac{6000}{4500}\right)^{1/60} - 1$ implies r = 6.7%

12. a. $r_E = \left(1+\frac{r}{4}\right)^m - 1 = \left(1+\frac{0.06}{4}\right)^4 - 1 = 6.14\%$

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b. $r_E = \left(1 + \frac{r}{4}\right)^m - 1 = \left(1 + \frac{0.04}{2}\right)^2 - 1 = 4.04\%$

13. a. $A = P(1 + r)^n$ implies $P = \frac{A}{(1 + r)^n} = A(1 + r)^{-n}$

$\therefore \quad P = 12000(1 + 0.048)^{-1} = 12000(0.954) = 1145.38$

b. $P = A(1 + r)^{-n} = 8800\left(1 + \frac{0.05}{4}\right)^{-5(4)} = 6864.10$

14. a. $S = R\left[\frac{(1 + i)^n - 1}{i}\right] = 9200\left[\frac{\left(1 + \frac{0.10}{2}\right)^{14} - 1}{0.05}\right] = 180307.42$

b. $S = 12500\left[\frac{(1 + 0.021)^{40} - 1}{i}\right] = 12500(61.73) = 771610.50$

15. $S = R\left[\frac{i}{(1 + i)^n - 1}\right]$ implies $R = 40000\left[\frac{0.07}{(1 + 0.07)^{60} - 1}\right] = 49$

16. $R = P\left[\frac{i}{(1 + i)^{-n}}\right] = 12000\left[\frac{0.01}{1 - (1 + 0.01)^{-16}}\right] = 9095$

b. $P = 5000\left[\frac{1 - (1 + 0.02)^{-40}}{0.02}\right] = 136777.40$

17. $R = P\left[\frac{i}{(1 + i)^{-n}}\right] = 12000\left[\frac{0.01}{1 - (1 + 0.01)^{-16}}\right] = 398.70$

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